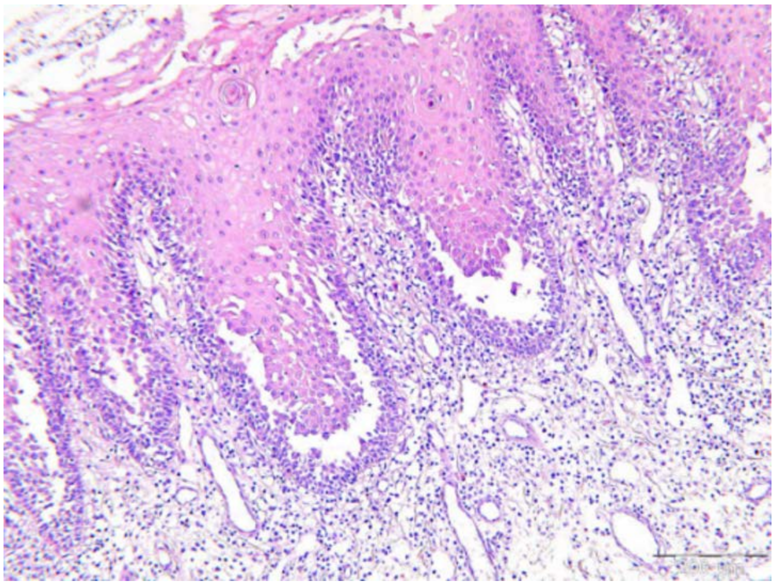
Dermatology AIIMS 2017

Question 1:

The histopathological slide given below is suggestive of:



a) Psoriasis vulgaris

b) Leishmaniasis

c) Mycosis fungoides

d) Pemphigus vulgaris

Question 2:

A 23-year-old patient came with complaints of non-scarring alopecia in the vertex region of his head. Histological examination shows loss of hair with haemorrhage into the follicular areas and mild perinuclear lymphocytic infiltrate. What is the diagnosis?

a) Alopecia areata

b) Telogen effluvium

c) Tinea infection

d) Trichotillomania

Question 3:

Which of the following diseases is unlikely to present with skin lesions on the shin?

a) Sarcoidosis

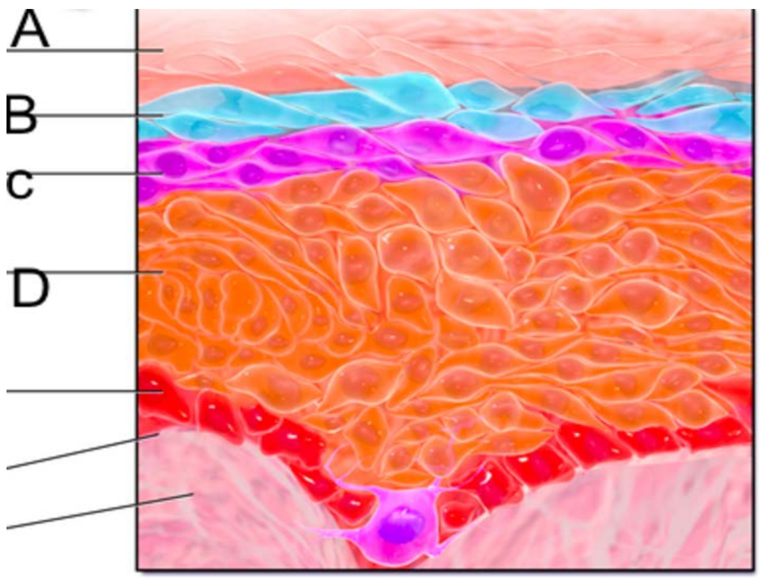
b) Diabetes

c) Hyperthyroidism

d) Hypothyroidism

Question 4:

Identify the layer of skin (in the image given below) in which desmosomes are abundant.



a) B

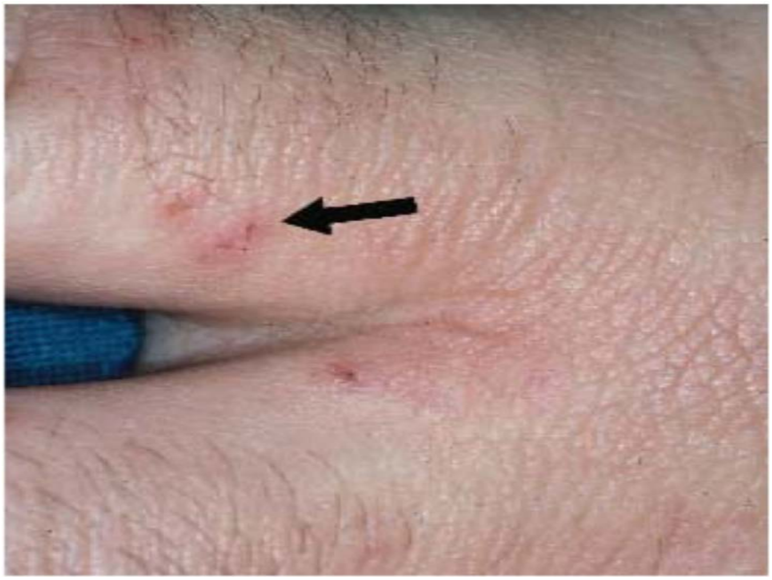
b) A

c) D

d) C

Question 5:

The following image depicts:



a) Scabies

b) Ichthyosis

c) Contact dermatitis

d) Urticaria

Question 6:

A female came with complaints of bone and joint pain. NSAIDs were prescribed. After a few days, she presents again complaining vesicobullous lesions healing to give a brownish discoloration over the face (upper lip area) as shown in the image. This most probably suggests a diagnosis of

a) Malaria

b) Dengue

c) Chikungunya

d) Fixed drug eruption

Question 7:

Dense irregular collagen is found in:

a) Ligaments

b) Dermis

c) Tendons

d) Bones

Question 8:

A patient presented with hair loss and itching of the scalp along with mild discharge. Which of the following is useful for diagnosis?

a) Tzanck smear

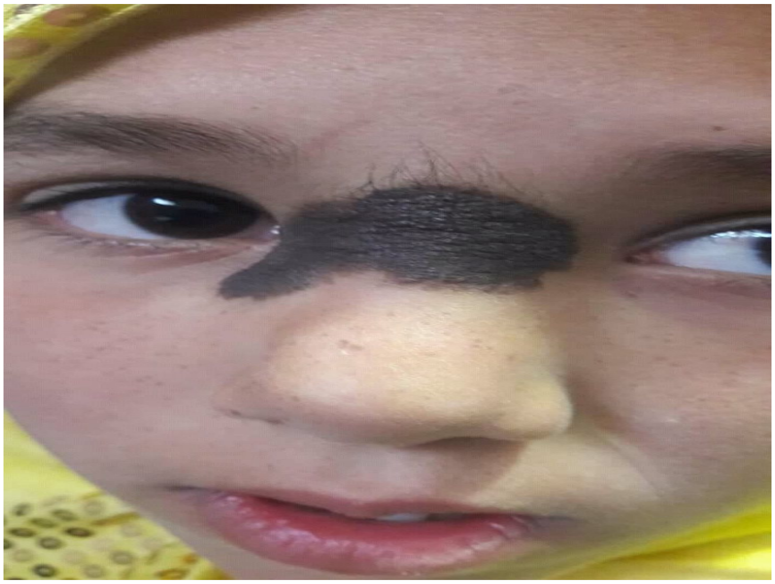
b) Gram stain

c) Slit skin smear

d) KOH mount

Question 9:

Identify the condition shown below:



a) Epidermal verrucous nevus

b) Congenital melanocytic nevus

c) Melanoacanthoma

d) Malignant melanoma

Answer Key

Question No.	Correct Option
1	d
2	d
3	d
4	c
5	a
6	d
7	b
8	d
9	b

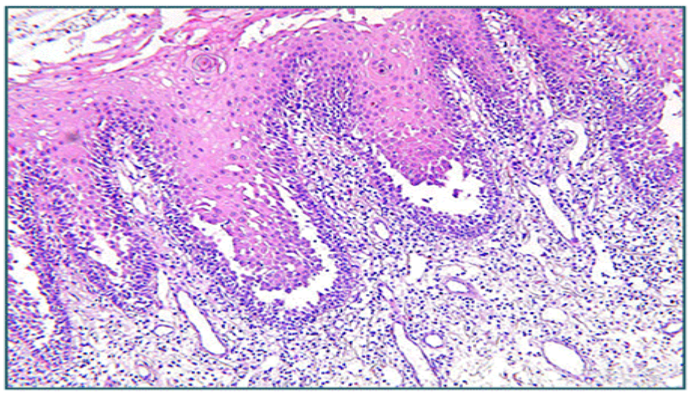
Detailed Explanations

Solution to Question 1:

The given slide shows a row of tombstones appearance, which is classically seen in pemphigus vulgaris.

In this condition, there is a suprabasal splitting of the epidermis leading to blister formation. The basal layer remains adherent to the basement membrane and gives the resemblance to a row of tombstones as shown in the histology image below.

Pemphigus Vulgaris -
Row of tombstones appearance

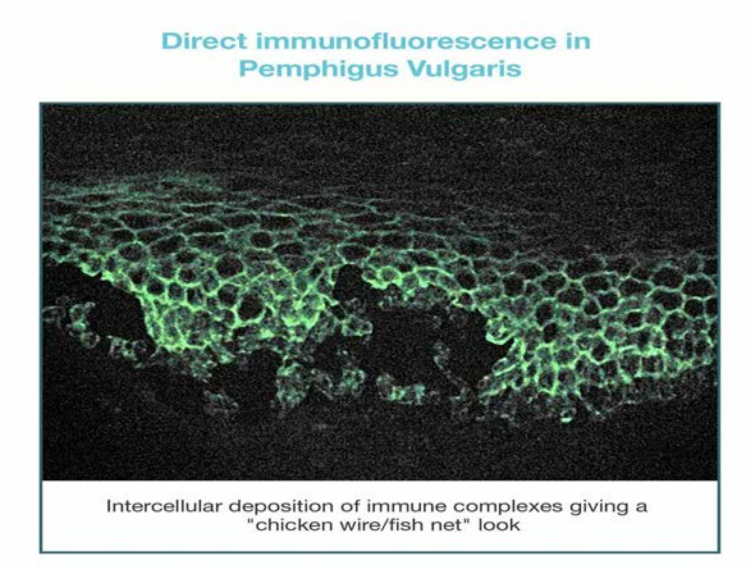


Pemphigus vulgaris is a chronic immunobullous disorder characterized by the presence of antibodies against desmosomal proteins, desmogleins 1 and 3 expressed in the skin and mucosa.

Clinically, the patients present with ill-defined, irregular, slow-to-heal oral lesions and erosions. Flaccid blisters filled with clear fluid may arise on normal skin or an erythematous base. The rupture of such blisters produces painful erosions and heals without scarring.

Nikolsky's sign is positive which is elicited by applying firm tangential sliding pressure with a finger on lesional or perilesional skin. This leads to the separation of the normal■looking epidermis from the dermis, producing erosion.

Direct immunofluorescence is considered the most accurate confirmatory test of this condition. It shows intercellular IgG and C3 deposition on the surface of keratinocytes throughout the epidermis in normal, perilesional skin, in a characteristic fishnet/chickenwire appearance as seen below.



Solution to Question 2:

The clinical scenario depicting a young adult presenting with non-scarring alopecia in the vertex region and histology showing perifollicular hemorrhage is suggestive of trichotillomania (trichotillosis).

Trichotillomania has to be differentiated from alopecia areata (Option A). Trichotillomania presents with incomplete hair loss within a patch with varying lengths of hair whereas alopecia areata shows smooth hairless patch with sparing of white hairs. Regular pitting of nails and exclamation mark hairs are seen in alopecia areata. When the diagnosis is uncertain, a scalp biopsy can be done. Perifollicular hemorrhage is characteristic of trichotillomania while peribulbar lymphocytic infiltrate is diagnostic of alopecia areata.

Trichotillomania is classified under obsessive-compulsive and related disorders. It is characterized by recurrent body-focused repetitive behavior (hair pulling) and repeated attempts to decrease or stop the behavior. It is commonly seen in females. The hair loss is commonly seen over the vertex but can involve eyelashes, eyebrows, and beard.

The most effective treatment for trichotillosis is habit-reversal training (HRT).

Other options:

Option B: Telogen effluvium is diffuse hair loss resulting in non-scarring alopecia due to loss of telogen hairs due to fever, pregnancy, hypothyroidism or anemia.

Option C: Tinea infection shows patchy hair loss, resulting in non-scarring/scarring alopecia. Easy pluckability, broken hairs and wood's lamp positivity are features.

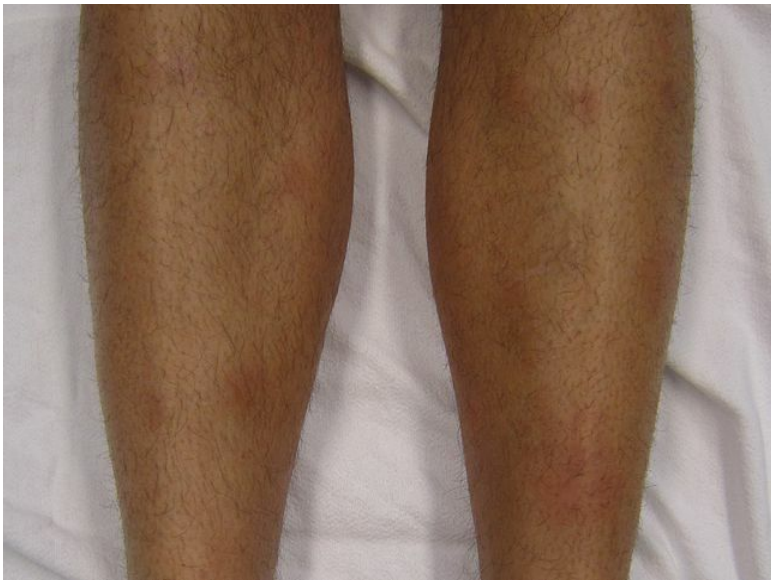
Solution to Question 3:

Hypothyroidism is unlikely to present with skin lesions on shin. Myxedema (severe hypothyroidism) and pretibial myxedema are different entities. Pretibial myxedema is the infiltrative dermopathy (mucin deposition) seen over shin in Grave's disease, a type of hyperthyroidism.

Other options :

Option B: Diabetes: Necrobiosis lipoidica diabeticorum is classically seen on the anterior leg.

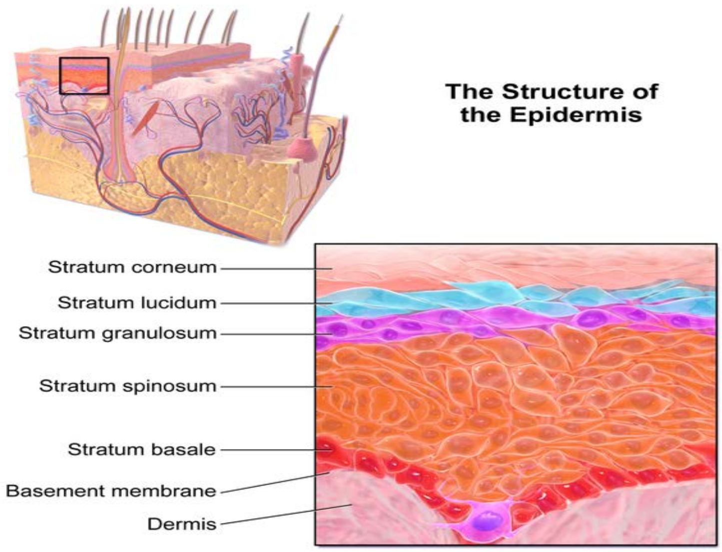
Option A: Sarcoidosis: Erythema nodosum may be seen on the anterior leg.



Solution to Question 4:

Desmosomes are abundant in the stratum spinosum (Option C, label D in the image) layer of the epidermis of the skin.

Desmosomes are adhesion complexes in epidermis which link adjacent keratinocytes. They appear like spines on microscopy and hence this layer is also called the prickle layer. Defects of desmosomal proteins, inherited or acquired, result in blistering disorders.



Solution to Question 5:

The given image shows raised, brownish thread-like lesions in the interdigital distribution (burrows), typically seen in scabies.

Scabies is an ectoparasitic infection caused in humans by Sarcoptes scabiei. It is transmitted by close personal contact. Pruritis with nocturnal exacerbation and contact cases within a family should raise the suspicion of scabies. Typical locations of lesions are the finger webs, the flexor surfaces of the wrists, the elbows, the axillae, the buttocks and genitalia and the breasts of women (Circle of Hebra). In adults, the face is often spared. Pruritic papules and nodules on male genitalia and breasts serve as valuable diagnostic clues in the absence of burrows.

Dermoscopy of the burrows shows jet■with■contrail appearance. Microscopy of scraping from burrows can be done to demonstrate mites.

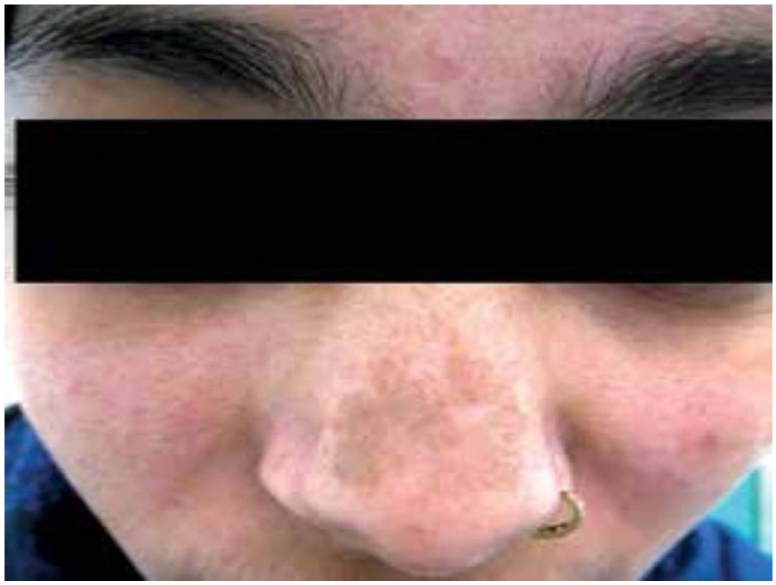
A single overnight application of 5% permethrin cream from neck to toe in adults and from head to toe in children is the drug of choice for the treatment of scabies in all ages including infants (at least 2 months old) and pregnant women. All family members, even if asymptomatic, need to be treated simultaneously for this condition to prevent reinfection. Oral ivermectin can also be used.

Solution to Question 6:

The patient has a history of consumption of NSAIDs, following which there is an appearance of vesiculobullous lesions on the lips, evolving to give a brownish discolouration. The most probable diagnosis is fixed drug eruption (FDE).

A strong differential diagnosis is post-chikungunya pigmentation (PCP) of the face seen on the nose and cheeks (chik sign). However, this is not the answer because there is no antecedent history of fever, myalgia or other symptoms suggestive of chikungunya. PCP often appears as the fever subsides and may persist for about 3-6 months. PCP mimics melasma and may be misdiagnosed if proper history is not taken.

The image below shows chik sign:



Fixed drug eruption (FDE) is a type of delayed hypersensitivity reaction. It is characterized by well-defined brown, hyperpigmented plaques that may follow vesiculobullous lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. They usually present 30 min to 8 h after drug exposure and recur at the same location on re-exposure. NSAIDs, Paracetamol and Co■trimoxazole are the most common drugs causing FDE.

Solution to Question 7:

Dense irregular collagen is found in dermis.

Irregular connective tissue is of three types: Loose, dense and adipose. Dense connective tissue is located in regions which are under consistent mechanical stress and where protection is required for the ensheathed organs. The abundant collagen fibres in the dermis run in a haphazard manner. This protects the skin.

It is usually present in :

• Reticular layer of the dermis

• Superficial connective tissue sheaths of the muscle and nerves

• Adventitia of large vessels

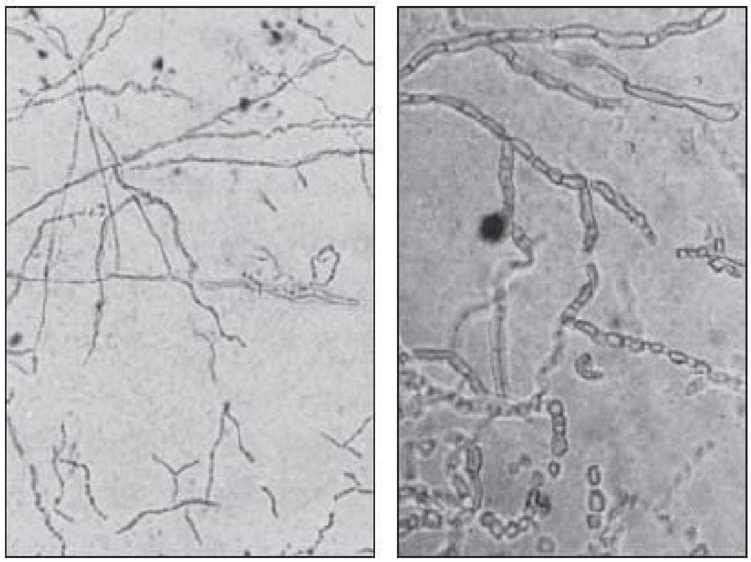
• Capsules of various organs and glands (e.g, testis, sclera, periosteum, perichondrium)

On the other hand, dense regular collagen fibres are found in tendons.

Solution to Question 8:

The patient has a history of itching and mild discharge and examination shows a well-demarcated patch of alopecia, with scaling and broken hair shafts. This is characteristic of tinea capitis. Though the diagnosis is clinical, a KOH mount is used for confirmation.

Scrapings from the edge of a lesion are transferred to a slide, to which KOH is added. It is then examined under the microscope for the presence of hyphae, as shown below.



Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

• Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.

• Ectothrix - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.

• Kerion - a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.

• Favus - is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii

Treatment:

• Terbinafine: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.

• Itraconazole: 2-4 mg/kg/day for 4-6 weeks.

Solution to Question 9:

The image shows a child with a well-circumscribed, black, heterogenous lesion covered with hair. This is characteristic of a congenital melanocytic nevus (CMN). They may be present at birth or develop shortly thereafter. They are benign and can be brown, black, purplish, or red, and are usually a palpable lesion.

They can be classified based on size: small (<1.5 cm), medium (1.5-19.9 cm), and large or giant (>20 cm). Larger CMN have a higher risk of developing malignant melanoma.

Management of CMN depends on the size and location of the lesion. Smaller nevi may be observed, while larger nevi or those at a higher risk for complications may require surgical excision, particularly if there is a concern for malignant transformation.

Complications of CMN include neurological abnormalities (intraparenchymal melanosis is the commonest finding), malignant melanoma, and other tumors like rhabdomyosarcoma.

The images below show different melanotic skin lesions:

Melanoacanthoma

Malignant melanoma

Epidermal verrucous naevus

Dermatology AIIMS 2018

Question 1:

Which of the following nerves is biopsied for the diagnosis of pure neuritic leprosy?

a) Median nerve

b) Superficial branch of radial nerve

c) Ulnar nerve

d) Lateral popliteal nerve

Question 2:

Which component of cement causes allergic dermatitis among construction workers in India?

a) Cobalt

b) Nickel

c) Iron

d) Chromate

Question 3:

Which of the following is a drug used in the treatment of tinea cruris for which widespread resistance is found to be developing?

a) Itraconazole

b) Griseofulvin

c) Terbinafine

d) Voriconazole

Question 4:

A patient came with a history of fever, body ache, and joint pain. He was given NSAIDs. A few days later, he developed brownish pigmentation over the nose. What is the diagnosis?

a) Melasma

b) Fixed drug eruption (FDE)

c) Dengue

d) Chikungunya

Question 5:

A mother has a pearly white papule on her forehead (as shown in the image below) which her infant also develops. What is the cause?

a) HPV

b) Pox virus

c) Adenovirus

d) Herpes virus

Question 6:

A disease associated with the occurrence of blistering and peeling of over >30% of body surface area is usually caused by:

a) Viral infection

b) Drugs

c) Fungal infection

d) Auto-immunity

Question 7:

A 23 year old patient presents with lesions in the groin as shown in the image below. Which of the following cannot be a cause of these lesions?

a) Trichophyton

b) Microsporum

c) Aspergillus

d) Epidermophyton

Question 8:

A patient presents with multiple nodulocystic lesions on the face. Which among the following is the drug of choice in this condition?

a) Steroids

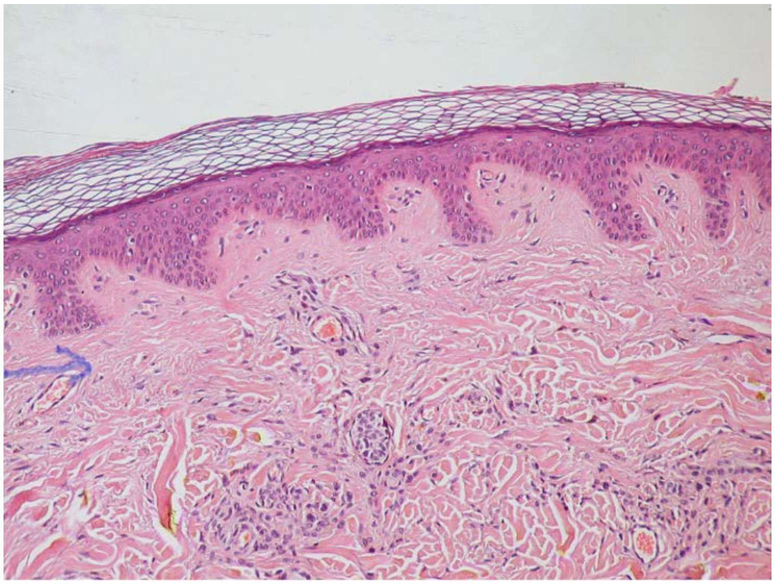
b) Isotretinoin

c) Erythromycin

d) Tetracycline

Question 9:

What is the type of connective tissue present in the area marked by the arrow in the given image?



a) Dense regular

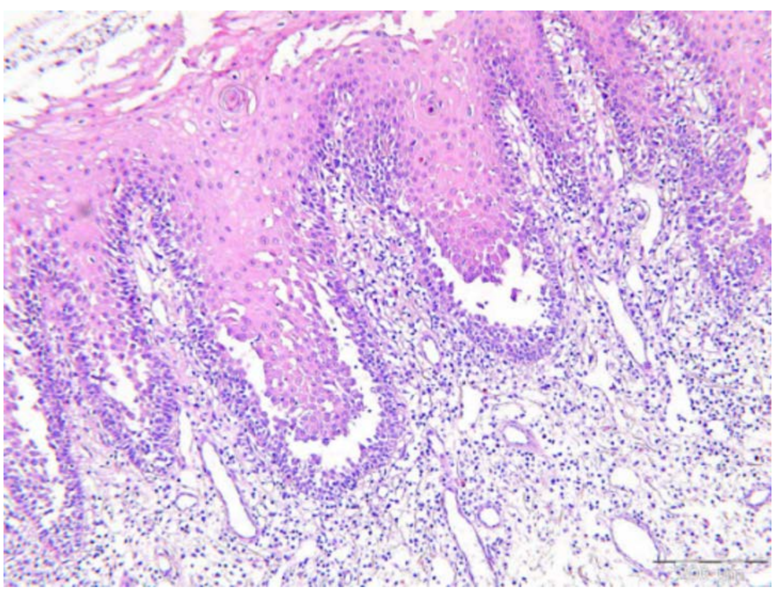
b) Dense irregular

c) Loose irregular

d) Specialised

Question 10:

The histopathological slide given below is suggestive of:



a) Psoriasis vulgaris

b) Leishmaniasis

c) Mycosis fungoides

d) Pemphigus vulgaris

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	d
5	b
6	b
7	c
8	b
9	b
10	d

Detailed Explanations

Solution to Question 1:

Superficial sensory branch of the radial nerve is most commonly used for nerve biopsy in neuritic leprosy.

Although any nerve involved would show the pathology, it is prudent to biopsy a branch of a sensory nerve, rather than a mixed nerve or major nerve trunk. So, the ulnar nerve will not be the best answer despite it being the most common peripheral nerve involved in neuritic leprosy.

The common nerves and sites where nerve biopsy is performed in leprosy are as follows:

• Superficial sensory radial nerve branch at the wrist

• Ulnar cutaneous nerve near the ulnar border of the base of the hand

• Sural nerve near the ankle

Note: Nerve biopsy is the gold standard for confirmation of pure neuritic leprosy.

Solution to Question 2:

Chromate is the most frequent allergen in cement causing occupational contact dermatitis. The most common allergen causing allergic contact dermatitis in India is nickel. It is commonly found in artificial jewelry. However, among construction workers, chromate is more common than nickel. The reported prevalence of allergic contact dermatitis (ACD) to chromate among construction workers is usually more than 10%.

Allergic contact dermatitis (ACD) is an inflammatory condition caused due to a delayed hypersensitivity reaction (Type 4 hypersensitivity) to a specific allergen that comes into direct contact with the skin. Some common allergens include nickel (found in artificial jewelry), PPD/Para phenylene diamine (in hair dye), chromate (in cement), rubber or latex, cosmetics, etc. Patch test is used for the diagnosis of ACD.

Patients usually present with pruritus, erythema, swelling, papules, and papulovesicles. In severe cases, large vesicles or blisters may form due to the disruption of the intercellular bridges. If exposure persists the affected area becomes dry, scaly, and thicker with lichenification and fissuring develops later. There is involvement of the covered areas as well as the exposed areas.

Solution to Question 3:

The use of terbinafine as the agent of choice for tinea cruris and corporis has been limited lately due to the development of resistance.

Tinea cruris (also known as jock itch) is a dermatophyte infection involving the crural fold. The most common cause is Trichophyton rubrum.

T. rubrum isolates have been found to exhibit resistance against terbinafine.

Note: The question is based on research papers about Trichophyton rubrum strains exhibiting primary resistance to terbinafine.

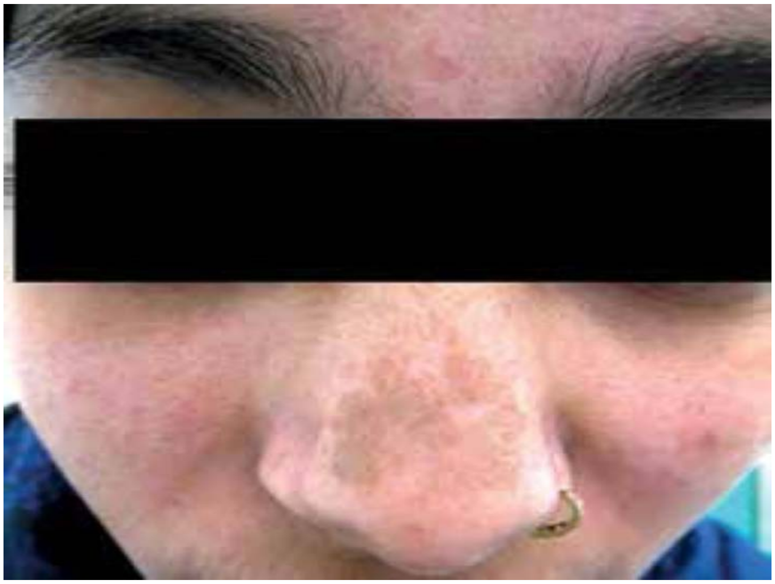
Solution to Question 4:

The given history of fever, joint pain, and centrofacial hyperpigmentation points toward a diagnosis of chikungunya. Post chikungunya pigmentation (PCP) usually appears after the fever subsides and persists for about 3-6 months. Hyperpigmentation associated with chikungunya fever is macular and most commonly affects the nose and cheeks.

Another close differential is fixed drug eruptions (FDE). FDE has a shorter latency, usually presents 30 min to 8 h after drug exposure, and recurs at the same location on re-exposure. It presents with hyperpigmented plaques that may be preceded by vesiculobullous lesions.

Hyperpigmentation of the centrofacial region (nose and cheeks) mimics melasma and may be misdiagnosed if proper history is not taken.

Nose pigmentation is striking in cases of chikungunya fever and is termed chik sign. The image below shows chik sign on the nose:



FDE is a type of delayed hypersensitivity reaction. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. NSAIDs, paracetamol and co■trimoxazole are the most common drugs causing FDE. NSAID■induced FDE mainly affects the genitals and lips.

The image below shows FDE on upper lip area:

Solution to Question 5:

The image shows pearly white umbilicated wart-like lesions with a characteristic dimple at the center, suggestive of molluscum contagiosum, which is caused by pox virus.

The age of peak incidence is between 2 and 5 years. In children, it is mainly caused by the type I molluscum contagiosum virus (MCV-1). There is a relatively higher incidence of MCV■2 type in adults and HIV infection. Sexually transmitted molluscum contagiosum may be present over the genital area.

It has an incubation period of 2 weeks to 6 months. Widespread and refractory lesions are seen most commonly in HIV or patients on immunosuppressive therapy.

It is characterized by the presence of intracytoplasmic eosinophilic inclusion bodies called Henderson-Patterson bodies or molluscum bodies. Pseudo-Koebner phenomenon is seen, that is, new lesions develop along lines of trauma.

Treatment is usually not required, and lesions are likely to resolve spontaneously. Stimulation of the immune response can be done after destructive or inflammatory therapies with salicylic acid, trichloroacetic acid, and imiquimod. Cryotherapy when done needs to be repeated at 3-4 week intervals. Extensive lesions are treated with an antiviral agent like cidofovir.

Solution to Question 6:

Blistering and peeling involving greater than 30% of the body surface area points to the diagnosis of toxic epidermal necrolysis (TEN). TEN is primarily drug■induced, with a culprit drug being demonstrated in approximately 85% of cases.

Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) are both severe mucocutaneous reactions, characterized by blistering and epithelial sloughing. The two terms describe phenotypes within a severity spectrum, where SJS is the less extensive form and TEN is the more extensive

Common drugs that can cause SJS/TEN are:

• Allopurinol

• NSAIDs- oxicam group

• Anti-epileptics - phenobarbital, phenytoin, carbamazepine, lamotrigine

• Antibiotics - penicillins, cephalosporins, sulfamethoxazole, sulfasalazine, nevirapine

Disease	Epidermal detachment	Purpuric macules and atypical target lesions
SJS	<10%	Present
Overlap SJS-TEN	10–30%	Present
TEN with spots	>30%	Present
TEN without spots	>30% (large sheets of epidermal loss)	Absent

Solution to Question 7:

The image shows a well-circumscribed, annular plaque in the groin region which is characteristic of tinea cruris. Tinea cruris is a dermatophytosis. Trichophyton, Microsporum and Epidermophyton are dermatophytes whereas Aspergillus is not.

Dermatophytes include 3 genera:

• Trichophyton species infect skin, hair, and nails and include T. rubrum, T. mentagrophytes, T. tonsurans, T. schoenleinii, and T. violaceum.

• Microsporum contains M. gypseum, M. canis, and M. nanum. These infect only hair and skin (not nails).

• Floccosum is the only species in the genus Epidermophyton, It infects only skin and nails (not hair).

Aspergillus spp. produce large necrotic lesions like ecthyma gangrenosum, and sometimes small papules and cold abscesses.

Solution to Question 8:

The presence of multiple nodulocystic lesions implies that this is a case of severe acne. Isotretinoin is the first-line drug for the treatment of severe acne (nodular acne and acne conglobata).

Systemic retinoids are highly teratogenic and are absolutely contraindicated in pregnant women, those planning to become pregnant, or breastfeeding mothers. Two negative pregnancy tests (on separate occasions) are required for females of childbearing age before starting acitretin or isotretinoin.

Baseline evaluation of serum lipids, transaminases, and complete blood counts should be done before starting any systemic retinoids. These values should be checked monthly for the first 3–6 months and once every 3 months thereafter. Monthly pregnancy tests should be done while on the drug.

Note: Female patients should avoid conception for 3 years after discontinuing acitretin treatment for cutaneous psoriasis.

Solution to Question 9:

The area marked is the reticular dermis. It is made up of dense irregular connective tissue.

It is found in the regions that are under considerable mechanical stress and where protection is given to ensheathed organs. It is usually present in:

• Reticular dermis

• Superficial connective tissue sheaths of the muscle and nerves

• Adventitia of large vessels

• Capsules of various organs and glands (e.g., testis, sclera, periosteum, perichondrium)

On the other hand, dense regular collagen fibers are found in tendons.

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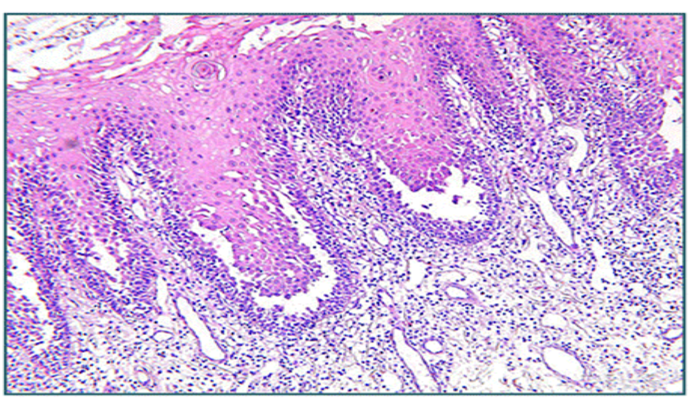
Row of tombstones appearance



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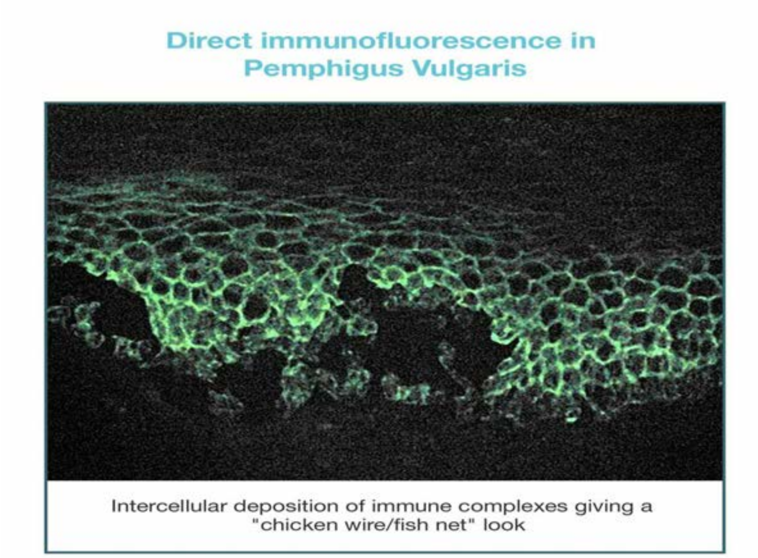
Direct immunofluorescence is considered the most accurate confirmatory test of this condition. It shows intercellular IgG and C3 deposition on the surface of keratinocytes throughout the epidermis in normal, perilesional skin, in a characteristic fishnet/chickenwire appearance as seen below.



Dermatology AIIMS 2019 (May & Nov)

Question 1:

A 5-year-old boy had painful sores in the mouth with a rash over the hands along with fever. Which of these is the causative agent?

- **a)** Pox virus
- **b)** Coxsackievirus
- **c)** HHV-7
- **d)** Parvovirus

Question 2:

The following lesion is seen in a 25-year-old man who came to a primary health centre 5 days after he had unprotected sexual intercourse. On palpation the ulcers were painful. Suppurative inguinal lymphadenopathy was present on the right side. The patient has no previous history of drug allergy. Which kit will be used for the management of this particular sexually transmitted disease?

a) Kit 6/Yellow

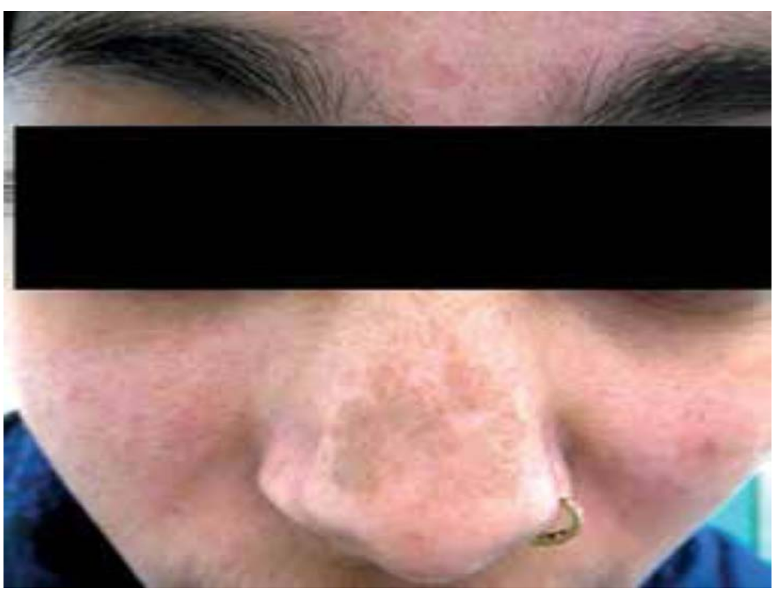
b) Kit 2/green

c) Kit 3/white

d) Kit 1/grey

Question 3:

A patient with fever and joint pain developed pigmentation on the nose after a few days of taking NSAIDs as shown below. What is the diagnosis?



a) Melasma

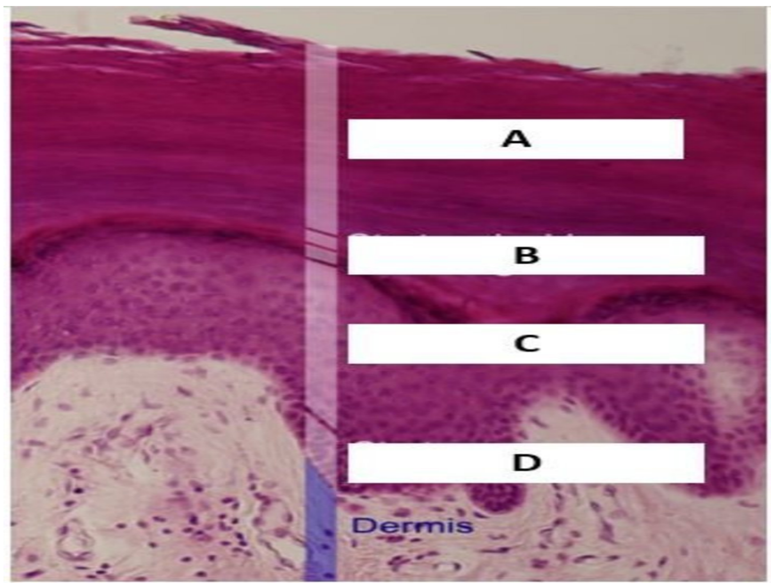
b) Fixed drug eruption

c) Lichen planus

d) Chikungunya

Question 4:

Which cell is not present in the layer marked D?



a) Keratinocyte

b) Merkel cell

c) Melanocyte

d) Langerhans cells

Question 5:

Topical steroids are most effective in:

a) Dermal atrophy

b) Eczematous dermatitis

c) Bullous lesions due to HSV

d) Rosacea

Question 6:

You have ordered a patch test for an 8-year-old boy with a history of allergic contact dermatitis. After the application of the patch, when should you ask him to return for removal?

a) 24 hours

b) 72 hours

c) 48 hours

d) 96 hours

Question 7:

A patient was treated with steroids for psoriasis. On stopping the treatment, he develops fever, malaise, and lesion as seen in the image below. What is the most probable diagnosis?

a) Pustular psoriasis

b) Staphylococcal infection

c) Acute generalised exanthematous pustulosis

d) Subcorneal pustulosis

Question 8:

Which of the following does not present with maculopapular rash?

a) Zoster (shingles)

b) Rubella

c) Dengue (during febrile phase)

d) Measles

Question 9:

A patient presents with flaccid bullous lesions involving the oral cavity and the skin. He has lesions as shown below. Acantholytic cells are seen on Tzanck smear. What is the most probable diagnosis?

a) Pemphigus foliaceus

b) Pemphigus vulgaris

c) Dermatitis herpetiformis

d) Bullous pemphigoid

Question 10:

A 10-year-old child presents with hair loss. The hair loss is patchy as shown in the image below. Black dots are noted in areas of hair loss. Which of the following types of tinea is this child suffering from?

a) Ectothrix

b) Endothrix

c) Kerion

d) Favus

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	d
5	b
6	c
7	a
8	a
9	b
10	b

Detailed Explanations

Solution to Question 1:

The presence of fever, along with painful sores in the mouth and small, tense vescicles with surrounding erythema on the palms and soles in a child point towards the diagnosis of hand-foot-mouth disease. The most common cause of which worldwide is coxackievirus A16.

Spread is by droplets or faecal contamination and the incubation period is 7 days. The fever is generally mild, and is followed by the appearance of lesions. The characteristic lesions are slightly oval, with a grey blister roof and a narrow rim of erythema. The oral lesions are aphthoid and are less painful than aphthous ulcers.

Only symptomatic treatment is given. The lesions generally disappear in 7 - 10 days.

Solution to Question 2:

The appearance of painful genital ulcers, along with unilateral suppurative inguinal lymphadenopathy, 5 days after unprotected sexual intercourse best fits the symptom cluster of a non - herpetic genital ulcer. The treatment of which is kit 3 / white kit in case of no penicillin allergy.

The above-shown clinical image and the findings are consistent with a diagnosis of chancroid.

Soft Chancre/Chancroid:

• The causative organism is Haemophilus ducreyi.

• The incubation period is 3 to 10 days.

• It presents with multiple painful ulcers on genitals.

• It may be accompanied by unilateral painful, suppurative inguinal lymphadenopathy.

• Treatment: Inj. Benzathine penicillin 2.4 million units + tab. Azithromycin 1 gm single dose

Note: In the syndromic approach to sexually transmitted infections, kit assignment based on the identification of a cluster of symptoms rather than making an actual diagnosis is given emphasis.

Solution to Question 3:

The given clinical history of fever and joint pain and the image showing hyperpigmented macules on nose (chik sign) is suggestive of Chikungunya.

Post-chikungunya pigmentation (PCP) most commonly affects the nose and cheeks. It often appears as the fever subsides and may persist for about 3-6 months.

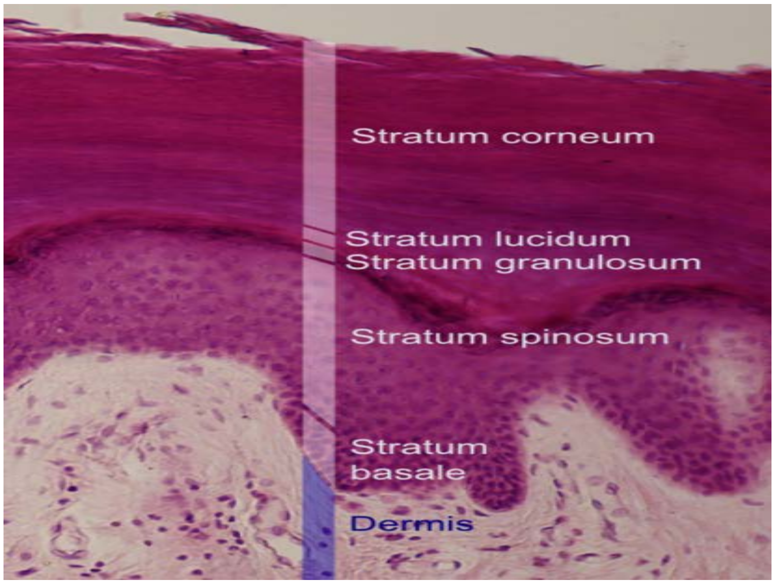
PCP mimics melasma and may be misdiagnosed if proper history is not taken.

Another close differential is fixed drug eruptions (FDE). FDE is a type of delayed hypersensitivity reaction. It is characterized by well-defined brown, hyperpigmented plaques that may follow vesiculobullous lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation.They usually present 30 min to 8 h after drug exposure and recur at the same location on re-exposure. NSAIDs, Paracetamol and Co■trimoxazole are the most common drugs causing FDE.

The image below shows FDE on upper lip area:

Solution to Question 4:

The layer marked as D is stratum basale. Langerhan's cells are not present in stratum basale. Langerhan's cells are predominantly present in stratum spinosum.



Keratinocytes, melanocytes, and Merkel discs are all seen in stratum basale:

(Option A) Keratinocytes are predominant cells of the epidermis and constitute the majority of the stratum basale.

(Option B) Merkel discs are type 1 slow-adapting touch receptors.

(Option C) Melanocytes are dendritic cells which are derived from the neural crest cells. The melanosomes synthesized by the melanocytes migrate to the adjacent keratocytes through the dendritic processes

Solution to Question 5:

Topical steroids are the drug of choice in the treatment of eczematous dermatitis.

Other options:

• Steroids cause atrophy and hence is contraindicated in patients with dermal atrophy.

• Herpes infection is associated with immunosuppression so steroids are not useful.

• Routine steroid use not recommended in rosea as it can worsen the condition.

Solution to Question 6:

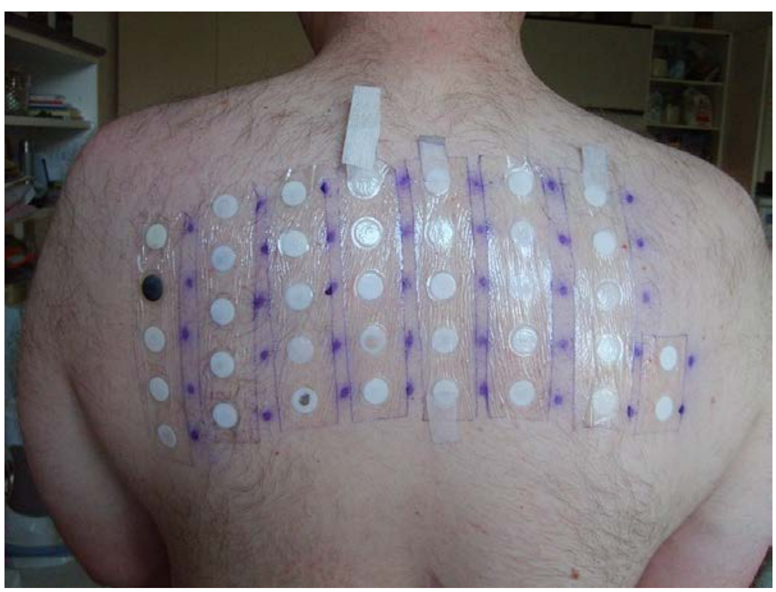
In patch testing, the patch is removed after 48 hours to examine the site of reaction.

Patch testing is a test to detect sensitivity to a specific allergen. A battery of suspected allergens is applied to the patient’s back under occlusive dressings or patches. It is allowed to remain in contact with the skin for 48 hours.

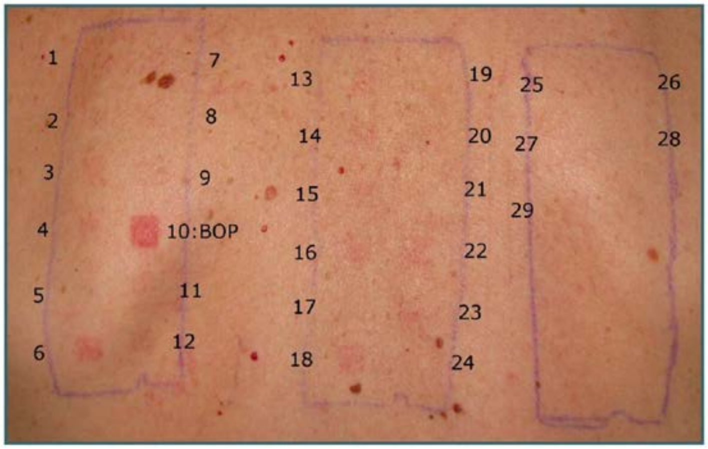
• 48 hours: Patch removal for initial reading to detect delayed hypersensitivity reaction.

• 72-96 hours: Final reading done for confirmation.

Note: Since the question is regarding the timing of patch removal and not regarding the final diagnosis, 48 hours is the answer.



Patch test



Solution to Question 7:

The given clinical scenario points towards the diagnosis of pustular psoriasis.

Pustular psoriasis is a type of psoriasis characterized by tiny pus-filled lesions (pustules) having a characteristic lakes or sheets of pus appearance (seen in the image).

The patient presents with a history of fever and malaise along with the presence of pustules on a circumscribed, fiery red, edematous or scaling plaque. The pus is usually sterile. Sudden withdrawal of systemic steroids is usually the trigger for this condition.

Pustular psoriasis can be divided into:

- Localized pustular psoriasis
- Generalized pustular psoriasis:
- Acute generalized pustular psoriasis (Von Zumbusch)
- Subacute annular and circinate pustular psoriasis
- Acute generalized pustular psoriasis of pregnancy (Impetigo herpetiformis)

Treatment: Acitretin (oral retinoid) is the drug of choice for pustular psoriasis. However, it is contraindicated in pregnancy. Systemic steroids are used instead. Cyclosporine is the second-line drug.

Note: Systemic steroids are never used for the treatment of psoriasis except in the case of Impetigo herpetiformis where they are first-line agents.

Solution to Question 8:

A maculopapular rash is not seen in zoster (shingles).

Shingles cause vesicular lesions that are grouped together and specific to a dermatome.

Shingles

Vesicular lesions grouped together in a dermatome

Maculopapular rashes are seen in:

- Rubella
- Dengue (febrile phase)
- Measles
- Chikungunya (acute infectious phase)
- Erythema infectiosum
- Zika virus
- West Nile virus
- Infectious mononucleosis

Type of rash	Description
Macule	Flatlesion with a well-circumscribed change in skin color < 1 cm
Papule	Elevated solidskin lesion < 1 cm
Vesicle	Smallfluidcontaining blister

Maculopapular rash

Solution to Question 9:

Flaccid lesions, mucosal involvement, and the presence of acantholytic cells are suggestive of pemphigus vulgaris.

Option A: There is no mucosal involvement in pemphigus foliaceus.

Option C: Dermatitis herpetiformis presents with multiple small vesicles and no acantholysis.

Option D: Bullous pemphigoid presents with tense bullae and no acantholysis.

Solution to Question 10:

The given clinical scenario depicting a young child with patchy hair loss and the image showing black dots is suggestive of endothrix type of tinea capitis.

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

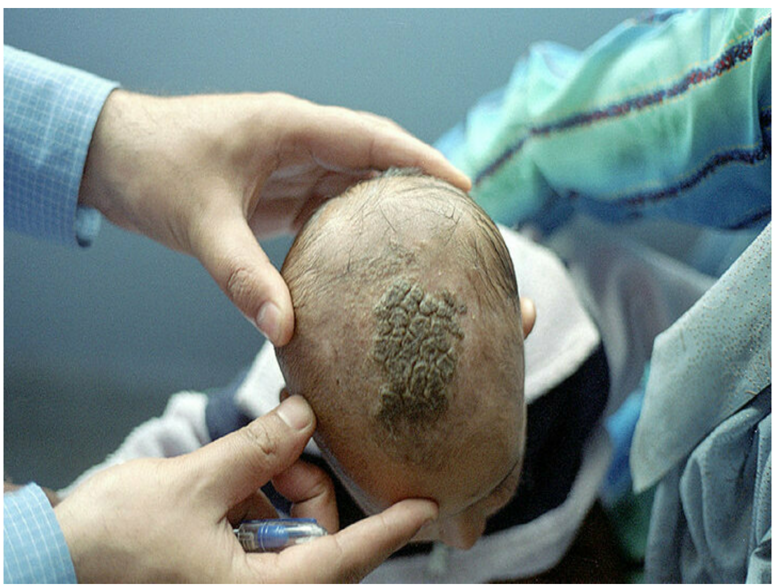
The clinical variants of tinea capitis are:

• Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.

• Ectothrix -a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.

• Kerion -a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.

• Favus - is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii



Treatment:

• Terbinafine: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.

• Itraconazole: 2-4 mg/kg/day for 4-6 weeks.

Dermatology AIIMS 2020 (May and Nov INI-CET)

Question 1:

A female patient presents with active genital warts with a history of recurrence, but her husband doesn't have it. What advice will you give to prevent transmission?

a) Give continuous prophylaxis to the patient

b) Give prophylactic anti-virals to the husband

c) Give anti-virals whenever active lesions are present

d) Don't involve in intercourse when active lesions are present

Question 2:

A male baby born to a non-consanguineous couple develops blisters on areas of friction. Their previous baby also had similar complaints and died at the age of 2 weeks. What is the diagnosis?

a) Epidermolysis bullosa

b) Chronic bullous disorder of childhood

c) Congenital syphilis

d) Neonatal pemphigus

Question 3:

A rose gardener with a history of thorn prick presents with the following lesions. What is the possible causative agent?

a) Streptococcus pyogenes

b) Sporothrix schenkii

c) Strongyloides stercoralis

d) Staphylococcus aureus

Question 4:

A school-going child who had applied henna comes to dermatology OPD with the following lesions. What is the diagnosis?

a) Contact dermatitis due to henna

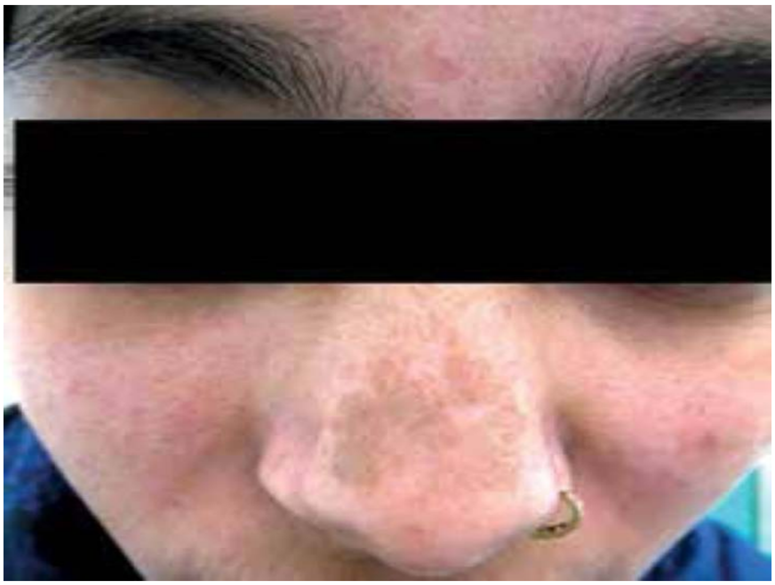
b) Dermatitis artefacta

c) Polymorphous light eruption

d) Vitiligo

Question 5:

A 40-year-old woman who presented with complaints of fever and joint pain develops the following lesion on her nose after a few days of taking NSAIDs. What is your diagnosis?



a) Dengue

b) Chikungunya

c) Melasma

d) Fixed drug eruption

Question 6:

Methylisothiazolinone (MI) in cosmetics is used as?

a) Surfactant

b) Humectant

c) Emulsifier

d) Preservative

Question 7:

The given condition is caused by which drug?

a) Actinomycin

b) Bleomycin

c) Mitomycin C

d) Doxorubicin

Question 8:

Which of the following does not present with maculopapular rash?

a) Zoster (shingles)

b) Rubella

c) Dengue (during febrile phase)

d) Measles

Question 9:

A child presents to OPD with tense bullae over the torso. Biopsy of the lesion showed a subepidermal level of blistering and neutrophil infiltration. What is the drug of choice?

a) Rituximab

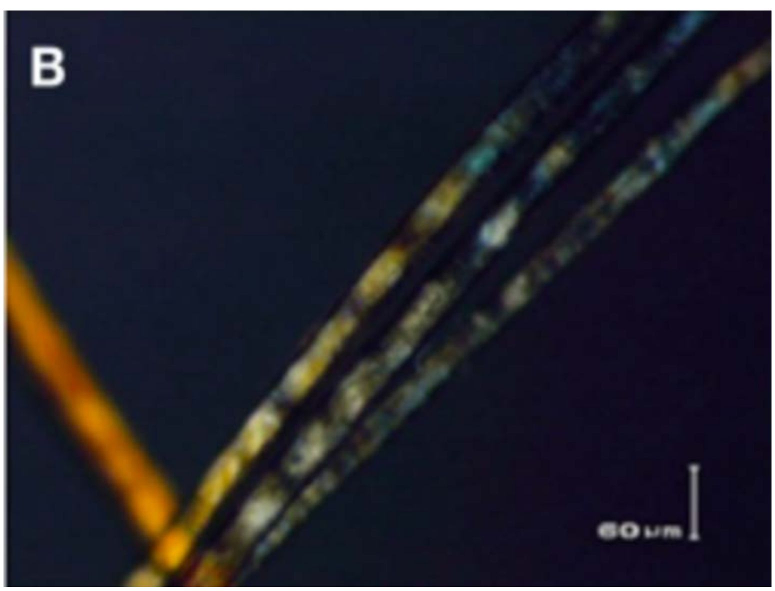
b) Dapsone

c) Cyclosporine

d) Azathioprine

Question 10:

What is the probable diagnosis of this condition affecting hair?



a) Trichorhexis nodosa

b) Trichorexis invaginata

c) Trichothiodystrophy

d) Monilethrix

Question 11:

A patient with a known case of chronic plaque psoriasis with more than 50% skin involvement presents to you. Which of the following will you not recommend?

a) Oral methotrexate

b) NB UVB

c) Oral steroids

d) Oral cyclosporine

Question 12:

Identify the type of skin lesion.

a) Nummular

b) Annular

c) Discoid

d) Target

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	b
5	b
6	d
7	b
8	a
9	b
10	c
11	c
12	b

Detailed Explanations

Solution to Question 1:

The advice to prevent transmission is to not involve in intercourse when active genital warts are present.

Genital warts are caused by HPV and transmitted through the sexual route. Antivirals are not very effective against anogenital warts. Biological treatment options like podophyllin, imiquimod, 5-fluorouracil are given. The best way to prevent transmission of infection is by avoiding sexual intercourse when active lesions are present.

Solution to Question 2:

The clinical history given in the question points to a diagnosis of epidermolysis bullosa congenita.

Epidermolysis bullosa is a mechanobullous disease characterized by blistering in areas of mechanical trauma. Blisters are commonly seen in trauma-prone areas like hands, feet, elbows, and knees. There are three types of epidermolysis bullosa:

• Epidermolysis bullosa simplex - defective K5/K14 protein

• Junctional epidermolysis bullosa - defective Laminin

• Dystrophic epidermolysis bullosa - defective Collagen-7

The image below shows epidermolysis bullosa lesions on the foot:

Other options:

Option B: Chronic bullous disorder of childhood - It typically presents with bullous lesions having a string of pearls/cluster of jewel appearance (image below). It is an IgA-mediated immunobullous disorder. It manifests in early childhood and is not a congenital condition.

Option C: Congenital Syphilis - Syphilitic pemphigus presents as a vesiculobullous disorder. The investigation of choice is VDRL.

Option D: Neonatal Pemphigus - It is an autoimmune blistering disease caused by the transfer of maternal IgG autoantibodies to desmoglein 3 to the neonate through the placenta when the mother is affected with pemphigus. It is clinically characterized by transient flaccid blisters and erosions on the skin and, rarely, the mucous membranes.

Solution to Question 3:

The given image shows the distribution of lesions along lymphatics. This distribution along with the history of thorn pick is suggestive of the causative agent Sporothrix schenkii.

Sporothrix schenkii causes rose gardener's disease. It is a saprophyte that lives on grass and plants. Hence a history of thorn prick in farmers or gardeners is an important clue.

Other causes of skin lesion distributed along lymphatics is fish tankgranuloma caused by Mycobacterium marinum.

Solution to Question 4:

The likely diagnosis in this case is dermatitis artefacta (factitious dermatitis).

Dermatitis artefacta is a psychodermatological excoriation disorder with a female preponderance, commonly affecting adolescents and young adults, where elaborate techniques (using sharp objects, cigarettes, lighters, chemicals etc.) are used to self induce skin lesions.

The morphology of the lesions are varied, with oddly linear, clear-cut and angular edges, surrounded by normal looking skin. These lesions appear with no prodromal stage or associated symptoms as they are self inflicted. They are distributed over easy to reach areas of the body and have a varied appearance based on the mode of injury.

• Option A: Contact dermatitis is due to contact with allergic agents such as paraphenylenediamine (PPD). Presentation can vary from erythema, swelling, papules, vesicles and in severe cases, blisters on the areas of contact with the allergen (henna, in this case). There exists a sequence of inflammatory changes here, unlike the lesions in dermatitis artefacta.

• Option B: Polymorphic light eruptions as the name suggests are seen over sunlight-exposed areas of the body. The lesions are polymorphic and are symmetrically distributed.

• Option C: Vitiligo is characterized by patchy involvement of skin with hypopigmentation.

Solution to Question 5:

The given history of fever and joint pain along with centrofacial hyperpigmentation shown in the image point towards a diagnosis of chikungunya. Hyperpigmentation associated with chikungunya fever is macular and most commonly affects the nose and cheeks.

Nose pigmentation is striking in cases of Chikungunya fever and is termed as Chik sign. It appears as the fever subsides and persists for about 3-6 months.

Hyperpigmentation of the centrofacial face (nose and cheeks) mimics melasma and may be misdiagnosed if proper history is not taken.

Another close differential is fixed drug eruptions (FDE). FDE is a type of delayed hypersensitivity reaction. It is characterized by well-defined brown, hyperpigmented plaques that may follow vesiculobullous lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. They usually present 30 min to 8 h after drug exposure and recur at the same location on re-exposure. NSAIDs, Paracetamol and Co■trimoxazole are the most common drugs causing FDE.

The image below shows FDE on upper lip area:

Solution to Question 6:

Methylisothiazolinone (MI) in cosmetics is used as a preservative to inhibit bacterial growth. It is permitted to be used up to the level of 100 ppm.

Several cases of allergic contact dermatitis due to MI were reported due to its presence in wet wipes, hair cosmetics, facial cosmetics, deodorants, and sunscreens.

Furthermore, cases of airborne exposure to MI causing severe airborne and generalized dermatitis from recently painted walls and even from toilet cleaners have been described.

Solution to Question 7:

The given image shows flagellate dermatitis. It is caused by bleomycin.

Flagellate dermatitis starts as bands of pruritic, flagellate, erythema over the trunk, and extremities within few hours to 6 months after administration. The rash heals with persistent flagellate hyperpigmentation.

No treatment is necessary since the lesions usually disappear when bleomycin is discontinued. If the patient is itchy, managing pruritus with an antihistamine may limit trauma to the skin, and thus avoid further hyperpigmentation.

Another side effect that one needs to watch out for with the use of bleomycin is the development of pulmonary fibrosis.

Solution to Question 8:

A maculopapular rash is not seen in zoster (shingles).

Shingles cause vesicular lesions that are grouped together and specific to a dermatome.

Shingles

Vesicular lesions grouped together in a dermatome

Maculopapular rashes are seen in:

- Rubella
- Dengue (febrile phase)
- Measles
- Chikungunya (acute infectious phase)
- Erythema infectiosum
- Zika virus
- West Nile virus
- Infectious mononucleosis

Type of rash	Description
Macule	Flatlesion with a well-circumscribed change in skin color < 1 cm
Papule	Elevated solidskin lesion < 1 cm
Vesicle	Smallfluidcontaining blister

Maculopapular rash

Solution to Question 9:

This clinical scenario with tense blisters showing subepidermal separation and neutrophilic infiltration at the basement membrane points towards a diagnosis of IgA pemphigoid or linear IgA disease or chronic bullous disease of childhood. The drug of choice is dapsone. It is bacteriostatic against M. leprae.

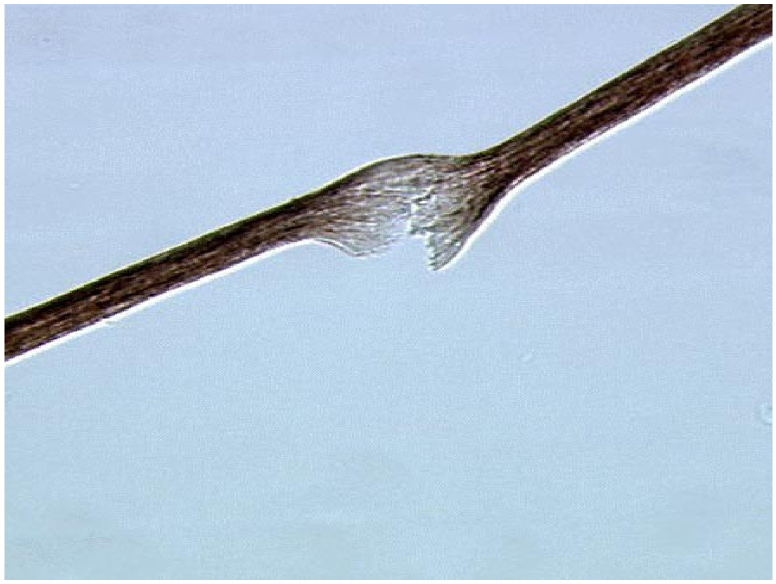
It also has an anti-inflammatory action by decreasing the action of neutrophil enzymes. Hence, it is useful in the treatment of IgA pemphigoid, linear IgA disease and chronic bullous disease of childhood.

Solution to Question 10:

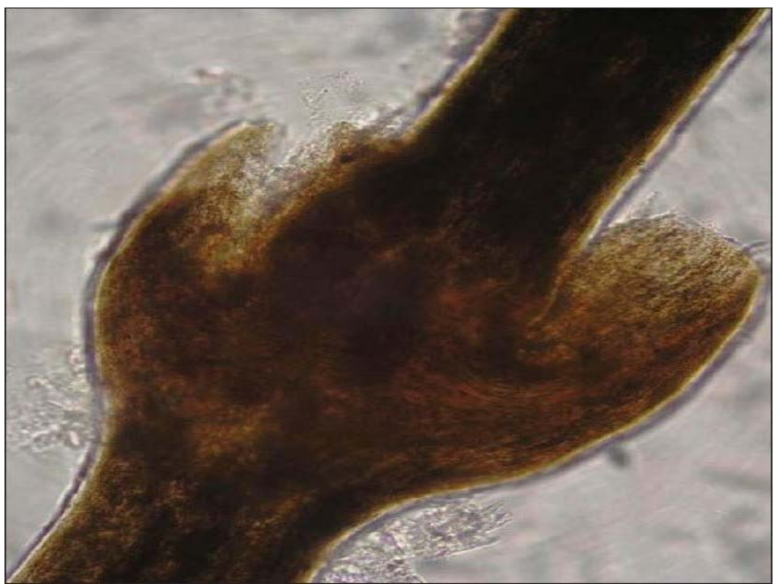
The given clinical image of hair seen under polarising microscopy reveals tiger tail banding (alternating light and dark bands) characteristic of trichothiodystrophy.

Trichothiodystrophy is an autosomal recessive condition causing defective DNA repair. It causes brittle sulfur-deficient hair, photosensitivity, ichthyosis and systemic involvement. Unlike xeroderma pigmentosum (a similar condition with defective DNA repair), there is no risk of skin cancer.

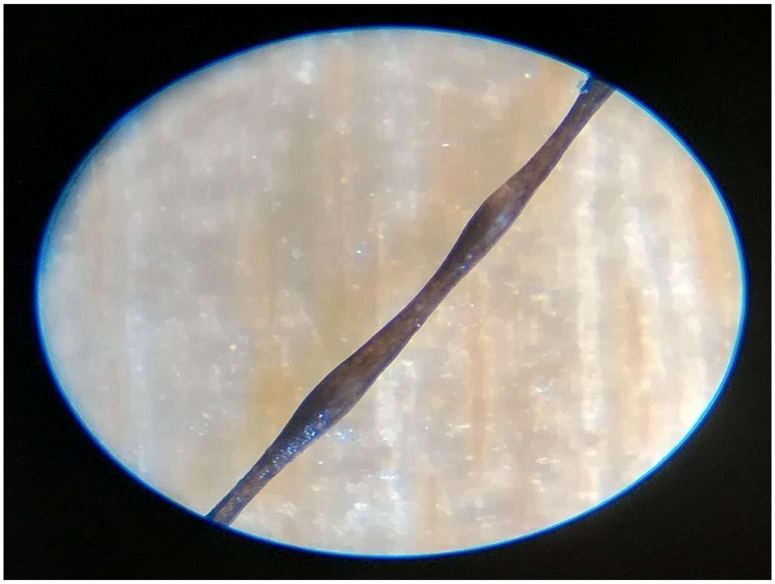
Option A: Trichorrhexis nodosa is associated with node formation and fractures. The classic appearance, shown in the image below, is known as thrust paintbrush appearance as if two paintbrushes are rubbing against each other. It is associated with trauma to the hair shaft and arginosuccinic aciduria.



B- Trichorrhexis invaginata or bamboo hair is characterized by invagination of the distal hair shaft into the proximal portion. It is seen in Netherton's syndrome. It is also called golf-tee or ball and socket appearance as seen below.



Option D: Monilethrix, seen in the image below, is characterized by beaded hair and is fragile at the constricted sites. It occurs as an isolated condition or can occur in Menkes disease.



Solution to Question 11:

Oral/systemic steroids are not used in the management of chronic plaque psoriasis. Topical steroids may be used.

Systemic steroids are avoided because the withdrawal of systemic steroids causes the development of pustular psoriasis.

The only indication of using systemic steroids in psoriasis is generalized pustular psoriasis in pregnancy (impetigo herpetiformis).

Other options maybe used in treatment of extensive (>30%) chronic plaque psoriasis.

Solution to Question 12:

The given image shows ring-shaped lesions with central clearing, which are characteristic of annular lesions.

In annular lesions, only the edges are active and appear raised, or in a different color. They are seen in tinea corporis, granuloma annulare, borderline leprosy, and psoriasis.

Dermatology NEET 2018

Question 1:

A 15-year-old girl presents with itchy lesions on her arm as shown. Her family history is positive for asthma. What could be the most probable diagnosis?

- **a)** Seborrhoeic dermatitis
- **b)** Atopic dermatitis
- **c)** Allergic contact dermatitis
- **d)** Erysipelas

Question 2:

Which of the following is a primary skin lesion?

- **a)** Crust
- **b)** Atrophy
- **c)** Purpura
- **d)** Induration

Question 3:

Which of the following statements is false regarding scleredema?

a) The dermis is 3–4 times thicker than normal

b) Erythema and peau d'orange appearance of the skin can be seen

c) Can occur in association with diabetes

d) Associated with sclerodactyly or Raynauds phenomenon

Question 4:

Identify the pigmentation abnormality in the image shown below.

a) Mongolian spot

b) Naevus of Ito

c) Congenital melanocytic naevus

d) Becker's naevus

Question 5:

What is the most common causative organism for verruca vulgaris?

a) Human papillomavirus-1

b) Human papillomavirus-3

c) Human papillomavirus-10

d) Human papillomavirus-2

Question 6:

Cutis marmorata occurs due to ______

a) Exposure to cold temperature

b) Humidity

c) Exposure to hot temperature

d) An adverse reaction to drugs

Question 7:

Varicella-zoster virus remains dormant in the:

a) Medulla oblongata

b) Dorsal root ganglion

c) Skin

d) Ventral root

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	d
5	d
6	a
7	b

Detailed Explanations

Solution to Question 1:

The clinical scenario depicting pruritic lesions at a flexural site along with a family history of atopy (asthma) is suggestive of atopic dermatitis (AD) or atopic eczema (AE). The scratch marks and excoriations shown in the image are due to pruritis, which is the hallmark feature of

AD.

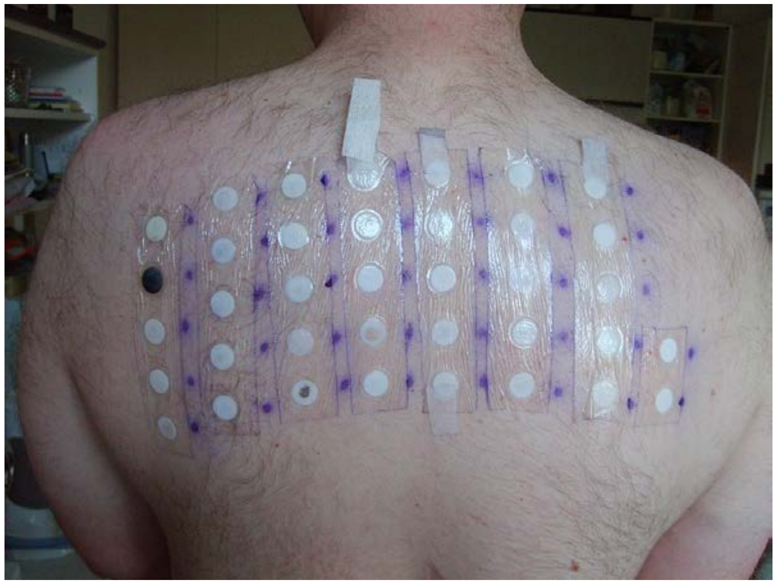
AD is a chronic relapsing inflammatory condition. It presents with erythematous macules, papules and vesicles that often start in early childhood, usually less than 2 years old. The lesions later become excoriated and lichenified. It is frequently associated with dryness of the skin and secondary infections. In infancy, the lesions initially appear on face (sparing the napkin area) and later on extensor aspect of knees and elbows as the child starts crawling. In later childhood and in adults, the principal site of involvement is flexures.

The filaggrin monomers in keratinocytes in the epidermis bind and aggregate keratin bundles and intermediate filaments to form the epidermal barrier. Any damage/dysfunction of this barrier is associated with an increased risk of developing AD. Damage to this barrier can be due to environmental factors (detergents, hard water, UV light) or local inflammatory reactions. Dysfunction of the barrier can be due to genetic mutation of the FLG gene leading to reduced filaggrin production or local immunomodulatory effects leading to reduced filaggrin expression.

Other options:

Option A: Seborrheic dermatitis is a chronic inflammatory skin condition characterized by erythema, pruritus, scaling, flaking, or crusting of the skin. It is often associated with Malasezzia fungal infection. A milder form of seborrheic dermatitis of the scalp is commonly known as dandruff.

Option C: Allergic contact dermatitis (ACD) is an inflammatory condition caused due to a delayed hypersensitivity reaction (Type 4 hypersensitivity) to a specific allergen that comes into direct contact with the skin.Patients usually present with pruritus, erythema, swelling, papules, and papulovesicles. Some common allergens include nickel (found in artificial jewelry), PPD/Para phenylene diamine (in hair dye), chromate (in cement), rubber or latex, cosmetics, etc. Patch test (image below) is used for the diagnosis of ACD.



Option D: Erysipelas is a bacterial skin infection caused by Streptococcus pyogenes (Group A Streptococcus) characterized by raised, erythematous sharply demarcated rapidly spreading skin lesions, along with systemic symptoms like fever and chills. The skin may also have a peau d'orange or orange peel-like appearance.

Solution to Question 2:

Purpura is a primary skin lesion. Crust, atrophy, and induration are secondary lesions.

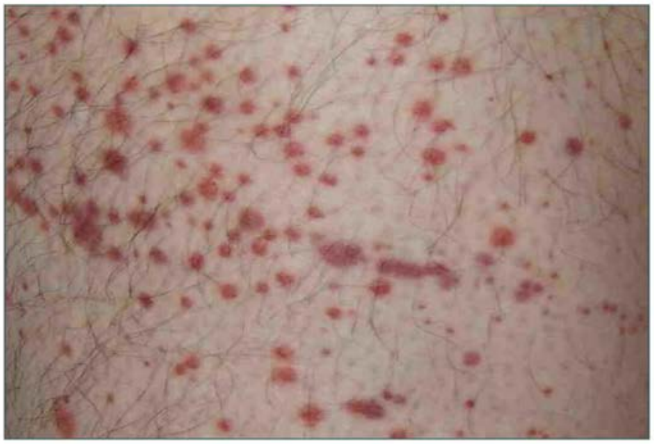
It is a condition with red or purple discolored spots on the skin that do not blanch on applying pressure. The spots are caused by bleeding underneath the skin. It is seen:

• Vasculitis

• Thrombocytopenia

• Vitamin C deficiency

Purpura



Option A: Crusts or scabs consist of dried-up discharge from the lesions.

Crusts (Scabs)

Option B: Atrophy refers to loss of tissue. There may be fine wrinkling and increased translucency if the process is superficial.

Option D: Induration refers to dermal thickening that clinically presents as skin that feels thicker and firmer than normal on palpation.

Solution to Question 3:

Scleredema is not associated with sclerodactyly or Raynauds Phenomenon. Scleredema and scleroderma are different entities. In contrast to systemic sclerosis (scleroderma), scleredema is not associated with sclerodactyly, Raynaud phenomenon, nail fold capillary changes, or serum autoantibodies.

In scleredema, the dermis is 3–4 times thicker than normal due to an increase in type 1 collagen synthesis by dysfunctional fibroblast. However, fibroblast proliferation is not seen.

Erythema and a peau d'orange appearance of the skin are commonly observed. The hands and feet are characteristically spared.

Diabetic scleredema is the most common type. In diabetic scleredema, the accumulation of collagen may be due to irreversible non■enzymatic glycosylation of collagen and resistance to degradation by collagenase. It can also occur post-streptococcal infections.

Solution to Question 4:

The given image showing hyperpigmented lesion on shoulder associated with hypertrichosis is suggestive of Becker's naevus.

It is an epidermal melanocytic naevus with aggregation of melanocyte clusters in the epidermis. It is often hairy. The onset is around adolescence when there is an increased sensitivity to androgens especially in the distribution of outer arms, scapular area and chest.

A triad of features can be seen:

- Hyperpigmentation
- Hypertrichosis
- Acne

Option A: Mongolian spots are slate blue areas in the lumbosacral region that spontaneously regress before puberty.

Option B: Naevus of Ito is unilateral hyperpigmentation in the acromioclavicular distribution. It is not known to have hypertrichosis or acne.

Option C: Congenital melanocytic nevi are common moles that become raised and have a cerebriform appearance and can be hairy. They can occur anywhere. The onset is at birth or soon after birth

Solution to Question 5:

Verruca vulgaris or common wart is most commonly caused by human papillomavirus-2 (HPV-2) but is also related to types 1, 4, 27, and 57.

Common warts are usually asymptomatic. They show acanthosis, hyperkeratosis, and papillomatosis and present as firm papules with a rough horny surface. The most commonly affected sites are on the backs of the hands and fingers, and, in children on the knees. Pseudo-Koebner phenomenon may be observed.

Treatment is usually with topical agents such as salicylic acid, caustics, glutaraldehyde, formalin, or with laser, cryotherapy, surgical removal.

verruca vulgaris.

Solution to Question 6:

Cutis marmorata refers to the pinkish-blue mottled or marbled appearance when subjected to cold temperatures.

It is seen throughout infancy and in 50% of children. It is caused by simultaneous dilation and contraction of the superficial capillaries. Rewarming restores the skin to normal. This is a physiological finding and requires no treatment. However, persistent cutis marmorata may be seen in trisomies 18 and 21.

Note: Cutis marmorata telangiectatica congenita is a distinct vascular developmental disorder and is easily distinguished as it is fixed (it does not disappear on rewarming), and it may be associated with a variety of other anomalies such as limb growth or renal anomalies.

Cutis marmorata

Solution to Question 7:

Varicella-zoster virus (VZV) is a human herpesvirus that causes varicella (chickenpox) as a primary infection and then remains in a latent form in the dorsal root ganglia for a variable period after which it can reactivate to cause herpes zoster (shingles).

Dermatology NEET 2019

Question 1:

A 15-year-old girl presents with itchy lesions on her arm as shown. Her family history is positive for asthma. What could be the most probable diagnosis?

- **a)** Seborrhoeic dermatitis
- **b)** Atopic dermatitis
- **c)** Allergic contact dermatitis
- **d)** Erysipelas

Question 2:

Bindi Leukoderma is caused by which chemical?

a) Mono-benzyl ether of Hydroquinone (MBH)

b) Crocein Scarlet MOO and Solvent Yellow 3

c) p-phenylenediamine (PPD)

d) Para Tertiary butylphenol (PTBP)

Question 3:

The most common triggering factor of the given condition is ________

a) Vaccination

b) Malignancy

c) Drugs

d) Infection

Question 4:

A young female presented with vaginal itching and green frothy genital discharge. Strawberry cervix is seen on examination. What will be the drug of choice?

a) Doxycycline

b) Oral fluconazole

c) Metronidazole

d) Azithromycin

Question 5:

What is correct regarding the diagnosis based on the given image?

a) The lesions are not infectious

b) Trigeminal dermatome is most commonly affected

c) Anterior nerve roots are more commonly involved

d) Mucous membranes within the affected dermatomes are involved

Question 6:

What is the diagnosis seen in the image below?

a) Candidal paronychia

b) Staphylococcal scalded skin syndrome

c) Candidal intertrigo

d) Diabetic foot ulcer

Question 7:

Which of the following is commonly associated with the condition shown below?

a) Oil drop sign

b) Leukonychia striata

c) Pterygium of nails

d) Pitting of nails

Question 8:

Identify the condition given in the image :

a) Dermatomyositis

b) Acanthosis nigricans

c) Pityriasis rotunda

d) Melasma

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	c
5	d
6	c

7	d
8	b

Detailed Explanations

Solution to Question 1:

The clinical scenario depicting pruritic lesions at a flexural site along with a family history of atopy (asthma) is suggestive of atopic dermatitis (AD) or atopic eczema (AE). The scratch marks and excoriations shown in the image are due to pruritis, which is the hallmark feature of AD.

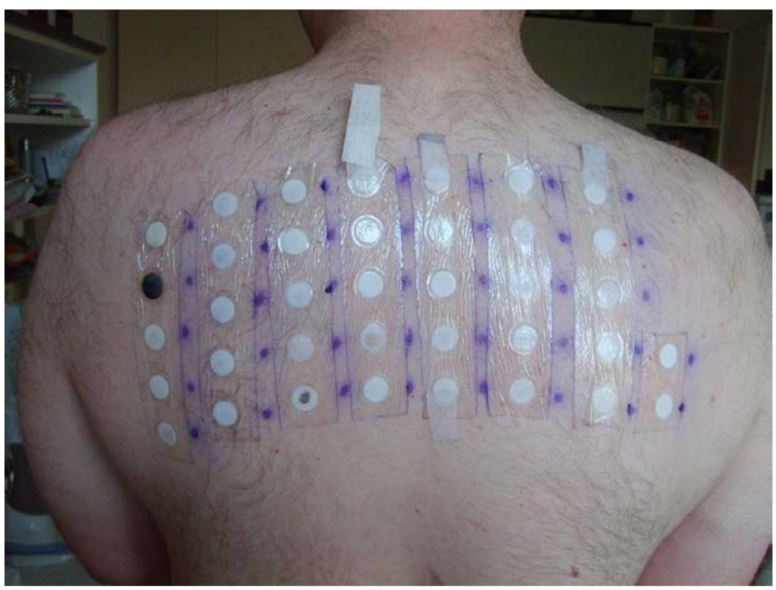
AD is a chronic relapsing inflammatory condition. It presents with erythematous macules, papules and vesicles that often start in early childhood, usually less than 2 years old. The lesions later become excoriated and lichenified. It is frequently associated with dryness of the skin and secondary infections. In infancy, the lesions initially appear on face (sparing the napkin area) and later on extensor aspect of knees and elbows as the child starts crawling. In later childhood and in adults, the principal site of involvement is flexures.

The filaggrin monomers in keratinocytes in the epidermis bind and aggregate keratin bundles and intermediate filaments to form the epidermal barrier. Any damage/dysfunction of this barrier is associated with an increased risk of developing AD. Damage to this barrier can be due to environmental factors (detergents, hard water, UV light) or local inflammatory reactions. Dysfunction of the barrier can be due to genetic mutation of the FLG gene leading to reduced filaggrin production or local immunomodulatory effects leading to reduced filaggrin expression.

Other options:

Option A: Seborrheic dermatitis is a chronic inflammatory skin condition characterized by erythema, pruritus, scaling, flaking, or crusting of the skin. It is often associated with Malasezzia fungal infection. A milder form of seborrheic dermatitis of the scalp is commonly known as dandruff.

Option C: Allergic contact dermatitis (ACD) is an inflammatory condition caused due to a delayed hypersensitivity reaction (Type 4 hypersensitivity) to a specific allergen that comes into direct contact with the skin.Patients usually present with pruritus, erythema, swelling, papules, and papulovesicles. Some common allergens include nickel (found in artificial jewelry), PPD/Para phenylene diamine (in hair dye), chromate (in cement), rubber or latex, cosmetics, etc. Patch test (image below) is used for the diagnosis of ACD.



Option D: Erysipelas is a bacterial skin infection caused by Streptococcus pyogenes (Group A Streptococcus) characterized by raised, erythematous sharply demarcated rapidly spreading skin lesions, along with systemic symptoms like fever and chills. The skin may also have a peau d'orange or orange peel-like appearance.

Solution to Question 2:

Bindi leukoderma is caused by para tertiary butylphenol (PTBP) present in adhesive sticker bindi.

• This is called as Contact Leukoderma or Chemical Leukoderma in which the chemical increases sensitivity and induces apoptosis of the melanocytes in the area of contact.

• There is also evidence that azo dyes, by virtue of structural similarity to tyrosine, can directly inhibit melanogenesis by competing with that amino acid for binding with tyrosinase enzyme.

Common chemicals in the Indian set up which cause Chemical leukoderma are:

Bindi Leukoderma	Para Tertiary butyl phenol (PTBP)
Synthetic leather, rubber slippers, Latex condoms, rubber gloves	Mono-benzyl ether of Hydroquinone (MBH)-Rubber Antioxidant
Hair dye	p-phenylenediamine (PPD)
Alta (red colour like henna)	Crocein Scarlet MOO and Solvent Yellow 3(azo dyes)

Solution to Question 3:

The given image shows characteristic target lesions of erythema multiforme. This most commonly occurs following a herpes simplex virus infection.

The common triggers are:

• Herpes simplex infection

• Mycoplasma infection

• Bacterial infections

• Fungal infections

• Vaccinations

• Malignancy- carcinoma, lymphoma, leukemia

• Drugs- sulfonamides, rifampicin, penicillins, hydantoin derivatives

• Miscellaneous- lupus erythematosus (Rowell Syndrome), sarcoidosis, polyarteritis nodosa

Erythema multiforme is best regarded as a self■limiting cytotoxic dermatitis resulting from cell■mediated hypersensitivity.

It presents clinically with a spectrum of macular, papular, or urticarial lesions, as well as the classic acral iris or ‘target lesions’.

Target lesions are less than 3 cm in diameter, rounded, and have three zones:

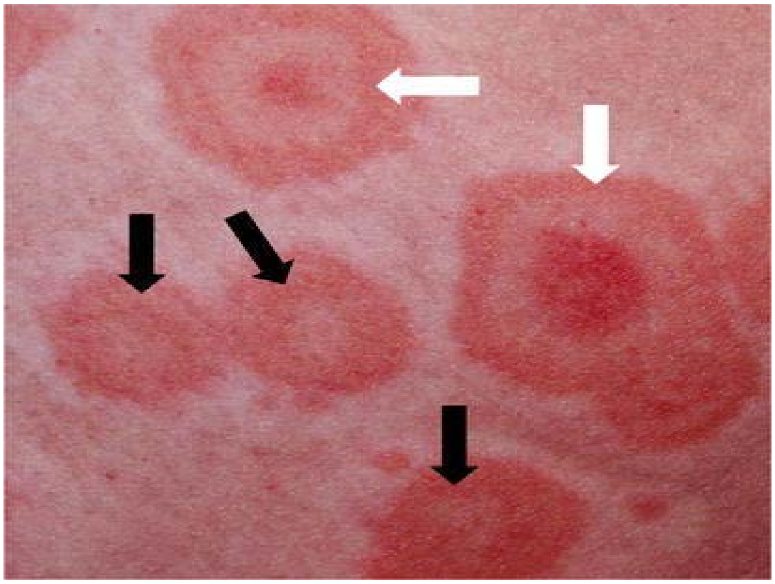
• Central area of dusky erythema or purpura

• Middle paler zone of edema

• Outer ring of erythema with a well■defined edge

Lesions may involve the palms and the trunk. Oral and genital mucosal membranes lesions are associated with erosions

The image below shows target lesions.



Solution to Question 4:

Vaginal itching and green frothy discharge in a young female with strawberry vagina points towards trichomoniasis. Metronidazole is the drug of choice for trichomoniasis.

It is is the most common parasitic infection in humans in the industrialized world.

It is caused by Trichomonas vaginalis, an anaerobic, flagellated protozoa, exists only as a trophozoite (no cyst stage).

Diagnosis is made by detection of motile flagella during microscopic examination of wet mounts. Direct immunofluorescent antibody staining is more sensitive (70-90%) than wet mount.

Culture is done on Feinberg-Whittington media.

Treatment: DOC- Metronidazole 2 gm single oral dose. All sexual partners must be treated to prevent reinfection, especially from asymptomatic males. Tinidazole can also be used for the treatment of trichomoniasis.

The image below shows strawberry vagina seen in trichomoniasis:

Solution to Question 5:

The above image shows unilateral, painful, grouped vesicles in a single dermatome suggestive of herpes zoster or shingles. The mucous membranes within the affected dermatomes are involved in shingles.

Herpes zoster/shingles is an acute cutaneous segmental eruption due to the reactivation of latent varicella-zoster virus from dorsal root ganglia. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually pain, 1-3 days after which the lesions appear. Occasionally, the pain may not be followed by the eruption (‘zoster sine eruptione’). The patients are infectious, both from the virus in the lesions and nasopharynx.

The most commonly involved dermatomes in decreasing order of frequency are thoracic > cervical > trigeminal > lumbosacral. The incidence of ophthalmic zoster increase with age. The posterior nerve roots and ganglia show inflammatory changes.

Herpes zoster ophthalmicus is a variant of zoster involving the ophthalmic division of the trigeminal nerve (V1). Herpes zoster oticus/Ramsay Hunt syndrome is another variant involving the geniculate ganglion and presents with ipsilateral facial nerve palsy, otalgia and vesicles over external ear.

Zoster is a self-limiting infection but it is painful and carries a risk of secondary infection and post-herpetic neuralgia. Analgesia and treatment of secondary infections are sufficient for mild infections.

Indications for antivirals in VZV (Acyclovir 800 mg):

• Painful zoster infections in adults

• Facial zoster at any age

• Immunocompromised

Solution to Question 6:

The condition shown in the image is candidal intertrigo.

It is a candidal infection of opposing skin surfaces, which presents as erythema with a little moist exudation starting deep in a fold. When the web spaces of the toes or fingers are affected, marked maceration with a thick, white, horny layer is usually prominent.

As the condition progresses, it spreads beyond the area of contact, usually developing the typical features of candidosis with a fringed, irregular edge and subcorneal pustules rupturing to give tiny erosions, and then further peeling of the stratum corneum.

Treatment:

Topical antifungal agents such as nystatin, ketoconazole, and clotrimazole are routinely used.

Solution to Question 7:

The condition shown in the image above is alopecia areata. Classically, it causes fine stippled pitting of the nails.

It is a chronic inflammatory disease which causes non-scarring patches of hair loss, especially in genetically predisposed.

The most characteristic lesion is a smooth circumscribed patch of alopecia, which has short extractable broken hairs called as exclamation mark hair. However, there is always sparing of white hairs in the region affected.

Other options :

Subungal oil drop sign is a highly specific pathognomonic sign seen in Nail psoriasis. It is a yellow-red discolouration in the nail bed resembling a drop of oil.

Leukonychia striata are lines that extend across the nail and lie parallel to lunula. They are seen in Hypoalbuminemic states such as nephrotic syndrome .

Nail Pterygium is a wing-shaped deformity of nail due to a central fibrotic band which divides the nail into two parts in the proximal.

• Dorsal pterygium is seen in lichen planus.

• Ventral pterygium is seen in trauma, systemic sclerosis, Raynaud phenomenon, lupus erythematosus, familial subungual pterygium, and infections.

Solution to Question 8:

The condition in the given image is acanthosis nigricans. It is characterized by thickened, gray-brown velvety plaques seen in flexural areas such as the back of the neck, axillae, inframammary creases, waist, and groin.

The causal mechanism is due to the secretion of epidermal growth factors in the region involved. It can be benign or malignancy-associated.

- Benign - commonly associated with obesity or insulin resistance, usually mild.
- Malignancy - much less common, associated with gastric, lung, and uterine cancers.

Acanthosis nigricans can also be seen in the following:

- Drugs such as nicotinic acid used to treat hyperlipidemia
- Sudden onset of multiple seborrhoeic keratoses

Dermatology NEET 2020

Question 1:

Identify the drug used to treat the condition shown below:

- **a)** Anti-fungal therapy
- **b)** Anti-tubercular therapy
- **c)** Topical steroids therapy
- **d)** Anti-leprosy treatment

Question 2:

Identify the finding associated with the condition shown in the image below.

a) Isomorphic phenomenon

b) Meyerson phenomenon

c) Gottron's papule

d) Nikolsky's sign

Question 3:

Identify the condition shown in the image below:

a) Herpes labialis

b) Herpangina

c) Molluscum contagiosum

d) Impetigo

Question 4:

Which of the following drugs used as nail lacquer belongs to morpholines?

a) Amorolfine

b) Oxiconazole

c) Ciclopirox olamine

d) Tioconazole

Question 5:

A child was born with membranes around the body and had ectropion and eclabium. He is brought to the OPD with lesions covering his face, trunk, and extremities. Which of the following is an unlikely diagnosis?

a) Ichthyosis vulgaris

b) Lamellar ichthyosis

c) Bathing suit ichthyosis

d) Harlequin ichthyosis

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	a
5	a

Detailed Explanations

Solution to Question 1:

The given image shows an annular plaque with central scarring points to lupus vulgaris, which is treated using anti-tubercular drugs.

It is the most common cutaneous tuberculosis in India. The characteristic lesions are annular, infiltrated plaques with serpiginous edges, central atrophy and scarring and are commonly seen on face, buttocks and extremities.

On diascopy, apple jelly nodules are seen.

Histologically, tubercles are seen in the superficial dermis showing scanty or absent central caseation with surrounding epithelioid histiocytes and giant cells.

Other differential diagnoses for annular plaque include:

• Tinea corporis shows central clearing, peripherally raised scaly margins and presents with an itch.

• Cutaneous leishmaniasis lesions have central crusting and are seen on the face in patients from areas of high leishmaniasis prevalence.

Solution to Question 2:

The image shows plaque psoriasis of the dorsum of the hand, and isomorphic or Koebner phenomenon is associated with the condition.

Option B: Meyerson phenomenon refers to the formation of an eczematous ring around a melanocytic nevus.

Option C: Gottron's papules are violaceous, erythematous papules overlying the dorsal interphalangeal or metacarpophalangeal, elbow, or knee joints.

Option D: Nikolsky sign refers to easy peeling of skin on applying tangential pressure over a bony prominence and is classically seen in pemphigus and staphylococcal scalded skin syndrome.

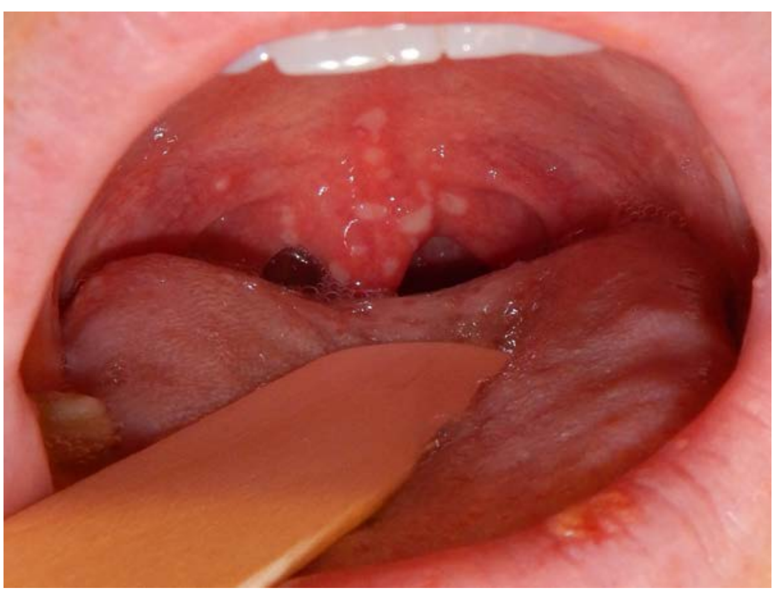
Solution to Question 3:

The above image shows multiple vesicles & crusts clustered on an inflamed base around the lips which is characteristic of herpes labialis or cold sore.

It is commonly caused by herpes simplex virus type 1(HSV-1) and rarely by herpes simplex virus type 2 (HSV-2). It begins with a prodrome of pain, burning, and tingling followed by the appearance of erythematous papules. The papules then become vesicular and may ulcerate.

It is treated with oral acyclovir 200 mg and 5% topical acyclovir.

Option B: Herpangina or vesicular pharyngitis is characterized by the presence of ulcerating small vesicles on the oral fauces and the posterior pharyngeal wall. It is commonly caused by the coxsackie virus.



Option C: Molluscum contagiosum is characterized by pearly white, wart-like umbilicated papules. It is caused by the pox virus.

Option D: Impetigo is characterized by golden yellow or honey-coloured crusts that form after the rupture of pustulating vesicles. Non-bullous impetigo is caused by both S. aureus and streptococcus. Bullous impetigo is caused by S. aureus.

Solution to Question 4:

Amorolfine is a morpholine that is available as a 5% nail lacquer. It is used in the treatment of onychomycosis.

Options B and D: Oxiconazole and tioconazole are classified as imidazoles

Option C: Ciclopirox olamine is a hydroxypyridone compound.

Solution to Question 5:

The clinical picture shows a neonate encased in a shiny, parchment-like membrane (collodion baby) suggestive of a disorder belonging to the autosomal recessive congenital ichthyosis spectrum.

Ichthyosis vulgaris is not a part of this disease spectrum. It does not present at birth and is therefore an unlikely diagnosis.

Ichthyosis vulgaris is an autosomal dominant condition caused due to a mutation in the filaggrin gene (FLG). It develops in the first few months of life. Patients exhibit accentuated palmar and plantar creases and have dry scaly skin. The scales appear light grey and cover the extensor surfaces of the limbs and the trunk. It usually spares the groin and larger flexures. Immunohistochemistry reveals an absent or markedly reduced filaggrin signal.

Autosomal recessive congenital ichthyosis spectrum includes the following:

• Harlequin ichthyosis (HI)

• Bathing suit ichthyosis (BSI)

• Lamellar ichthyosis (LI)

• Congenital ichthyosiform erythroderma (CIE)

• Self- improving congenital ichthyosis (SICI)

• Transient manifestations, such as collodion baby.

Dermatology INI-CET 2021

Question 1:

A forest officer develops the lesion as shown in the image. Which of the following is not a differential to consider?

a) Scrub typhus

b) KFD

c) Cutaneous anthrax

d) Healing brown recluse spider bite

Question 2:

A patient presents with following lesions on the genitals. Scrapings from the lesion would show?

a) Tzanck cells

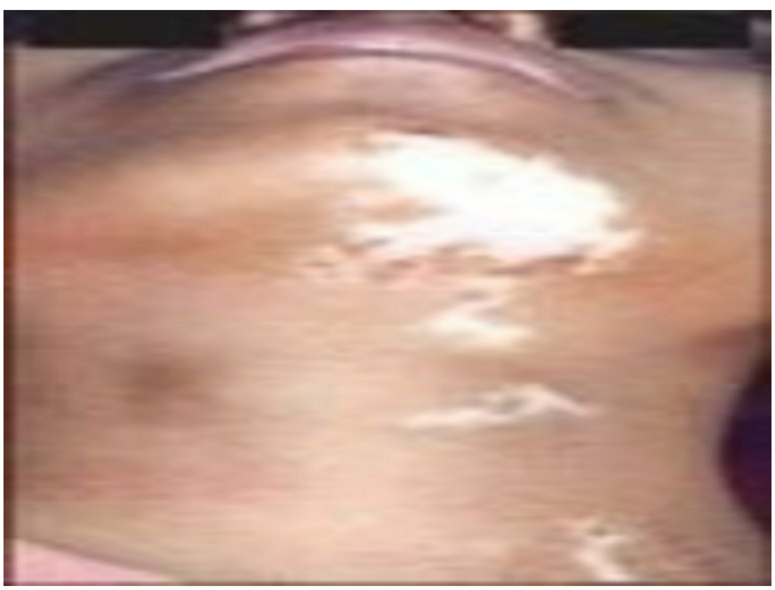
b) Cells with clear halo around the nucleus.

c) Molluscum bodies.

d) HP bodies.

Question 3:

A young girl presents with leukotrichia and lesions as shown in the image. What is the most likely diagnosis?



a) Focal vitiligo

b) Segmental vitiligo

c) Piebaldism

d) Nevus depigmentosus

Question 4:

The most specific and typical sign of juvenile dermatomyositis in a 17-year-old patient is

a) Photosensitive skin rash.

b) Nail bed capillary changes.

c) Gottron's papules.

d) Malar rash.

Question 5:

A child presents to you with itchy lesions on the scalp with hair loss as shown in the image below. What would be the next best step?

a) KOH mount

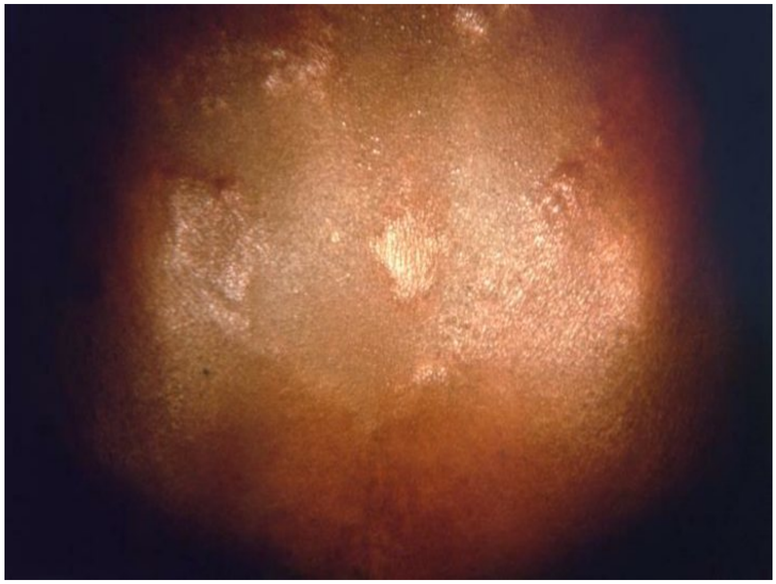
b) Slit skin mount

c) Gram staining

d) Biopsy

Question 6:

The below-given image is characteristically seen in which of the following type of leprosy?



a) Borderline lepromatous leprosy

b) Lepromatous leprosy

c) Borderline leprosy

d) Histoid leprosy

Question 7:

A child presented with itchy plaques over the neck, the bilateral popliteal and cubital fossa. What could be the diagnosis?

a) Atopic dermatitis

b) Dermatitis herpetiformis

c) Pemphigus vegetans

d) Psoriasis

Question 8:

A 65-year-old man presents with itchy tense bullae over the limbs. The roof of the bullae collapsed after a few days. A clinical image is shown below, what is the likely diagnosis?

a) Bullous pemphigoid

b) Pemphigus vulgaris

c) Dermatitis herpetiformis

d) Linear IgA dermatosis

Question 9:

A young female presented with the following findings on the face. Which among the following will not be useful for its treatment?

a) Topical benzoyl peroxide

b) Topical fluocinolone

c) Topical tazarotene

d) Topical antibiotics

Question 10:

A 45-year-old male patient from Bihar presented with normoanesthetic hypopigmented lesion as shown below. He had a history of fever for a long duration in his childhood. No thickened nerves were present. What is the probable diagnosis?

a) Tuberculoid leprosy

b) Lepromatous leprosy

c) Mycosis fungoides

d) Post kala-azar dermal leishmaniasis

Question 11:

A child presented with asymptomatic lesions on the forearm and on the shaft of the penis. The lesions on the forearm are shown below. What is the most likely diagnosis?

a) Lichen planus

b) Lichen nitidus

c) Scrofuloderma

d) Scabies

Question 12:

Which among the following is correct regarding the healing of leprosy if left untreated?

a) No spontaneous remission is seen

b) Half of the cases resolve spontaneously after bacilli exposure

c) Spontaneous remission is more common in children than in adults

d) Indeterminate leprosy have more remission than tuberculoid

Question 13:

A young man presented with painful lesions as shown in the image below. What is the likely diagnosis?

a) Herpes Zoster

b) Molluscum contagiosum

c) Herpes Simplex

d) Lymphangioma circumscriptum

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	c
5	a
6	c
7	a
8	a
9	b
10	d
11	b
12	d
13	a

Detailed Explanations

Solution to Question 1:

The given image shows the presence of eschar and Kyasanur Forest Disease (KFD) is not a differential diagnosis for the presence of eschar.

Kyasanur Forest Disease is a hemorrhagic fever found in the state of Karnataka, India. KFD has an abrupt onset of fever, headache, conjunctivitis, myalgia, and severe prostration. Characteristic skin lesions that are associated with KFD are petechial lesions. Some cases may develop hemorrhages in the skin, mucosa, and viscera, but there is no eschar formation.

Infection is transmitted by the bite of ticks, the principal vector being Haemaphysalis spinigera

Other options:

A) Scrub typhus:

• Scrub Typhus is also known as chigger-borne typhus.

• Scrub typhus is caused by Orientia tsutsugamushi.

• The vector is trombiculid mite.

• Humans are infected when they are bitten by the mite larvae known as chiggers.

The incubation period is 1- 3 weeks. The patient typically develops a characteristic eschar at the site of the mite bite as shown in the image below, with regional lymphadenopathy and maculopapular rash.

C) Cutaneous anthrax:

• In cutaneous anthrax, the lesion starts as a papule 1-3 days after infection and becomes vesicular, containing fluid that may be clear or blood-stained.

• The whole area is congested and edematous and several satellite lesions filled with serum or yellow fluid are arranged around a central necrotic lesion which is covered by a black eschar.

• This lesion is called a malignant pustule.

• The disease used to be common in dock workers, carrying loads of hides and skins on their bare backs and hence known as hide porter's disease.

D) Healing brown recluse spider bite:

• The bite of the Loxosceles reclusa, the brown recluse spider is usually relatively painless.

• After minutes or hours, severe pain develops at the site, accompanied by erythema, edema, and a central bulla.

• In severe envenomation, a ‘target lesion’ is seen- central blue/ purple discoloration surrounded by an ischemic halo and an outer ring of erythema (the ‘red, white and blue’ sign).

• After 3-4 days, the central area becomes necrotic and an eschar develops.

• The eschar sheds eventually, leaving behind an ulcer, which takes time to heal.

Solution to Question 2:

The given image shows clusters of vesicles seen on an erythematous base suggesting a diagnosis of genital herpes. Tzanck cells or multinucleated giant cells with intranuclear bodies are present on microscopy. Intranuclear inclusion bodies are seen when the smears are stained with Giemsa stain.

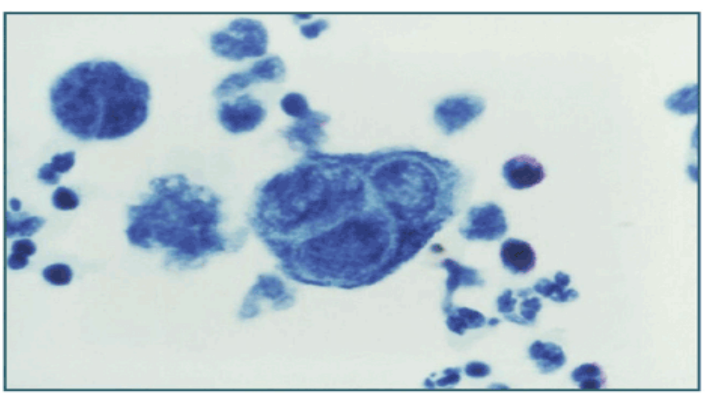
The typical herpes lesions are thin-walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers. They heal without scarring. The most common site of cutaneous lesions in the face - on the cheeks, chin, around the mouth, or on the forehead.

The Tzanck smear technique is a rapid and inexpensive diagnostic method. Scrapings (preferably from the base of vesicles) are obtained from the lesions and smears are prepared. When stained with 1% aqueous solution of O-toluidine blue for 15 seconds, multinucleated giant cells with faceted nuclei and homogenously stained ground glass chromatin known as ‘ Tzanck cells’ are seen.

Genital herpes:

Herpes Simplex Virus (HSV) 2 is more responsible for genital herpes than HSV 1. The vesiculo-ulcerative lesions are very painful. In men, it occurs mainly on the penis or urethra. In women, the cervix, vagina, vulva and perineum can all be involved. The primary infection causes serious symptoms accompanied by fever and malaise, followed by recurrent episodes.

Tzanck smear



©MARROW

Solution to Question 3:

The image shows areas of localized, unilateral depigmented macules. This feature, along with a history of leukotrichia suggest that the patient is suffering from segmental vitiligo. It occurs due to a lack of melanocytes in the basal layer of the epidermis.

It is a localised depigmentation that occurs due to a progressive loss in the melanocytes. It is classified as

• Segmental vitiligo- which appears as unilateral maculae in a segmented or band-shaped pattern

• Non-segmental vitiligo- which appears as bilateral maculae distributed in an acrofacial pattern or scattered symmetrically over the entire body

The lesions become apparent between 20-30 years of age.

Hypopigmented macules in vitiligo are commonly found in the hips, dorsum of the hands/fingers, feet, elbows, knees, and ankle areas. These are the areas of repeated friction. Rarely there may be complete vitiligo (vitiligo universalis), or trichrome vitiligo where there are areas of depigmentation, hyperpigmentation, along with the normal skin are present. The Koebner phenomenon is the development of lesions at the site of trauma on the uninvolved skin.

Several autoimmune conditions can be associated with vitiligo. They are:

• Thyroid disease (hyperthyroidism and hypothyroidism)

• Pernicious anemia

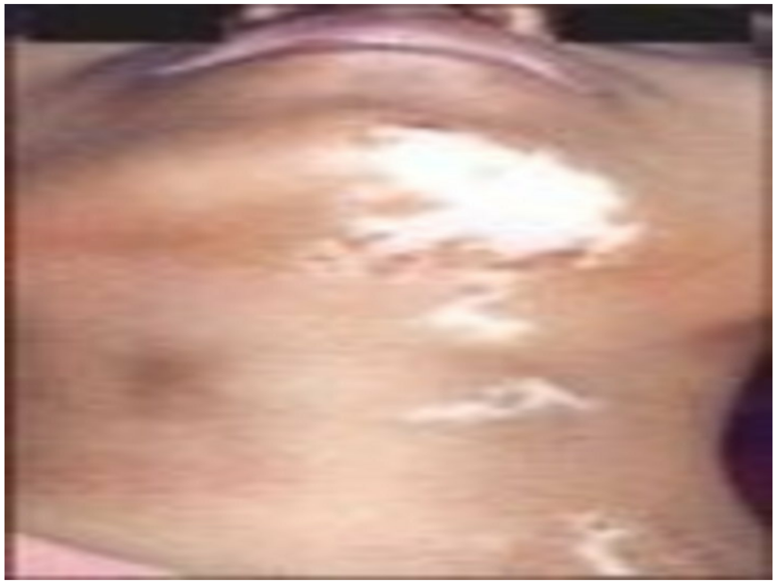
• Addison disease

• Diabetes

• Hypoparathyroidism

Under the Wood's lamp examination the depigmented areas emit a bright blue-white fluorescence and appear sharply demarcated.

Treatment involves cosmetic camouflage and the use of sunscreen. The topical application of corticosteroids is the first line of treatment. Topical calcineurin inhibitors like pimecrolimus, tacrolimus have also shown to be successful. Psoralen photochemotherapy, khellin, selective ultraviolet B therapy are used as second-line treatment. In patients who do not respond to this treatment, surgical grafting techniques can be used, where the melanocytes from a normal pigmented area are transferred to a non pigmented area. In patients with extensive vitiligo where only a few pigmented areas are left, depigmentation can be done.



Other options:

A) Focal vitiligo:

These are small isolated depigmented lesions that are neither segmentally distributed nor evolved into non-segmental lesions after 1-2 years. The focal vitiligo along with isolated mucosal lesions on one site are considered as undetermined /unclassified vitiligo.

C) Piebaldism:

It commonly presents with a white V-shaped forelock. it is caused due to the impaired migration of melanocytes to the skin, resulting in hypopigmented patches.

D) Nevus depigmentosus:

It is a congenital pigmentary disorder that presents with a non-progressive hypopigmented macule. It usually is present from birth or early childhood and remains stable throughout life.

Solution to Question 4:

Gottron papules are typical and specific sign of juvenile dermatomyositis.

Gottron papules and a heliotropic rash of the eyelids are the classical criteria that are required for the diagnosis of juvenile dermatomyositis along with three of the following:

Weakness	SymmetricProximal
Muscle enzyme elevation (≥1)	Creatine kinaseAspartate aminotransferaseLactate dehydrogenaseAldolase
Electromyographic changes	Short, small polyphasic motor unit potentialsFibrillations Positive sharp wavesInsertional irritabilityBizarre, high-frequency repetitive discharges
Muscle biopsy	NecrosisInflammation

Gottron’s papules are violaceous (of violet color), lichenoid (red-purple), inflammatory, flat-topped papules on proximal interphalangeal joints, distal interphalangeal joints, and metacarpophalangeal joints.

This should not be confused with Gottron's sign.

Gottron's sign is erythematous or violaceous, macules or patches in the same symmetric distribution pattern but sparing the interphalangeal spaces and sometimes involving elbows and knees.

The heliotrope rash is a violet discoloration of the eyelids that may be associated with periorbital edema. Rarely, an erythematous, scaly rash develops on the palms of the children called ‘mechanic hands’. Nail bed capillary changes are visible as individual thickened and telengiectatic capillary loops, which can be well seen with the help of capillaroscopy. Facial erythema mimicking malar rash of systemic lupus erythematosus, but with the involvement of nasolabial folds are also common.

Solution to Question 5:

The above image shows tinea capitis, which presents with a well-demarcated scaly patch in which hair shafts are broken off just above the skin resulting in alopecia. KOH mount is useful for the depicted condition.

The scrapings should be taken from the edge of a lesion and transferred to a slide, to which KOH is added and examined under the microscope for the presence of hyphae. Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area.

The clinical variants of tinea capitis are:

• Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.

• Ectothrix - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.

• Kerion - a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.

• Favus - is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii.

Treatment:

• Terbinafine: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.

• Itraconazole: 2-4 mg/kg/day for 4-6 weeks.

Solution to Question 6:

The given image shows a case of borderline tuberculous leprosy, or dimorphous leprosy, as evidenced by the cutaneous, rash-like discoloration of the patient's skin. This generally overlies the back, and crosses the midline. So, the best choice would be Borderline leprosy.

Microscopically, there is diffuse epithelioid cell granuloma with very scanty lymphocytes and no giant cells; the papillary zone is clear and the nerves are swollen slightly by cellular infiltrate. Acid-fast bacilli are present in moderate numbers.

The borderline leprosy patients are immunologically unstable and are at an increased risk of developing immune-mediated reactions. Type 1 lepra reactions that occur most commonly in mid-borderline leprosy patients, are delayed hypersensitivity reactions caused by increased recognition of mycobacterial leprae antigens in skin and nerve sites.

Solution to Question 7:

The given clinical history of itchy plaques over the neck, cubital fossa and popliteal fossa suggests the diagnosis of ‘atopic dermatitis’.

Atopic dermatitis is the cutaneous expression of the atopic state, characterised by a family history of asthma, allergic rhinitis or eczema.

The etiology is only partially defined, but there is a clear genetic predisposition. A characteristic defect is impaired epidermal barrier. In many patients a mutation in gene encoding filaggrin (a structural protein in the stratum corneum) is responsible. The lesions are distributed in the flexural aspects.(as shown in the image).

Patients may display many immunoregulatory abnormalities, including increased IgE synthesis, increased serum IgE levels, delayed type hypersensitivity reactions.

The cardinal symptom is itching, along with rash.

Atopic Eczema can be divided into 3 phases: Infantile, Childhood, and Adult phases.

Infantile phase: The lesions of Atopic Eczema most frequently start on the face but may occur anywhere on the skin surface. Often, the napkin area is relatively spared.

• When the child begins to crawl, the exposed surfaces, especially the extensor aspect of the knees and elbows are most involved.

• The lesions consist of erythema and discrete or confluent oedematous papules. The papules are intensely itchy and may become exudative and crusted as a result of rubbing.

• Secondary infection and lymphadenopathy are common.

Childhood Phase: From 18 to 24 months onwards, the sites most characteristically involved are the elbow and knee flexures, a mixture of erythema, crusting, excoriation, hyper■ and hypopigmentation, and warty lichenification may be seen depending on the skin type on the sides of the neck, wrists, and ankles.

• The sides of the neck may show a striking reticulate pigmentation, sometimes referred to as atopic dirty neck.

• The erythematous and oedematous papules tend to be replaced by lichenification.

Adult Phase: The picture is essentially similar to that in later childhood, with lichenification, especially of the flexures and hands.

Other options:

B) Dermatitis herpetiformis:

• Dermatitis herpetiformis is a chronic, intensely pruritic, skin condition associated with gluten■sensitive enteropathy.

• It typically appears in the 4th decade, and the patients present with itching and rash, most typically over the extensor surfaces of the elbows, knees, buttocks, and scalp.

• They may also have symptoms associated with gluten-sensitive enteropathies, like bloating, diarrhea and other gastrointestinal complaints.

• On examination, multiple erythematous papules and vesicles can be seen over the extensor aspects. They appear symmetrical and heal without scarring. The oral cavity may show mucosal abnormalities and enamel pits.

C) Pemphigus vegetans:

It is a rare form of the bullous disorder that presents with large, moist, verrucous, vegetating plaques studded with pustules on the groin, axillae and flexural surfaces.

Predominant cell	Name of microabscess	Disease
Neutrophilic microabscess	Papillary tip microabscessMunroe's microabscess (superficial)Spongiform pustules of Kogoj (deep)	Dermatitis HerpetiformisPsoriasisPsoriasis
Lymphocytic microabscess	Pautrier's microabscess	Mycosis fungoides
Eosinophillic microabscess		Pemphigus vegetans

D) Psoriasis:

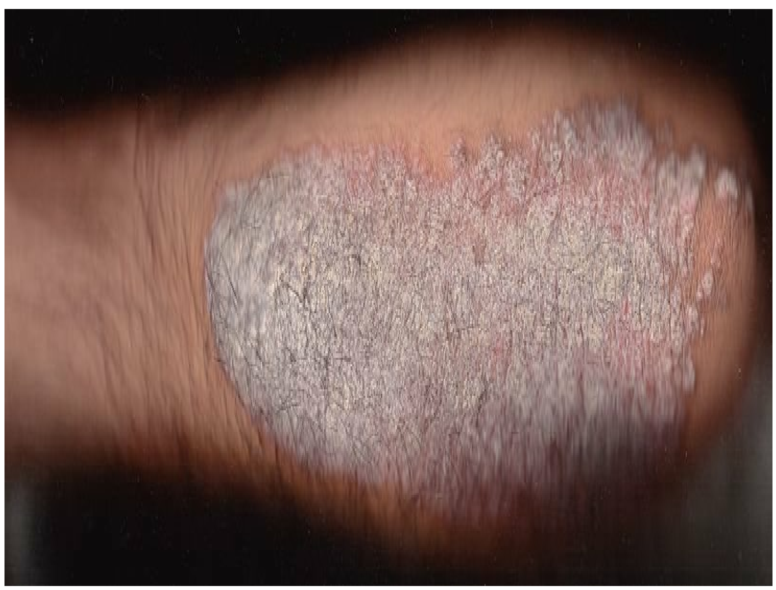
• Plaque psoriasis is the most common type of psoriasis and is seen in 80-90% of patients.

• The peak on occurrence is between 16 to 22 years and between 57 to 62 years. They usually present with itching and a papulosquamous rash.

• The lesions are red, scaly, and have well demarcation from the normal skin.

• They can also be encircled by a clear peripheral zone which is called the Woronoff ring.

• They commonly appear in the scalp, trunk, and extensor surfaces of the limbs.



Solution to Question 8:

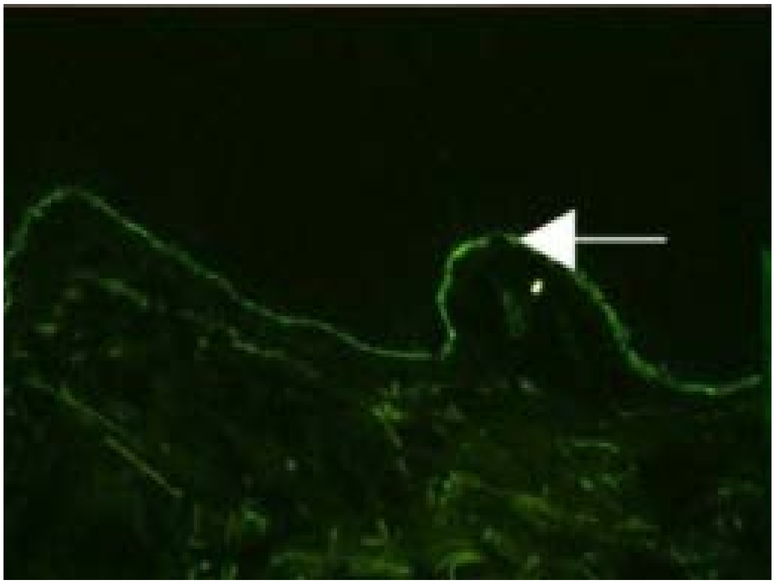
The given clinical scenario is highly suggestive of bullous pemphigoid.

Bullous pemphigoid is characterized by tense itchy bulla in the flexor areas and trunk which do not rupture easily. Elderly aged people are usually affected. It is a subepidermal immunobullous disorder, which occurs due to the production of autoantibodies (IgG) against BP180 (BPAG2) and BP 230 (BPAG1) antigens on hemidesmosomes.

Mucosal membrane involvement is rare in this disease. Most of the lesions heal without a scar. Since these bullae are subepidermal, pressure over the lesions does not separate the epidermis from the dermis. (Nikolsky sign is negative).

Immunofluorescent microscopy will show the presence of subepidermal split and deposition of IgG antibodies at the dermo-epidermal junction. Histopathology will show the presence of subepidermal bullae with eosinophils.

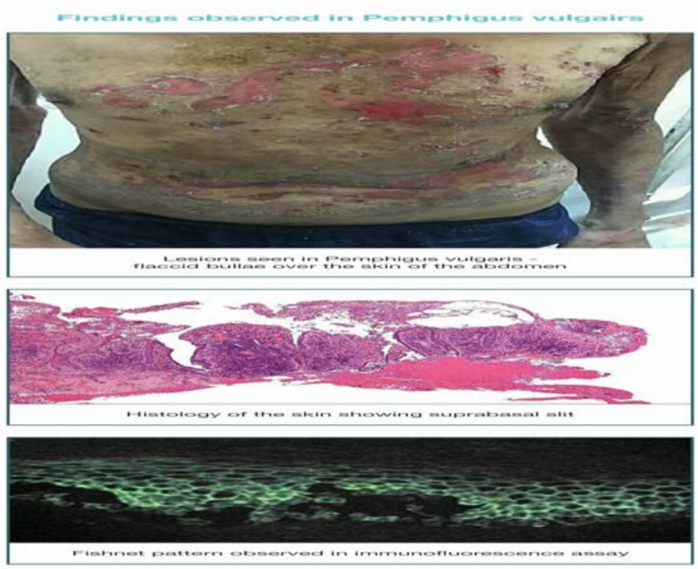
It is mainly treated with corticosteroids. Other drugs used are azathioprine, dapsone, doxycycline, etc.



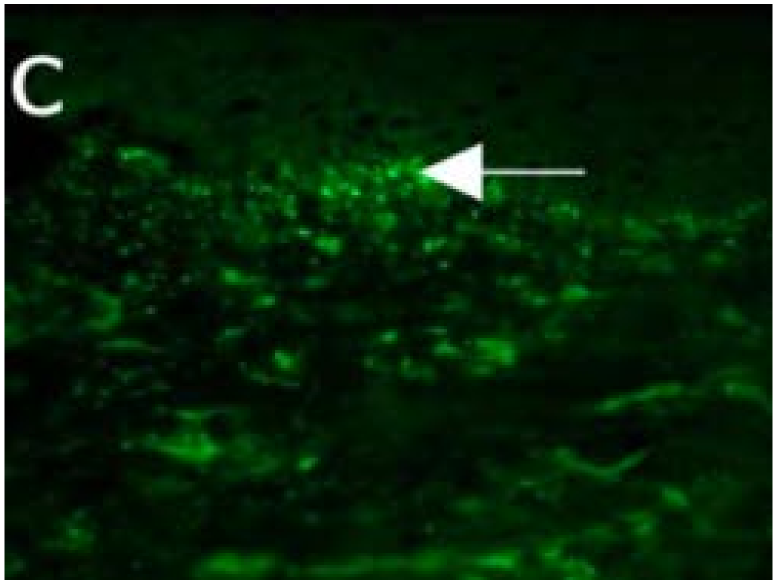
The above image (immunofluorescence microscopy) shows linear IgG deposits at the dermo-epidermal junction seen in the bullous pemphigoid.

Other options:

Option B: Pemphigus Vulgaris is an intraepidermal immunobullous disease. It occurs commonly between the 4th to 6th decade. They present with flaccid blisters, unlike bullous pemphigoid. Firm pressure over the lesion separates the epidermis from the dermis. Nikolsky's sign is positive. Mucosal involvement is seen in the form of buccal and palatal erosions.



Option C: Dermatitis herpetiformis typically appears in the 4th decade, and the patients present with itching and rash, most typically over the extensor surfaces. The majority of the patient have symptoms related to gluten-sensitive enteropathies, like bloating, diarrhea, and other gastrointestinal complaints. Immunofluorescence microscopy will show the deposition of IgA antibodies in the papillary dermis in a granular or fibrillar pattern(as shown in the below image).



Option D: Linear Ig A dermatosis (Chronic bullous dermatosis of childhood) is an autoimmune blistering disease seen in infants and children. The disease can also be seen in adults and is known as Linear IgA disease of adults or IgA bullous pemphigoid. The blisters are arranged in an annular pattern commonly known as the crown of jewels appearance (image below).

Solution to Question 9:

The age of the patient and the presence of papules and pustules with an erythematous base on the face are suggestive of acne. All the above agents are useful in the treatment of acne except topical fluocinolone (topical steroid). Corticosteroids may provoke an acneform reaction irrespective of their route of administration.

Acne is a chronic inflammatory disease of the pilosebaceous unit. It is commonly seen in teenagers. It can vary from very mild comedonal acne to severe aggressive pustular acne. These eventually may lead to scarring due to sinus formation.

Acne results from the combination of:

• Increased sebaceous gland activity under the influence of androgens - oily skin

• Abnormal follicular differentiation with increased keratinization- hypercornification of the duct due to increased IL-1

• Microbial hypercolonization of the follicular canal leading to its inflammation.

Management-

• Mild comedonal acne is treated with a topical retinoid. Mild papular and pustular acne is treated with topical retinoid and topical antibiotics.

• Moderate papular/pustular acne and moderate nodular acne are treated with oral antibiotic + topical retinoid with or without benzoyl peroxide.

• Severe nodular and conglobate acne are treated with oral isotretinoin.

Tazarotene is a third■generation retinoid, which is available for the treatment of acne.

Solution to Question 10:

The case presenting with hypopigmented lesions with normal sensations and a history of fever in childhood suggests a diagnosis of post-kala-azar dermal leishmaniasis.

Post-kala-azar dermal leishmaniasis (PKDL) is a condition when Leishmania donovani is found in the skin and manifests as dermal lesions. It is usually seen in the kala-azar endemic zones i.e. Bihar, Jharkhand, U.P, West Bengal (Northeastern states of India)

These patients present with erythematous rash and hypopigmented macules similar in appearance and distribution to those of lepromatous leprosy but normally less than 1cm. After a variable period of years or months, diffuse nodulation begins to develop in these macules.

The rash is progressive over many years and seldom heals spontaneously.

The image below shows post-kala-azar cutaneous leishmaniasis.

Post-kala-azar Cutaneous Leishmaniasis

Other options:

Options A and B: The presence of hypopigmented macules over the trunk and face, suggests a differential of leprosy, but the nerves and sensations are normal, hence ruling it out.

Option C: Mycosis fungoides is characterized by polymorphic erythematous patches and plaques which are usually associated with severe pruritus. They could be asymptomatic too. The image below shows characteristic lesions in mycosis fungoides.

Solution to Question 11:

The image shows an aggregate of pinhead-sized papules over the forearm of a child which is suggestive of lichen nitidus.

Lichen nitidus presents with tiny pinhead-sized flesh-colored papules and has a flat or dome■shaped, shiny surface. They are found on any part of the body but the sites of predilection are the forearms, penis, abdomen, chest, and buttocks.

Lichen nitidus is considered to be a variant of Lichen planus as both are histologically similar in early stages. It has a "claw clutching a ball' appearance on histopathological examination. It refers to a sharply circumscribed inflammatory infiltrate in the dermis, that spans 4-5 dermal papillae.

The diagnosis can be made clinically, but skin biopsy provides a definitive diagnosis. Lesions are usually asymptomatic and rarely require any treatment. Symptomatic or generalized lesions can be treated with fluorinated topical steroids, PUVA, narrow■band UVB phototherapy, astemizole, or acitretin.

Lichen nitidus

Pinhead-sized flesh-colored papules seen.

Other options:

Option A: Lichen planus is an idiopathic inflammatory disease affecting the skin and mucous membrane. It usually presents with pruritic, purple, polygonal, papules. The lesions have thin white lines on their surface known as Wickham's striae.

Option C: Scabies has typical distribution in the interdigital spaces and burrows are the characteristic lesions. The below image shows a magnified view of a burrowing trail of the scabies mite.

Option D: Scrofuloderma results from the direct invasion of the tubercle bacillus into the skin from underlying tuberculous foci.

Solution to Question 12:

Among the given options, Indeterminate leprosy have more remission than tuberculoid is the most appropriate answer.

The manifestations of leprosy depend on the immune response of an individual to the bacillus Mycobacterium leprae. About 80-90% of the people do not develop any disease after bacilli exposure. The remaining patients develop indeterminate leprosy. The lesions of indeterminate leprosy tend to be hypopigmented and ill-defined. Among those patients, 75% undergo spontaneous remission. Very few patients present clinically between tuberculoid or lepromatous leprosy. Therefore, more patients with indeterminate leprosy undergo spontaneous remission compared with those having tuberculoid leprosy. (Option D).

Leprosy (Hansen's disease) is a chronic infective granulomatous disease caused by the bacillus Mycobacterium leprae.

Leprosy is clinically characterized by one or more of the following features :

• Hypopigmented patches

• Partial or total loss of cutaneous sensation in the affected areas

• Presence of thickened nerves, and

• Presence of acid-fast bacilli in the skin or nasal smears.

Option A is incorrect because spontaneous remission is seen with leprosy. Even in the absence of specific treatment, a majority of patients tend to get cured spontaneously.

Option B is incorrect because around 80-90% of the cases (not half) resolve spontaneously after bacilli exposure.

Option C is incorrect because spontaneous remission in children is variable. Even though few studies mentions that children have more remission, however, the observation period was not long enough to establish its permanence.

Solution to Question 13:

The above image shows unilateral, red, closely grouped vesicles in a single dermatome associated with severe pain is suggestive of herpes zoster or shingles.

It is a segmental eruption due to the reactivation of latent varicella-zoster virus from dorsal root ganglia. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually pain and it may be accompanied by fever, headache, malaise, and tenderness localized to areas of one or more dorsal roots. The mucous membranes within the affected dermatomes are involved. The lymph nodes draining the affected area are enlarged and tender.

The patients are infectious, both from the virus in the lesions and nasopharynx. The most commonly involved dermatomes in decreasing order of frequency are thoracic > cervical > trigeminal > lumbosacral. The posterior nerve roots and ganglia show inflammatory changes.

Zoster is a self■limiting infection but it is painful and carries a risk of secondary infection and post■herpetic neuralgia. Analgesia and treatment of secondary infections are sufficient for mild infections.

Indications for antivirals in VZV (Acyclovir 800 mg):

• Painful zoster infections in adults

• Facial zoster at any age

• Immunocompromised

Other options:

Option B: The lesions of the molluscum contagiosum are shiny, white, and hemispherical. It is umbilicated with a central pore. It is caused by poxvirus.

The below image shows the lesions seen in Molluscum contagiosum

Option C: Herpes simplex virus are classified into two major antigenic types,

• Type 1 (HSV-1) which is classically associated with facial infections;

• Type 2 (HSV-2) which is typically genital.

Below image shows small, closely grouped vesicles on an inflamed base and is characteristic of Herpes Labialis or Cold sore.

Option D: Lymphangiomas are dilated dermal lymphatics that blister onto the skin surface. The fluid is usually clear, but occasionally maybe blood-stained. Lymphangiomas may thrombose and get fibrosed, to form hard nodules in the long term. This may raise concerns about malignancy.

If lymphangiomas are <5 cm across, they are called lymphangioma circumscriptum (shown below), and they are called lymphangioma diffusum if they are more widespread.


J. Menzies

Dermatology NEET 2021

Question 1:

A 10-year-old male child complains of mild painful swelling over the scalp for 3 months as shown in the image below. He also has a pet dog at his home. What is the diagnosis of the given condition?

a) Furuncle

b) Epidermoid cyst

c) Kerion

d) Folliculitis

Question 2:

A child with a history of night-blindness presented to you with the following findings. What is the diagnosis?

- **a)** Phrynoderma
- **b)** Keratosis pilaris
- **c)** Darier disease
- **d)** Follicular eczema

Question 3:

A 30-year-old presented with flaccid bullae on her skin which are easy to rupture. The biopsy revealed a suprabasal split. What is the most likely diagnosis?

- **a)** Pemphigus vulgaris
- **b)** Pemphigus foliaceus
- **c)** Pemphigus vegetans
- **d)** Erythema multiforme

Question 4:

What is the treatment of choice for a patient with paronychia of the index finger on the right hand, as shown in the image below?

a) Amoxiclav

b) Metronidazole

c) Amikacin

d) Norfloxacin

Question 5:

A patient taking multi-drug therapy (MDT) presents with worsening of existing lesions and nerve involvement. What will be your next best step of action?

a) Stop MDT, start systemic corticosteroids

b) Continue MDT, start systemic steroids

c) Stop MDT give thalidomide

d) Continue MDT, start thalidomide

Question 6:

A 30-year-old male patient presents to you with inflammatory alopecia. Which of the following variants of tinea capitis infection commonly result in this type of alopecia?

a) Endothrix and Favus

b) Ectothrix and Endothrix

c) Favus and Kerion

d) Kerion and Endothrix

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	a
5	b
6	c

Detailed Explanations

Solution to Question 1:

The clinical vignette of a young boy in contact with a zoophilic organism, presenting with a boggy inflammatory mass, with thick crusting and matting of hair is suggestive of a kerion. Kerion is a severe inflammatory type of tinea capitis.

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

• Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.

• Ectothrix - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.

• Kerion - a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.

• Favus - is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii.

Treatment:

• Terbinafine: <10 Kg- 62.5 mg; 10-20 kg - 125 mg; >20 kg- 250 mg, all given daily for 4 weeks.

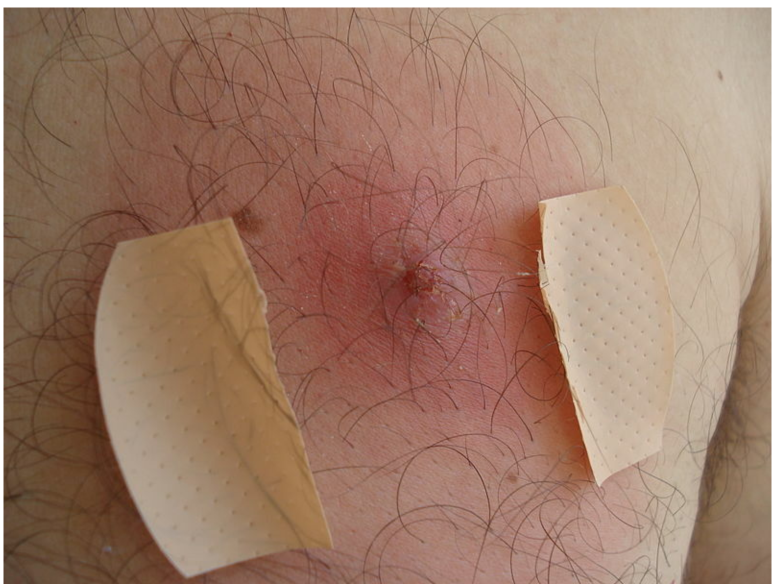
• Itraconazole: 2-4 mg/kg/day for 4-6 weeks.

The image below shows a kerion:

The image below shows the endothrix variant of tinea capitis:

Option A: Furuncle/ boil is caused by Staphylococcus. It is painful and usually situated on the neck area, and the shedding of loosened hairs is generally much less evident than in kerion. It is pustular and later becomes necrotic and forms a scar.

The image below shows a furuncle- pustule over the hair follicle with a necrotic base:



Option B: Epidermoid cyst/ sebaceous cyst results from squamous metaplasia in a damaged sebaceous gland. Many are the result of inflammation around a pilosebaceous follicle. The most common site of a sebaceous cyst is the postauricular sulcus or below and behind the ear lobule. They are fixed to the skin and usually have a central punctum. Treatment depends on the clinical state of the cyst. When inflamed or infected, they should be incised and drained initially, total surgical excision should be done once the inflammation has subsided.

The image below shows an epidermoid cyst of the post-auricular sulcus- cyst fixed to the skin, shows a central punctum:

Option D: Folliculitis would involve specific inflammation in the pilosebaceous units. It is caused by Staphylococcus aureus. They heal without a scar.

The image below shows folliculitis- inflammation of the pilosebaceous units leading to pustule formation:

Solution to Question 2:

A child having night-blindness, and multiple keratotic papules on elbows as shown in the image suggests a diagnosis of phrynoderma or toad-skin. This can occur due to the deficiency of vitamin A.

Phrynoderma is not specific to vitamin A deficiency. It may also be associated with deficiencies of B complex, riboflavin, vitamin C, vitamin E, and essential fatty acids. It manifests as groups of papules with a central keratotic plug. The commonest sites are elbows and knees. They are present around the follicles and give the skin a rough texture. Treatment involves the correction of poor diet and administration of a multivitamin preparation containing vitamin A.

Option B: Keratosis pilaris is an inherited abnormality of keratinization. It affects the follicular orifices leading to keratotic follicular plugging, perifollicular erythema, and follicular atrophy. It usually presents during puberty. They present as small, keratotic papules on the extensor aspects of the limbs, particularly the arms which can lead to ingrowing of the hair. Treatment is purely reserved for cosmetic purposes.

Keratosis pilaris

Keratotic follicular plugging with surrounding perifollicular erythema

Option C: Darier disease is an autosomal dominant genodermatosis characterized by keratotic papules that show acantholysis and dyskeratosis. ATP2A2 gene mutations are present in Darier disease. Histologically, acantholysis (separation of keratinocytes) and blistering are seen in the suprabasal epidermis.

It presents with discrete or confluent rough, greasy, skin-colored papules in seborrheic areas (trunk, face scalp margins) and flexures (perineum, axillae, groin). There may also be isolated nipple hyperkeratosis. Nail findings include fragility, painful longitudinal splits, or distinctive red and white longitudinal bands terminating in V-shaped nicks.

Darrier's disease

Discrete or confluent rough, greasy, skin-colored papules in seborrheic areas

Darrier's disease

Greasy papules over the flexural areas like thigh and groin

Nail finding in Darrier's disease

Distinctive red and white longitudinal bands terminating in V-shaped nicks.

Option D: Follicular eczema is a form of atopic dermatitis that presents with a discoid patch formed due to the coalescence of the follicular lesions with peripheral excoriation marks. The biopsy shows spongiotic dermatitis.

Solution to Question 3:

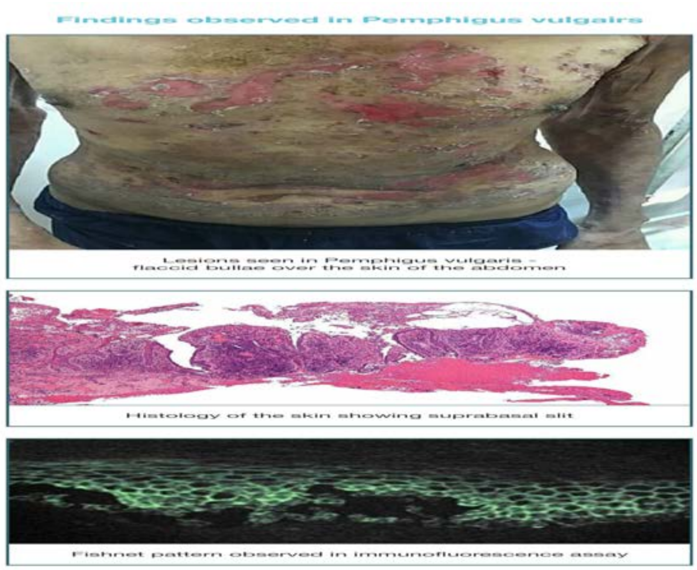
The given clinical vignette of a bullous disorder with the histologic picture pointing towards a suprabasal blister suggests the diagnosis of pemphigus vulgaris.

It is caused due to the production of antibodies mainly against desmoglein 3. Rarely antibodies against desmoglein 1 can also be found. The antibodies are commonly IgG antibodies. Desmoglein 3 is mainly present in the basal and suprabasal layers of the skin and mucosa.

The patients, therefore, present with flaccid bullae over the skin and mucosal ulcerations. Oral mucosa is commonly affected. Other areas where ulceration can be seen are conjunctiva, nasopharynx, larynx, etc. Cutaneous lesions are more common over the scalp, face, neck, upper chest, and back.

Direct immunofluorescence will show the deposition of antibodies around the keratinocytes in the epidermis. It presents with a characteristic fish-net appearance in the immunofluorescence assay It is treated with steroids, azathioprine, mycophenolate mofetil, rituximab, etc.

The images attached below depict the lesions, the suprabasal split, and the fish-net pattern respectively seen in a case of pemphigus vulgaris.



Option B: Pemphigus foliaceus is a benign form of pemphigus, an inflammatory blistering disorder. This is a form of a subcorneal blister, where the stratum corneum forms the roof of the bulla. It is caused by autoantibodies that result in the dissolution of intercellular attachments within the epidermis. Antibodies against desmoglein 1 are mainly seen in pemphigus foliaceous. There is no involvement of the mucosal epithelium in pemphigus foliaceus.

Lesions are most common on the scalp, face, chest, and back. Bullae are superficial and mainly present as areas of erythema and crusting. These represent superficial erosions at sites of blister rupture.

Option C: Pemphigus vegetans is a rare form of the bullous disorder that presents with large, moist, verrucous, vegetating plaques studded with pustules on the groin, axillae, and flexural surfaces. It is associated with eosinophilic microabscess. It has two types, a severe Neumann type, and a milder Hallopeau type.

Option D: Erythema multiforme is a self■limiting cytotoxic dermatitis resulting from cell■mediated hypersensitivity most commonly to infection or drugs. The common triggers are

• Herpes simplex infection

• Mycoplasma infection

• Bacterial infections

• Fungal infections

• Vaccinations

• Malignancy: carcinoma, lymphoma, leukemia

• Drug reactions: sulfonamides, cotrimoxazole, penicillins

• Miscellaneous: Lupus erythematosus (Rowell Syndrome), Sarcoidosis, Polyarteritis Nodosa

It presents clinically with macular, papular, or urticarial lesions, along with the classic acral iris or target lesions. The lesions are distributed on the distal extremities, palms or trunk, and oral and genital mucous membranes.

Target lesions are less than 3 cm in diameter, rounded, and have three zones:

• The central area of dusky erythema or purpura

• Middle paler zone of edema

• The outer ring of erythema with a well■defined edge

Atypical lesions will have only 2 zones. They fade in 1-2 weeks leaving dusky discoloration.

Eythema multiforme

Classical acral iris/ Target sign

Solution to Question 4:

The above image depicts paronychia associated with a reddish erythematous streak-like lesion on the hand suggestive of paronychia associated with lymphangitis. The treatment of choice for this patient is amoxiclav.

Acute paronychia is an acute infection of the nail beds and folds. It is most commonly caused by Staphylococcus aureus. It presents as a tender, erythematous swelling around the nail.

Treatment of pyogenic paronychia:

• Protection against trauma

• Keep affected fingernails dry.

• Abscesses - Incision and drainage. (abscess is often opened by pushing the nail fold away from the nail plate)

• Antibiotics- A penicillin in combination with a β-lactamase inhibitor, e.g. amoxicillin with clavulanic acid (Amoxiclav)

Chronic paronychia is usually due to infection with Candida albicans and responds to topical antifungal therapy.

Lymphangitis:

Lymphangitis represents acute inflammation elicited by the spread of bacterial infections into lymphatics, with group A β-hemolytic streptococci being the most common causative agent. It is manifested by red, painful subcutaneous streaks (the inflamed lymphatics) and painful enlargement of the draining lymph nodes (lymphadenitis). It is managed by elevating the affected extremity and broad-spectrum antibiotics.

Solution to Question 5:

The given scenario of worsening of existing lesions in the context of leprosy treatment with multi-drug therapy (MDT) is suggestive of a type 1 lepra reaction. The next best step in the management would be to continue MDT and start systemic steroids.

Type 1 lepra reactions present with inflammation of existing skin lesions. The lesions become red, painful, and edematous. They may rarely ulcerate. New lesions may occur but are very rare. It is associated with acute neuritis, with the tenderness and loss of sensory and motor function.

MDT is never stopped in patients with lepra reactions. For pain, NSAIDs can be supplemented along with oral steroids.

The use of thalidomide is restricted to severe forms of type-2 lepra reaction or erythema nodosum leprosum. These are characterized by systemic effects like iritis, orchitis, and eruption of new skin lesions.

The summary of lepra reactions is attached in the pearl below.

Solution to Question 6:

Favus and kerion are the inflammatory variants of tinea capitis.

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

• Kerion is a severe inflammatory type of tinea capitis which presents with an inflammatory mass with thick crusting and matting of hair.

• Favus is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles.

- Endothrix is a non inflammatory type in which black dots form around swollen hair shafts.
- Ectothrix is a mild inflammatory type in which dull grey coated broken hairs is seen.

Dermatology INI-CET 2022

Question 1:

A 10-year-old boy presents with fever, joint pain, and a lesion over his hand as seen in the image below. Which of the following is the clinical finding given and what is the likely diagnosis?

a) Gottron's papules - Juvenile dermatomyositis

b) Rheumatoid nodules - Juvenile rheumatic arthritis

c) Sclerodactyly - Systemic sclerosis

d) Mechanic's hand - Systemic lupus erythematosus

Question 2:

A 12-week pregnant woman who is already on multidrug therapy for leprosy presents with type 2 lepra reaction. What is the treatment of choice for this patient?

a) Stop MDT and start oral steroids

b) Antibiotics

c) Thalidomide

d) Continue MDT and add oral steroids

Question 3:

A patient presents to the outpatient department with a satellite skin lesion. On examination, he has one hypopigmented lesion and one thickened nerve. Images of two blister packs are given below. The treatment best suited for this patient would be:

Packet 1 Packet 2

- **a)** Packet 1 for 12 months
- **b)** Packet 2 for 12 months
- **c)** Packet 1 for 6 months
- **d)** Packet 2 for 6 months

Question 4:

Chemoprophylaxis for leprosy should be given to which of the following persons?

- **a)** 1, 2, 4
- **b)** 1, 3, 4
- **c)** 2, 3, 4
- **d)** 1, 2, 3

Question 5:

Match the following types of scales with their associated conditions.

- **a)** 1-b, 2-c, 3-d, 4-a

b) 1-a, 2-b, 3-d, 4-c

c) 1-d, 2-c, 3-a, 4-b

d) 1-c, 2-b, 3-d, 4-a

Question 6:

A 60-year-old patient presents with the lesion given in the image below. He also describes having the sensation of spiders crawling in the area of the lesion when he woke up. Which of the following is the most likely diagnosis?

a) Herpes zoster infection

b) Irritant contact dermatitis

c) Allergic contact dermatitis

d) Herpes simplex infection

Question 7:

A 60-year-old diabetic female patient presents with progressive skin lesions, as shown in the image. What is the diagnosis?

a) Squamous cell carcinoma

b) Basal cell carcinoma

c) Discoid lupus eryhtematosus

d) Melasma

Question 8:

A diabetic patient presents with the following lesions in the armpit. Which of the following would be seen on Wood's lamp examination?

a) Yellow fluorescence

b) Coral red fluorescence

c) Blue fluorescence

d) Green fluorescence

Question 9:

A 10-year-old boy is brought with weakness in his lower limbs and rash over the eyelids. The biopsy shows inflammation of the muscle. What is the finding shown in the image below?

a) Gottron's papules

b) Shawl sign

c) Heliotrope rash

d) Scleroderma

Question 10:

Consider the causes of alopecia. Which among the following cause(s) non-scarring alopecia?

a) 3 and 4

b) 1, 2, and 3

c) 2, 3, and 4

d) Only 4

Question 11:

Match the following therapeutic methods used in the management of psoriasis with their possible side effects.

- **a)** 1-c, 2-b, 3-a, 4-d
- **b)** 1-c, 2-a, 3-b, 4-d
- **c)** 1-b, 2-d, 3-c, 4-a
- **d)** 1-b, 2-c, 3-a, 4-d

Question 12:

Skin manifestations caused by the drug administered for amebic dysentery are due to ______.

- **a)** Fixed drug eruption
- **b)** Urticaria
- **c)** Maculopapular rash
- **d)** Atopic dermatitis

Question 13:

A 40-year-old man presents with erythematous painful lesions on the trunk and face, joint pain, and glove and stocking pattern of anesthesia. Which of the following investigations is used for the diagnosis of this condition?

- **a)** 1, 2
- **b)** 1, 2, 3
- **c)** 1, 3, 4
- **d)** 1, 2, 3, 4

Question 14:

Identify the condition shown in the image below.

a) Paget disease

b) Lichen sclerosus

c) Lichen planus

d) Angioma of vulva

Question 15:

A patient comes to the OPD with the following skin manifestation of a disease associated with Raynaud phenomenon. What is the manifestation?

a) Poikiloderma

b) Salt and pepper pigmentation

c) Leukotrichia

d) Vitiligo

Question 16:

A female patient came to the hospital with a history of relapsing, remitting, mildly itchy erythematous plaques over the scalp, sacral area, and knees. She has a history of joint pain and her nail changes are shown in the image below. What is your diagnosis?

a) Adult onset Still's disease

b) Pachyonychia congenita

c) Psoriasis

d) Lichen planus

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	a

5	a
6	a
7	b
8	b
9	a
10	b
11	a
12	a
13	b
14	b
15	b
16	c

Detailed Explanations

Solution to Question 1:

The given image showing erythematous papules over dorsal metacarpophalangeal and interphalangeal joints (Gottron's papules) is a characteristic skin lesion seen in dermatomyositis (DM). Fever and joint pain (arthralgia) are non-specific complaints.

DM is an inflammatory myopathy affecting skin and skeletal muscle. DM can be seen in children (juvenile DM) or in adults. It presents with symmetric, proximal more than distal muscle weakness associated with characteristic skin manifestations:

• Heliotrope rash: Lilac erythema of eyelids with periorbital edema

• Gottron’s papules: Violaceous, flat■topped papules on the skin overlying the interphalangeal joints and metacarpophalangeal joints

• Gottron sign: Erythematous rash over the extensor surfaces of joints such as the knuckles, elbows, knees, and ankles

• V-sign: Erythematous macules over anterior chest

• Shawl sign: Erythematous macules over posterior neck and upper back

• Mechanic's hand: Roughening and cracking of the skin of the tips and sides of the fingers, resulting in irregular, dirty-appearing lines that resemble those of a mechanic or manual laborer

• Nail bed telangiectasias

• Subcutaneous calcium deposits

DM is associated with interstitial lung diseases (ILD) and internal malignancies, most commonly adenocarcinomas. Adult onset DM has a stronger association with malignancies.

Laboratory evaluation of patients with dermatomyositis will reveal elevated serum creatinine kinase and antinuclear antibodies. Electromyography will reveal features of myopathy. Histology will show perifascicular atrophy. Immunohistochemical staining will reveal myxovirus resistance protein A, a sensitive and specific marker.

Treatment of dermatomyositis is include:

• Immunotherapy with high-dose glucocorticoids - first-line drug

• Methotrexate - second line drug. Indications include -

• When patient presents with severe weakness, or other organ system involvement

• Patients with risk of steroid complications like patients with diabetes, osteoporosis, or postmenopausal women

• Patients who do not show improvement after 2-4 months of treatment with prednisone or those who cannot be maintained on a low dose

• Triple therapy with intravenous immunoglobulin or rituximab may be required

The prognosis is favourable in the absence of associated ILD or malignancies.

Solution to Question 2:

The treatment of choice for a pregnant lady with type 2 lepra reaction is to continue multidrug therapy (MDT) and start oral steroids. MDT is not stopped in lepra reactions. MDT is proven to be safe in pregnancy. Thalidomide is contraindicated (especially in the 1st trimester) due to the risk of teratogenic changes in the fetus (phocomelia).

Type 2 lepra reaction or erythema nodosum leprosum (ENL) is an immune complex-mediated syndrome (type 3 hypersensitivity) affecting the skin, nerves, and other organs. It is mainly seen in the multibacillary form of leprosy and is commonly recurrent. It usually occurs during treatment, but can also be seen before or after treatment.

ENL presents with acute onset, painful, erythematous nodules on the face and extensor surfaces of limbs that blanch on pressure. Systemic manifestations include fever, malaise, anorexia, neuritis, uveitis, dactylitis, arthritis, myositis, etc. can also be seen.

Treatment includes:

• Thalidomide is the drug of choice in young men with severe or recurrent ENL. It is contraindicated during pregnancy due to its high teratogenicity causing phocomelia. Even in women of reproductive age, it is given with caution.

• Glucocorticoids (high doses) are routinely used in all patients and help in suppressing the acute signs and symptoms of inflammation.

• Clofazimine is useful as long-term maintenance therapy for preventing recurrent reactions. Clofazimine is not given to this patient because it is already a part of the MDT regimen containing dapsone, rifampicin, and clofazimine.

Solution to Question 3:

The treatment best suited for this patient would be packet 2-maroon for 6 months. The clinical scenario of one hypopigmented satellite skin lesion and one thickened nerve suggests paucibacillary leprosy. The treatment of paucibacillary leprosy (PB) is multidrug therapy (MDT) with rifampicin, dapsone, and clofazimine which is the maroon blister packet for 6 months.

The current recommendation by WHO is to treat PB leprosy with a combination of 3 drugs - rifampicin, dapsone, and clofazimine (maroon blister pack) for 6 months. Meanwhile, multibacillary (MB) leprosy is treated using the same 3 drugs for a duration of 12 months. While the NLEP guidelines earlier mentioned a 2-drug MDT for PB leprosy, the National Strategic Plan and Roadmap for Leprosy (2023-2027) now recommends a uniform MDT of 3 drugs for both PB and MB leprosy.

Management of leprosy is based on the bacterial load. WHO defines PB disease as the presence of no bacilli on smears or biopsy and less than or equal to 5 skin lesions with no or only one peripheral nerve involved. MB leprosy is characterized by bacilli on smears or biopsies of more than 5 skin lesions and more than 1 peripheral nerve involvement.

The image given below shows a hypopigmented satellite skin lesion seen in leprosy.

Chemoprophylaxis for leprosy should be given in those

• Age more than 2 years

• Close contact for more than 20 hours per week for more than 3 months

A single dose of rifampicin is recommended for the close contact of a patient with leprosy provided they do not develop any signs or symptoms pertaining to leprosy.

Solution to Question 4:

Chemoprophylaxis for leprosy should be given to contacts 1,2 and 4.

Chemoprophylaxis for leprosy should be given to the following contacts:

• A person who has been in close contact with the index case for 20 hours or more per week for more than 3 months.

• Close contact if the age is more than 2 years

The contact mentioned in option 4, is with the patient for more than 3 months. Therefore, even this contact will require chemoprophylaxis.

Chemoprophylaxis cannot be given if there is/are:

• Possible signs of leprosy

• Signs or confirmed tuberculosis

• Liver or kidney disease

• Pregnancy

• Received rifampicin in the last 2 years

A single dose of rifampicin is used in chemoprophylaxis. Chemoprophylaxis must be initiated only 4 weeks after treatment for leprosy has begun in the patient so as to ensure the patient is non-infectious.

Dose of rifampicin in chemoprophylaxis of leprosy	
Age	Dose
Children 2-5 years	10-15 mg/kg
Children 6-9 years (less than 20kg)	150 mg
Children 6-9 years (more than 20kg)	300 mg
10-14 years	450 mg
15 years and above	600 mg

Solution to Question 5:

The correctly matched option is option A: 1-b, 2-c, 3-d, 4-a.

Solution to Question 6:

The given image showing unilateral, red, grouped vesicles in a single dermatome associated with crawling sensation is highly suggestive of Herpes zoster infection.

Herpes zoster infection or shingles is a segmental eruption due to the reactivation of latent varicella-zoster virus from the dorsal root ganglia. An earlier infection with varicella is essential before zoster can occur.

The first symptom of zoster is usually pain. It may also be accompanied by fever, headache, malaise, and tenderness localized to areas of one or more dorsal roots. The mucous membranes within the affected dermatomes are involved.

The patients are infectious, both from the virus in the lesions and the nasopharynx. The most commonly involved dermatomes in decreasing order of frequency are thoracic > cervical > trigeminal > lumbosacral. The posterior nerve roots and ganglia show inflammatory changes.

Zoster is a self■limiting infection but it is painful and carries a risk of secondary infection and post■herpetic neuralgia. Analgesia and treatment of secondary infections are sufficient for mild infections.

Other options:

Option B: Irritant contact dermatitis is the cutaneous response to the physical/toxic effects of a wide range of environmental exposures. The main clinical features are usually dryness or chapping. The lesions could commonly develop into patchy or diffuse erythema with scaling and fissuring.

Option C: Allergic contact dermatitis is an eczematous reaction following exposure to a substance to which the immune system has previously been sensitized.

Option D: Herpes simplex lesions are thin-walled, umbilicated vesicles, the roof of which breaks down, leaving tiny superficial ulcers. They heal without scarring. The most common site of cutaneous lesions in the face - is on the cheeks, chin, around the mouth, or on the forehead.

Solution to Question 7:

The image above shows a central ulcer with raised borders and rolled-out edges (rodent ulcer) suggesting a diagnosis of basal cell carcinoma.

Basal cell carcinoma is a slow-growing malignant tumor with locally invasive characteristics. It affects the pilosebaceous skin.

Ultraviolet radiation exposure is the strongest predisposing factor. Other risk factors include exposure to arsenic, coal tar, aromatic hydrocarbons, and ionizing radiation.

Basal cell carcinoma mainly affects elderly or middle-aged men with excessive skin exposure, commonly in the age group of 40–80 y and in white-skinned people.

Macroscopically, it is categorized as localized and generalized. Nodular, nodulocystic (nodular and nodulocystic together account for 90% of basal cell carcinoma), pigmented, and nevoid are the localized cancers, while superficial multifocal and superficial spreading, infiltrative, and cicatrizing are the generalized types. The nodular type is described as pearly waxy nodules at the margins. A central depression (umbilication) or ulceration and rolled-out edges are characteristic of basal cell carcinoma.

The below gross image shows pearly, rolled-out borders suggestive of basal cell carcinoma.

Sites near the eye, nose, and ear and of size >2 cm are considered to be high risk. Direct invasion at these sites can reach the cranium.

Management is by Mohs microscopic surgery. Non-surgical measures, such as the administration of topical 5-fluorouracil and imiquimod, are useful. Metastasis is extremely rare.

Other options:

Option A: Squamous cell carcinoma (SCC) is the second most common skin cancer. UV radiation exposure is the primary risk factor for SCC. Classically, it appears as a scaly or ulcerated papule or plaque that can bleed with minimal trauma, but the pain is rare. It can exhibit in situ (confined to the epidermis) or invasive subtypes. Wide surgical excision including subcutaneous fat is the treatment of choice for SCC.

Option C: Discoid lupus erythematosus (DLE): DLE is typically located on the nose, cheeks, ear lobe, and concha. It may involve lips, oral mucosa, nose, or eyelids. Initial lesions of DLE are dry red patches, which then progress to indurated red or hyperpigmented plaques with adherent scale. Follicular keratosis (plugs of keratin within hair follicles) is observed when the surface scales are removed (carpet-tack sign). Older lesions are hyperpigmented, especially on the plaque edges. There is scarring, which results in central loss of pigment, white patches, and skin atrophy.

Option D: Melasma is the most common cause of facial melanosis that manifests as hyperpigmented macules on the face. It is exacerbated by sun exposure and oral contraceptive pills. The condition is predominantly seen in females. Melasma is very common in the 3rd trimester of pregnancy and is most marked in brunettes. It occurs due to increased levels of estrogen and progesterone, stimulating the activity of melanocytes. It resolves on its own after pregnancy. Hypermelanosis affects mainly the upper lip, the malar regions, forehead, and the chin.

Solution to Question 8:

The image showing brown patches in axilla in a diabetic patient is suggestive of erythrasma. Coral■red fluorescence under Wood's lamp is seen in erythrasma.

Erythrasma is a skin infection caused by Corynebacterium minutissimum. It is common in diabetics and immunodeficient patients and is associated with obesity, sweating, and poor hygiene. It affects the axillae, groins, toe webs, inter-gluteal and submammary regions. Lesions may be symptomless or show mild discomfort and itching.

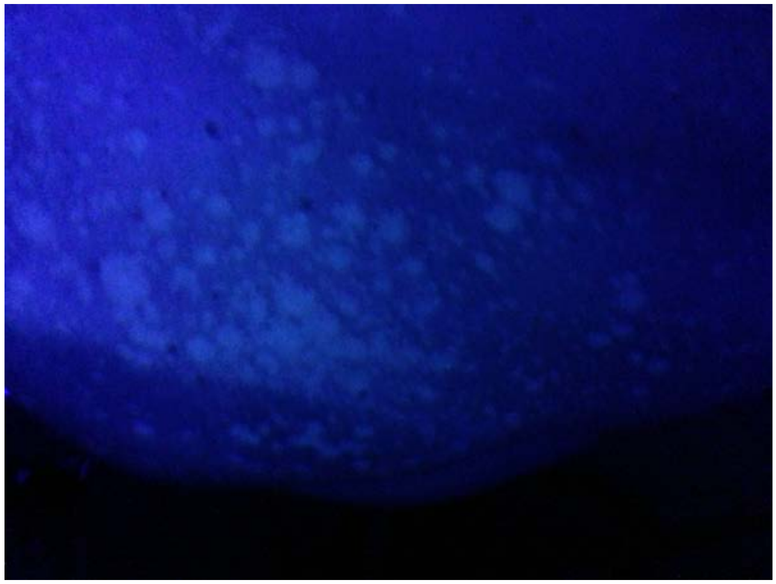
The lesions may be of irregular shape with sharp margins. They may start as smooth, red patches and then become brown. The lesions are more likely to be pruritic in tropical regions and may be asymptomatic in a temperate climate.

Coral■red fluorescence with Wood's light is due to coproporphyrin III and suggests erythrasma. The fluorescence may persist after the eradication of the disease.

Treatment of erythrasma is by using topical clotrimazole and miconazole for two weeks. For more extensive lesions, erythromycin can be administered. Relapse can occur. In such cases, long-term topical povidone-iodine application with drying agents can be used. Photodynamic therapy can also be attempted.

Green and blue-green fluorescence is seen in tinea capitis infections associated with microspora species and favus.

Pityriasis versicolor is caused by Malassezia species and the typical lesions are asymptomatic hypopigmented or hyperpigmented macules and patches and produce fine scales. The color of the scales may vary from pale ochre to medium brown. Pale yellow fluorescence seen in Wood's lamp examination is characteristic of tinea versicolor or pityriasis versicolor.



Solution to Question 9:

The finding shown in the image is Gottron's papules. The given scenario of a patient with weakness, rash, muscle inflammation, and presence of Gottron's papules on the skin is highly suggestive of dermatomyositis.

Dermatomyositis (DM) is an autoimmune disorder predominantly affecting the skin and skeletal muscle. The following are various manifestations of dermatomyositis:

• Proximal myopathy: Patients have trouble performing everyday activities like climbing stairs, getting up, and lifting their arms above their heads

• Gottron's papules: Violaceous, erythematous papules overlying the dorsal interphalangeal or metacarpophalangeal, elbow, or knee joints, as shown in the image below

• Gottron's sign: Symmetric, non-scaling, violaceous, erythematous macules or plaques, often atrophic, in the same distribution as Gottron's papules

• Shawl sign/V-sign: Fixed macular erythema, across the shoulders and outer arms (Shawl distribution)

• Periungual telangiectasias: Dilated capillaries of the nail folds

• Mechanic's hand: Roughening and cracking of the skin of the tips and sides of the fingers, resulting in irregular, dirty-appearing lines that resemble those of a mechanic or manual laborer

• Less common manifestations include: Facial swelling, malignancy, erythroderma, lichen planus

Laboratory evaluation of patients with dermatomyositis will reveal elevated levels of serum creatine kinase and antinuclear antibodies. Electromyography will reveal features of myopathy. Histology will show perifascicular atrophy. Immunohistochemical staining will reveal myxovirus resistance protein A, a sensitive and specific marker.

Treatment of dermatomyositis:

• Immunotherapy with high-dose glucocorticoids - first-line drug

• Methotrexate - second-line drug; indications include the following

• When the patient presents with severe weakness, or other organ system involvement

• Patients with a risk of steroid complications like patients with diabetes, osteoporosis, or postmenopausal women

• Patients who do not show improvement after 2–4 months of treatment with prednisone or those who cannot be maintained on a low dose

• Triple therapy with intravenous immunoglobulin or rituximab may be required

Solution to Question 10:

Non-scarring alopecia is caused by androgenic alopecia, alopecia areata, and telogen effluvium (1, 2 and 3). Frontal fibrosing alopecia is a type of primary cicatricial (scarring) alopecia.

Alopecia areata is a chronic inflammatory disease that causes non-scarring hair loss. It is a T-cell-mediated inflammatory disease. It is associated with several autoimmune diseases, including thyroiditis, lupus erythematosus, vitiligo, and psoriasis. It mainly affects the scalp, but can also affect the beard, eyebrows, and eyelashes. In some cases, this progresses to a total loss of scalp hair (alopecia totalis) or a loss of all hair on the body (alopecia universalis).

Telogen effluvium is a type of nonscarring alopecia that is marked by diffuse shedding of normal hair. This occurs due to premature termination of the anagen phase of the hair cycle. It can be either acute or chronic.

• Acute telogen effluvium occurs 2–3 months after a stressful event such as high fever, surgery, starvation, and hemorrhage. Hormonal fluctuations during the postpartum period can trigger it (telogen gravidarum). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally resolves spontaneously.

• Chronic telogen effluvium is used to describe the diffuse shedding of telogen hair for more than 6 months. Common causes include thyroid disorders, iron-deficiency anemia, and malnutrition.

Androgenetic alopecia is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. Dihydrotestosterone, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes frontal hairline recession and thinning of the vertex. Female-pattern hair loss involves the mid-frontal scalp diffusely. There is no or minimal involvement of the temporal region.

Solution to Question 11:

The correctly matched pairs are as follows:

PUVA is highly effective in clearing psoriasis with clearance rates of over 90%. It is associated with both acute (erythema, blistering) and chronic side effects. Long-term PUVA use is associated with the development of non-melanoma skin cancer (squamous cell cancer and basal cell cancer).

Cyclosporine is associated primarily with renal impairment (decreased GFR), metabolic disturbances, and an increased risk of malignancy. Due to its nephrotoxic effects, it is now uncommonly used for the maintenance of psoriasis.

Both acitretin and methotrexate are teratogenic.

Acitretin is associated with dose-dependent mucocutaneous side effects (mucositis) and hyperlipidemia. To prevent retinoid-induced embryopathy, female patients should avoid conception for 3 years after discontinuing acitretin treatment.

Acitretin is approved for use in the cutaneous manifestations of psoriasis.

Absolute contraindications:

• Pregnant women

• Women who are planning to conceive

• Breastfeeding

Methotrexate is also associated with mucositis and teratogenicity, it has been used as an abortifacient and is used in the management of molar pregnancies. Methotrexate is also associated with hepatotoxicity.

Column A	Column B
1. PUVA	c. Long-term use increases the risk of skin cancer
2. Cyclosporine	b. Effective but is highly nephrotoxic
3. Methotrexate	a. Abortifacient
4. Acitretin	d. Mucositis, teratogenic

Solution to Question 12:

The skin manifestations are most commonly due to metronidazole, which is associated with fixed drug eruption (FDE).

Although maculopapular rashes can also be seen with metronidazole, the incidence of a fixed drug eruption is higher with metronidazole.

Fixed drug eruptions/ reactions are characterized by sharply demarcated, dull red to brown lesions. They occur on lips, hands, legs, face, genitalia, and oral mucosa, and can cause a burning sensation. They are usually solitary or few in number and recur at the exact location. But, new lesions can occur on different body parts on re-exposure to the offending drug.

FDE is a type of delayed hypersensitivity reaction. Paracetamol/NSAIDs are some of the most common drugs causing FDE.

Drug■specific clinical patterns of FDE include:

Diagnosis is based on history and lesion morphology. Discontinuation of the offending drug is the first step in management.

The image below shows a well-defined hyperpigmented patch of a fixed drug eruption.

Drugs	Drug-specific clinical pattern of FDE
NSAIDs	Lips and genitals
Tetracycline/ Trimethoprim-sulfamethoxazole	Genitals
Metamizole	Trunk and extremities
Carbocysteine	Face

Solution to Question 13:

The clinical scenario points to a case of leprosy. The diagnosis of this condition requires nerve examination, skin biopsy, and a slit skin smear (option B- 1, 2, 3).

Leprosy is a chronic granulomatous condition caused by Mycobacterium leprae. It affects the skin, cutaneous and peripheral nerves, and mucous membranes. The most commonly affected peripheral nerves are the posterior tibial nerve, followed by the ulnar, median, and lateral popliteal nerves. The most commonly involved cranial nerve is the facial nerve.

Leprosy has a long incubation period, usually ranging from 2 to 12 years. It commonly presents as an area of numbness on the skin or a visible skin lesion which could be a hypopigmented or erythematous macule. Lesions usually involve the face, extensor aspect of the limbs, trunks, and buttocks. The lesions have impaired nerve function and poor hair growth. The patients may also present with burns or ulcers of the anesthetic limb.

Classification systems of leprosy:

• Ridley-Jopling classification:

• Tuberculoid (TT)

• Borderline tubercular (BT)

• Borderline (BB)

• Borderline lepromatous (BL)

• Lepromatous (LL)

• WHO classification:

• Paucibacillary

• Multibacillary

Cardinal signs for the diagnosis of leprosy:

• Characteristic skin lesions- hypopigmented or erythematous lesions with definite impairment of sensation.

• Peripheral nerve involvement- thickening of the nerve with sensory impairment. Nerve tenderness can also be seen.

• Detection of acid-fast bacilli in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases with low bacterial load.

Solution to Question 14:

The given image shows lichen sclerosus.

Lichen sclerosus usually presents in postmenopausal women. Multiple etiologies have been suggested (infectious, hormonal, genetic, and autoimmune). Symptoms include itching, excoriations, vulval skin thinning and crinkling, dysuria, and dyspareunia. Perianal involvement is frequently seen in the form figure of 8 appearance. The typical appearance is white, atrophic papules may coalesce into plaques and distort normal anatomy. Labia minora regression, clitoral concealment, urethral obstruction, and introital stenosis can develop. A biopsy is needed to exclude malignancy.

Topical corticosteroids remain the first line of treatment.

Premalignant lesions of vulva:

• Vulvar intraepithelial neoplasia (VIN)

• Paget's disease

• Lichen sclerosus

• Squamous cell hyperplasia

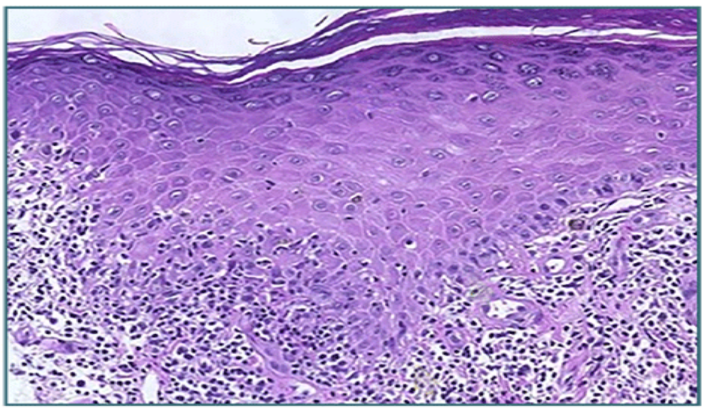
• Condyloma accuminata

• Bowen's disease (erythroplasia of Queyrat)

Lichen planus is an inflammatory disorder affecting the skin and mucosal membranes. Nail involvement includes exaggeration of the longitudinal lines and linear depression. It manifests as 6Ps pruritic, purple, polygonal, planar, papules, and plaques. It also exhibits Wickham striae (thin white lacy lines on the lesion), and the Koebner phenomenon (new lesions at sites of skin trauma). Mucosal involvement is common including mouth and tongue, rarely it can also be found on the anus and genitalia

The histological changes in lichen planus include increased Langerhans cells in the epidermis, spongiosis, and sawtooth appearance of Rete ridges. A fully developed lichen planus papule shows acanthosis of the epidermis, thickening of the granular layer, and hyperkeratosis. The image below shows the histopathological changes in lichen planus.

HPE Lichen Planus



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Vulvar Paget disease is an intraepithelial neoplasia that presents as an eczematoid, red, weeping area on the vulva. It is associated with invasive Paget disease or adenocarcinoma of the vulva or carcinoma at another non-vulvar location. It can be managed by local excision or vulvectomy. This condition needs long-term surveillance as recurrence is very common.

Angioma of vulva - Hemangiomas and angiokeratomas are benign lesions of the vulva. Angiokeratomas are benign tumors characterized by hyperplasia of the epidermis along with multiple dilated vessels in the superficial dermis. They present as well-defined, erythematous, hyperkeratotic papules. Senile (cherry) hemangiomas usually present as red macules that rapidly progress to well-circumscribed, raised, red, and soft lesions. Excision biopsy of bleeding hemangiomas, cryosurgery, or carbon dioxide laser can be used for management.

The image below shows a cherry angioma:

Solution to Question 15:

The clinical scenario is suggestive of systemic sclerosis, and the skin manifestation shown is a salt-and-pepper appearance.

Systemic sclerosis is an autoimmune disease of unknown etiology, complex pathogenesis, and variable clinical presentations. It is associated with significant disability and mortality. Virtually every organ is affected. Thickened and indurated skin, called scleroderma, is the distinguishing hallmark of systemic sclerosis.

Cutaneous features of systemic sclerosis:

• Raynaud phenomenon, with the typical white discoloration of the fingers caused by vasospasm and bluish discoloration of the fingertips

• Loss of pigmentation except in the perifollicular areas is known as the salt and pepper appearance of the skin, which is mostly seen in the upper back, chest, and scalp

• Puffy fingers in early disease

• Sclerodactyly and sclerosis extending proximal to the metacarpophalangeal joints with or without contractures

• Vasculopathic ulcers over the bony prominences

• Loss of fingertip pulp

• Severe flexion contractures

• Facial features include expressionless facies, mat■like telangiectases, microstomia, perioral furrowing, and a beak■like nose

Other options:

Option A: Poikiloderma is a skin disorder characterized by hypopigmentation and hyperpigmentation, wrinkling following atrophy, and telangiectasias. It is seen in ionizing radiation-damaged skin and dermatomyositis.

Option C: The hair in the amelanotic lesions of vitiligo can sometimes lose color and become white. This is called leukotrichia.

Option D: Vitiligo is characterized by depigmented macules that have sharp demarcation. It is due to the loss of melanocytes.

Solution to Question 16:

The diagnosis in this patient with the above clinical features is psoriasis.

The image shows nail pitting, oil drop sign, and salmon patches in fingernails

Psoriasis is a chronic inflammatory condition of the skin with systemic manifestations. The skin lesions can have exacerbations and relapse. They include red, silvery-white well-demarcated scales. A peripheral clearing known as the ring of Woronoff may be present. It is usually seen on the scalp, trunk, and extensor surfaces of the limbs. The appearance of a smooth, glossy red membrane with small bleeding points on the removal of the supra-papillary epidermis by gentle scraping is known as Auspitz's sign.

Psoriatic arthritis presents as early morning stiffness and joint swelling, heel pain due to enthesitis of the Achilles tendon, or plantar fasciitis. The clinical examination may reveal evidence of dactylitis (sausage fingers) or swollen or tender joints. The distal interphalangeal joint (DIP) is classically involved in causing pencil-in cup deformity.

Punctate pitting is the most common nail finding and others include the oil-drop sign, subungual hyperkeratosis, splinter haemorrhages, and distal onycholysis.

The topical application of steroids, retinoids, vitamin D, and emollients is the mainstay of management. Phototherapy and oral methotrexate, acitretin, and cyclosporine are used for more extensive and systemic diseases.

Dermatology NEET 2022

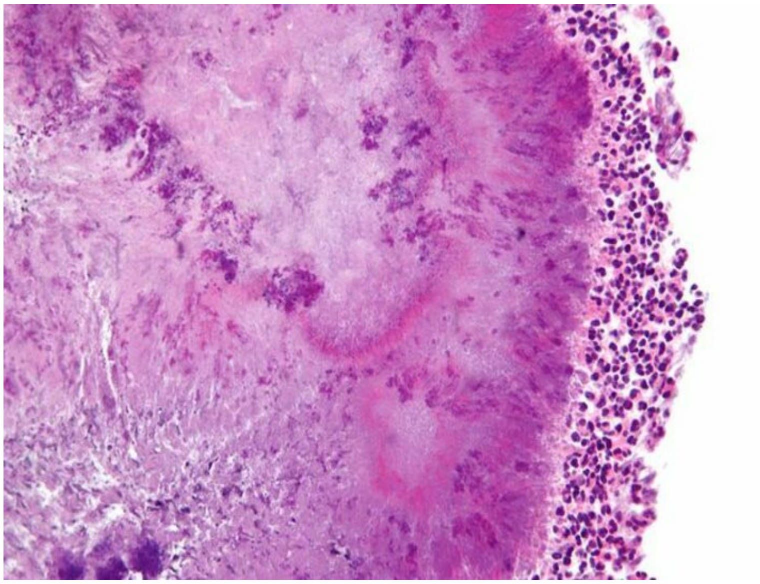
Question 1:

A patient presents to you with multiple anogenital warts. The biopsy of these lesions showed squamous atypia. Which of the following human papillomavirus types are considered high-risk?

- **a)** HPV 2
- **b)** HPV 18
- **c)** HPV 6
- **d)** HPV 11

Question 2:

A farmer presents to you with a swelling on the foot and multiple discharging sinuses in the lower limb. Granules from the discharge were examined under the microscope, which is shown below. Which of the following is true regarding this condition?



a) Both bacteria and fungi can be causative

b) Undergoes lymphatic spread

c) There is lymphocyte accumulation

d) Involves only superficial tissues

Question 3:

A farmer presents to you with a cauliflower-shaped mass on the foot, which developed after a minor injury. Microscopy shows copper penny bodies. What is the most likely diagnosis?

a) Chromoblastomycosis

b) Blastomycosis

c) Sporotrichosis

d) Phaeohyphomycosis

Question 4:

A patient presents with an anesthetic patch in the areas on the face as shown in the image below. Which of the following nerves is the most commonly involved in this condition?

a) Abducens nuclei

b) Facial nerve

c) Optic nerve

d) Trigeminal nerve

Question 5:

A female patient with a body mass index of 30 kg/m2 presents to you with a lesion on the neck as shown below. Which of the following condition is she most likely to be suffering from?

a) Hypothyroidism

b) Metabolic syndrome

c) Addison's disease

d) Hyperparathyroidism

Question 6:

Irregular pitting of nails with subungual hyperkeratosis is seen in _______.

a) Lichen planus

b) Psoriasis

c) Atopic dermatitis

d) Alopecia areata

Question 7:

A 35-year-old woman presents to you with hair loss for the past three months. She tested positive for COVID-19 eight months ago. What is the most likely diagnosis?

a) Tinea capitis

b) Telogen effluvium

c) Trichotillomania

d) Female-pattern androgenic alopecia

Answer Key

Question No.	Correct Option
1	b
2	a
3	a
4	b
5	b
6	b
7	b

Detailed Explanations

Solution to Question 1:

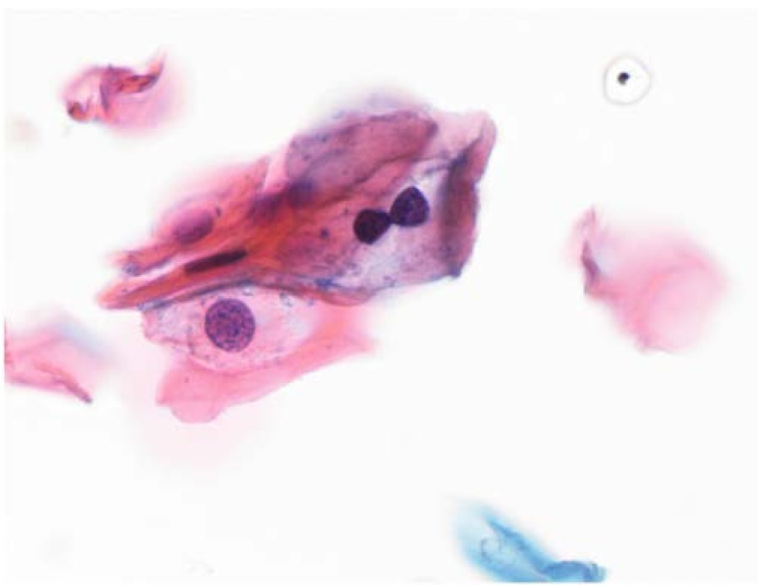
Human papillomavirus (HPV) types 16 and 18 are considered high-risk types that are commonly implicated in cervical, vaginal and vulval cancer. Other high-risk types include HPV types 31, 33, and 35.

The E6 protein of high-risk types binds to the p53 gene while the E7 protein inactivates the pRb gene. p53 and pRb are tumor suppressor genes that when inactivated results in uncontrolled cell proliferation. The most common type of cancer seen is squamous cell carcinoma. The image below depicts vulval carcinoma which is also associated with HPV infection.

The image given above shows anogenital warts pointing to condyloma acuminata (venereal warts). It is commonly caused by HPV types 6 and 11. However, rarely HPV 16 and 18 can cause it too. It is sexually transmitted. Patients are advised not to involve in intercourse when active genital warts are present. It can appear discrete or may coalesce to resemble cauliflower-like growths. 20% of cases may develop into cervical, vaginal, or vulval cancers. Risk factors include oral contraceptives and pregnancy. The condition is diagnosed by colposcopy.

The presence of koilocytes on histopathological examination is pathognomonic of the virus. These are cells with irregular and hyperchromatic nuclei, perinuclear halo, and peripheral cytoplasmic condensation. These cells disappear when there are dysplastic changes.

The image below depicts koilocytes.



Other options:

Option A - HPV type 2 is responsible for common warts in non-genital regions.

Options C and D - HPV types 6 and 11 are responsible for anogenital warts and laryngeal papillomatosis.

Solution to Question 2:

The clinical vignette along with the image showing sulfur granules is consistent with the diagnosis of mycetoma. Both bacteria and fungi can be causative agents of the disease.

Mycetoma is a chronic, slow-progressing subcutaneous infection that commonly affects the foot. Other sites which can be affected include the hands, upper back, and posterior part of the neck.

The mode of infection is usually traumatic. Agricultural workers are more prone to infection.

There are three types - eumycetoma, actinomycetoma, and botryomycosis. Eumycetoma is caused by fungal species such as Madurella mycetomatis and Madurella grisea. Adjacent fascia and bones may be involved as the disease progresses.

Actinomyces israelii and Nocardia asteroides result in the bacterial form - actinomycetoma. It is generally more invasive and can spread via the fascial planes to the underlying muscle or bone. Dissemination is rare but can occur in lesions of the head and neck.

Staphylococcus aureus and other pyogenic bacteria can cause botryomycosis.

Initially, it starts off as a small subcutaneous swelling of the foot, which then enlarges to form multiple discharging sinuses. The discharge contains granules which are microcolonies of the causative organism. Different causative organisms produce granules that are characteristic in color, consistency, and size, which helps in delineating the agent in each case.

The granules in eumycetoma are generally hard, while that in actinomycetoma is soft in consistency.

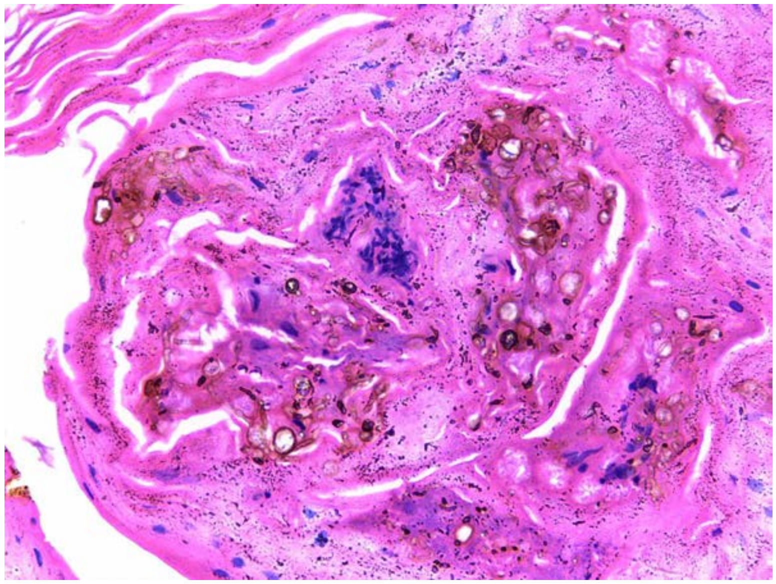
Medical management of eumycetoma includes ketoconazole and miconazole, while that of actinomycetoma includes dapsone or rifampicin. It should be continued for 18-24 months even after symptomatic recovery or the absence of laboratory evidence for infection.

Solution to Question 3:

The most likely diagnosis based on the clinical vignette is chromoblastomycosis.

Chromomycosis is a term used to describe various conditions caused by pigmented (dematiaceous) fungi. Chromoblastomycosis, the most common chromomycosis, is more commonly seen in tropical areas among agricultural workers and woodcutters who walk barefoot. Traumatic implantation is the mode of infection.

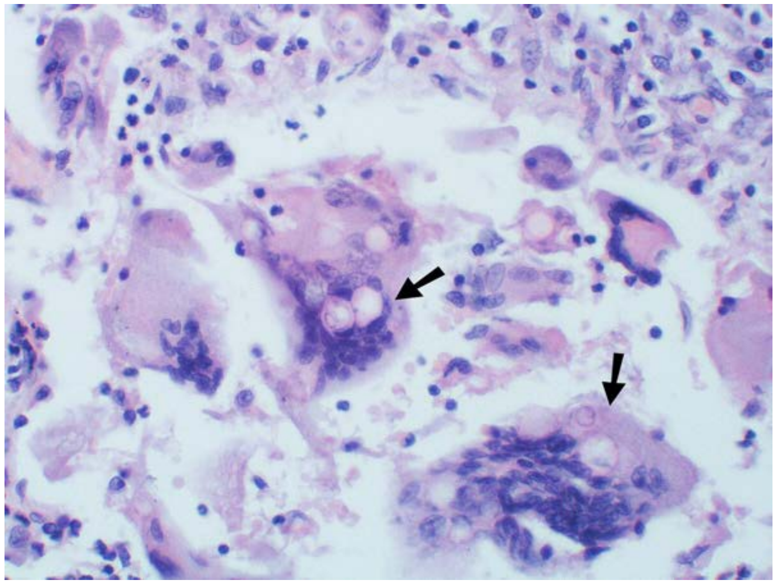
Patients present with warty cauliflower-like cutaneous nodules on the feet and lower legs. Species of the fungal genus Fonsecaea are implicated as the causative agents. Diagnosis is made by the demonstration of sclerotic bodies (copper-penny bodies) in tissue sections. Sclerotic bodies appear as round or irregular, dark brown bodies with septae.



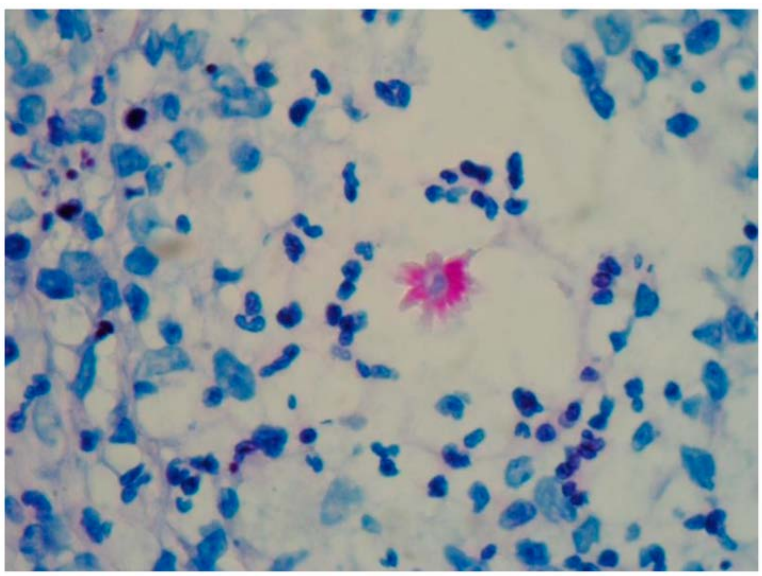
It can be managed with itraconazole (100–200 mg daily) or terbinafine (250 mg daily). For small lesions, surgical excision is the therapy of choice.

Other options:

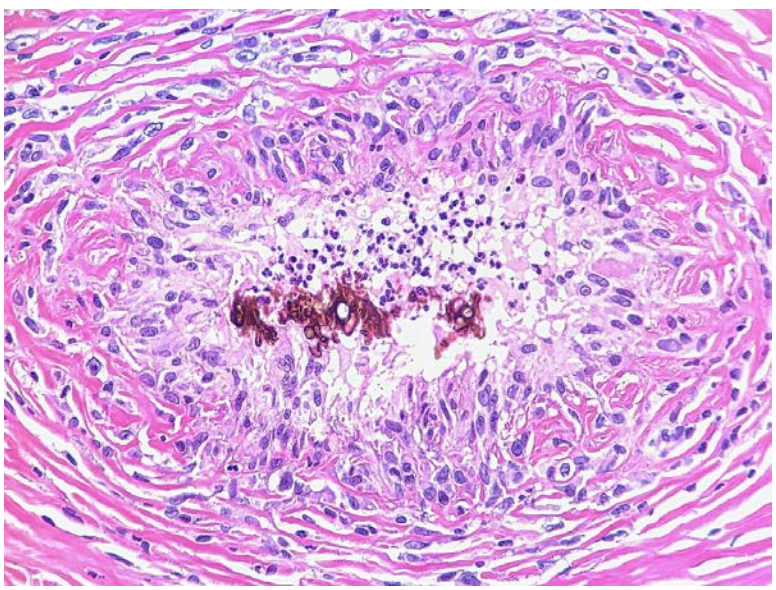
Option B - Blastomycosis is an infection caused by Blastomyces dermatitidis. It causes a chronic infection with granulomatous and suppurative lesions, which starts in the lungs but may disseminate to other organs. Symptoms of an acute lower respiratory tract infection with pulmonary infiltrates are the most common clinical presentations. Microscopic examination of stained specimens shows broad-based budding yeast cells.



Option C - Sporotrichosis caused by the dimorphic fungus Sporothrix schenckii, is commonly seen in gardeners followed by minor trauma. Patients present with skin nodules or lymph node ulceration. Microscopy shows asteroid bodies. It can be seen as an oval, basophilic structure with eosinophilic radiation.



Option D - Phaeohyphomycosis caused by dematiaceous fungi is frequently encountered in immunodeficient individuals. Clinical manifestations include brain abscesses, intramuscular abscesses, and cysts. On microscopy, the fungi appear as distorted hyphal strands. In contrast to chromoblastomycosis, sclerotic bodies are not found.



Solution to Question 4:

The clinical scenario points to a case of leprosy and the most commonly involved cranial nerve is the facial nerve.

Leprosy is a chronic granulomatous condition caused by Mycobacterium leprae. It affects the skin, cutaneous and peripheral nerves, and mucous membranes. The most commonly affected peripheral nerves are the posterior tibial nerve followed by ulnar, median, and lateral popliteal nerves. The most commonly involved cranial nerve is the facial nerve.

Leprosy has a long incubation period, usually ranging from 2 to 12 years. It commonly presents as an area of numbness on the skin or a visible skin lesion which could be a hypopigmented or erythematous macule. Lesions usually involve the face, extensor aspect of the limbs, trunks, and buttocks. The lesions have impaired nerve function and poor hair growth. The patients may also present with burns or ulcers of the anesthetic limb.

Classification systems of leprosy include:

• Ridley Jopling classification:

• Tuberculoid (TT)

• Borderline tubercular (BT)

• Borderline (BB)

• Borderline lepromatous (BL)

• Lepromatous (LL)

• WHO classification:

• Paucibacillary

• Multibacillary

The cardinal signs for the diagnosis of leprosy include:

• Characteristic skin lesions- hypopigmented or erythematous lesions with definite impairment of sensation.

• Peripheral nerve involvement- thickening of the nerve with sensory impairment. Nerve tenderness can also be seen.

• Detection of acid-fast bacilli in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases in low bacterial load.

Solution to Question 5:

The patient is most likely to be suffering from metabolic syndrome based on the clinical scenario given and the image depicting acanthosis nigricans.

Metabolic syndrome (syndrome X) is a group of conditions that increases the risk for cardiovascular disease. These include hypertension, hyperglycemia, hypertriglyceridemia, and central obesity. Increasing age, genetics, and sedentary lifestyle are considered risk factors.

Insulin resistance is the most accepted hypothesis used to explain the pathophysiology of the condition. An elevated level of circulating fatty acids is considered a major factor in the development of insulin resistance. It is associated with acanthosis nigricans which are velvety hyperpigmented lesions in skin flexures such as the axilla, groin, and back of the neck. Acanthosis nigricans can also be seen in malignancies, especially those of the gastrointestinal tract, acromegaly, and Cushing's syndrome.

Other conditions associated with metabolic syndrome include polycystic ovary syndrome, obstructive sleep apnea, and non-alcoholic fatty liver disease. The condition is managed by dietary and lifestyle modifications. Individual components of the syndrome also need to be addressed by the use of statins, antihypertensives, or hypoglycemic agents.

Other options:

Option A - Hypothyroidism is more common in women and can be either congenital or acquired. Autoimmune and iatrogenic hypothyroidism are the common causes of the latter. Symptoms include dry coarse skin, weight gain, poor appetite, fatigue, and alopecia.

Option C - Addison's disease (primary adrenal insufficiency), most commonly caused by autoimmune adrenalitis presents with hyperpigmentation of the skin, especially in areas exposed to sunlight or prone to friction. This is due to excessive stimulation of melanocytes by the raised adrenocorticotropic hormone (ACTH). This is seen in chronic cases.

Option D - Hyperparathyroidism is a disorder that arises due to excessive secretion of parathyroid hormone. Solitary adenomas are the most common cause. It primarily affects the renal and skeletal systems. Sestamibi scan is the imaging of choice to delineate parathyroid abnormalities.

Solution to Question 6:

Irregular pitting of nails with subungual hyperkeratosis is seen in psoriasis.

Psoriasis is a chronic inflammatory condition of the skin with systemic manifestations. Plaque psoriasis is the most common type. The nails are frequently involved, and punctate pitting is the most common finding seen. Nail beds can also be affected, giving rise to the oil-drop sign—a highly specific sign seen in this condition. Other manifestations include subungual hyperkeratosis, splinter hemorrhages, and distal onycholysis.

The skin is involved in the form of red/ salmon pink, scaly, and well-demarcated lesions. A peripheral clearing (ring of Woronoff) may be present. It is usually seen on the scalp, trunk, and extensor surfaces of the limbs. The lesions are surrounded by silvery-white scales. The appearance of a smooth, glossy red membrane with small bleeding points on the removal of the supra-papillary epidermis by gentle scraping (Auspitz's sign) is characteristic.

Psoriasis is associated with Koebner or isomorphic phenomenon. It describes the appearance of new lesions along the site of trauma (linear lesions). It is also observed in lichen planus.

Topical application of steroids, retinoids, vitamin D, and emollients is the mainstay of management. Phototherapy and oral methotrexate, acitretin, and cyclosporine can also be used. Both methotrexate and acitretin are teratogenic drugs. It is recommended that women do not conceive for three years after cessation of the acitretin.

Other options:

Option A - Lichen planus is an inflammatory disorder affecting the skin and mucosal membranes. Nail involvement includes exaggeration of the longitudinal lines and linear depression. This is due to nail plate thinning. If the nail bed is involved, it can result in hyperpigmentation, subungual hyperkeratosis, and onycholysis. In some cases, pterygium unguis can be seen due to adhesion between the dorsal nail fold and the nail bed.

Option C - Atopic dermatitis is a relapsing chronic inflammatory skin disease. Patients present with itchy erythematous papules/papulo-vesicles in flexural regions. Nail involvement is in the form of coarse pitting and ridging.

Option D - Alopecia areata is a condition that results in non-scarring hair loss. It can cause fine stippled pitting of the nails, roughening of the nail plate (trachyonychia), or atrophic dystrophy.

Solution to Question 7:

The most likely diagnosis based on the clinical scenario is telogen effluvium.

The hair follicle goes through three phases known as the hair cycle:

- Anagen: Phase of active hair growth
- Catagen: Involutionary phase
- Telogen: Time period between catagen and next anagen phase

Telogen effluvium is a type of nonscarring alopecia that is marked by diffuse shedding of normal hair. This occurs due to premature termination of the anagen phase of the hair cycle. It can be either acute or chronic.

Acute telogen effluvium occurs 2-3 months after a stressful event such as high fever, surgery, starvation, and hemorrhage. Hormonal fluctuations during the postpartum period can trigger it (telogen gravidarum). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally resolves spontaneously.

Chronic telogen effluvium is used to describe the diffuse shedding of telogen hair for more than 6 months. Common causes include thyroid disorders, iron-deficiency anemia, and malnutrition.

Other options:

Option A - Tinea capitis is used to describe an infection of the scalp by the dermatophyte group of fungi. The most common pathogenic species are Trichophyton tonsurans and Trichophyton violaceum. The presentation can vary from hair breakage associated with scaling to widespread inflammatory mass on the scalp.

Clinical variants of tinea capitis:

• Endothrix - A non-inflammatory type in which multiple black dots are present within the areas of alopecia

• Ectothrix - A mild inflammatory type in which single or multiple scaly patches with hair loss are seen

• Kerion - A severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation

• Favus - Characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles

Observation under KOH mount is used for diagnosis.

Endothrix variant of tenia captitis

Kerion

Option C - Trichotillomania (trichotillosis) is considered an obsessive-compulsive disorder (OCD) spectrum disorder, marked by gratification from pulling one's own hair. It could be associated with underlying anxieties or depression. The vertex region of the scalp is most commonly involved. Scalp histology will show perifollicular hemorrhages and intrafollicular pigment casts.

Option D - Androgenic hair loss is used to describe a progressive decrease in hair fiber production. Dihydrotestosterone, an androgenic metabolite, is implicated in the etiology. Female-pattern hair loss involves the mid-frontal scalp diffusely. There is no or minimal involvement of the temporal region.


Stage 1
Stage 2
Stage 3
Stage 4
Stage 5

Dermatology NEET 2023

Question 1:

A lady developed a skin reaction as shown in the image after using a hair dye. Which of the following chemicals is responsible for this condition?

- **a)** Pollen
- **b)** Chromates
- **c)** Balsam of Peru
- **d)** p-Phenylenediamine

Question 2:

A female patient presented with acne that is not resolving on oral isotretinoin and antibiotics therapy. Which of the following is the next best investigation?

- **a)** Look for dietary triggers
- **b)** Evaluate for hyperandrogenism
- **c)** Check for antibiotic resistance
- **d)** Look for drug triggers

Question 3:

A young woman presents with complaints of a painless ulcer in the genital area. It is associated with non-tender inguinal lymphadenopathy. What is the most likely diagnosis?

a) Chancroid

b) Syphilis

c) Herpes genitalis

d) Granuloma inguinale

Question 4:

Which of the following is associated with the clinical condition shown in the image?

a) Cataract

b) Glaucoma

c) Malignant melanoma

d) Basal cell carcinoma

Answer Key

Question No.	Correct Option
1	d
2	b

3	b
4	c

Detailed Explanations

Solution to Question 1:

This is most likely a case of hypersensitivity reaction caused by p-phenylenediamine (PPD), which is present in hair dyes.

Hair dyes such as PPD have the potential to cause contact dermatitis. 10% of PPD users tend to develop contact dermatitis. A patch test is recommended 24-48 hours before its usage either on the forearm or behind the ear. The appearance of any redness, blistering, or swelling implies allergy; therefore, the application of dye is not recommended. Cross-sensitization with other aromatic benzenes is also noted, which needs to be avoided.

Hair dyes can trigger acute edema and intense pruritus, followed by desquamation. Hair dye can elicit an allergic reaction on the forehead. PPD in the hair dye can also cause an erythema multiforme–like reaction.

In PPD-associated allergies, there is edema and weeping of the scalp margins, eyes, and ears. There may be relative sparing of the scalp. Immediate hypersensitivity reactions present as urticaria, and contact anaphylaxis have also been noted to occur in patients using PPD. To minimize the risk of an allergic reaction to hair dye, individuals should follow the manufacturer's instructions, perform a patch test, and avoid over-processing the hair. Treatment for PPD-induced contact dermatitis may include topical corticosteroids, oral antihistamines, and in severe cases, oral corticosteroids.

Other options:

Option A: Pollen is not typically found in hair dyes and is not responsible for this condition. Pollen allergies usually present as hay fever or seasonal allergic rhinitis.

Option B: Chromates, which can be found in cement, leather, and paints, are not typically present in hair dyes and are not responsible for this skin reaction.

Option C: Balsam of Peru, a fragrant resin, is found in various cosmetic and food products but is not a common component of hair dyes. It is less likely to be responsible for this condition compared to PPD.

Solution to Question 2:

The next best step for a female patient with acne that is resistant to oral isotretinoin is to check for hyperandrogenism.

Hyperandrogenism usually manifests as hirsutism, acne, and/or androgenic alopecia. Acne that is persistent, moderate to severe, or late-onset raises concern for PCOS or other causes of hyperandrogenism.

The pathogenesis of acne vulgaris involves

• Blockage of the follicular opening by hyperkeratosis

• Sebum overproduction

• The proliferation of commensal Propionibacterium acnes

• Inflammation

Management is directed at minimizing inflammation, decreasing keratin production, antibiotics, and antiandrogens to diminish sebum production.

Other options:

Option A: Looking for dietary triggers could be considered, but there is limited evidence to support a direct relationship between specific food items and acne.

Option C: Since the patient has not shown improvement even with antibiotic therapy, it is reasonable to consider checking for antibiotic resistance. However, evaluating for hyperandrogenism should be prioritized due to its potential role in the patient's condition.

Option D: Looking for drug triggers may be relevant in some cases, but it is less likely to be the primary cause of acne that is not responding to isotretinoin and antibiotics. Drug-induced acne typically resolves once the offending drug is discontinued.

Solution to Question 3:

The given clinical scenario of a painless ulcer and non-tender lymphadenopathy is suggestive of primary syphilis.

The incubation period of syphilis is 9-90 days and varies inversely with the size of the Spirochaete inoculum.

Stages of syphilis:

• Primary chancre: Painless, indurated, button-like ulcer with painless lymphadenopathy.

• Secondary syphilis is the stage when generalized manifestations of syphilis appear on the skin and mucous membranes. A macular rash is commonly seen and this stage may be preceded by constitutional symptoms such as fever, sore throat, and malaise.

• Latent syphilis is the stage when there are no clinical features of the active disease but serological testing for the disease is positive.

• Late syphilis (tertiary syphilis) commonly presents as superficial or nodular punched-out ulcers or gummas (granulomas appearing as cutaneous plaques or nodules with central ulceration) that appears 3-10 years after infection. It may also present as red nodules (nodular syphilides), neurosyphilis, and cardiovascular syphilis.

Serological testing is done using standard non-treponemal antibody tests like VDRL and rapid plasma reagin (RPR). Specific treponemal antibody tests like Treponema pallidum hemagglutination assay (TPHA) and fluorescent treponemal antibody absorption test (FTA-ABS) are further required. Microscopically, T. palladium is identified with characteristic corkscrew morphology and motility on darkfield microscopy.

Parenteral penicillin G is the drug of choice for all stages of syphilis. Treatment of recent sexual contacts is recommended.

Other options:

Option A: Chancroid presents with a painful genital ulcer and tender inguinal lymphadenopathy. It is caused by the bacterium Haemophilus ducreyi.

Option C: Herpes genitalis is characterized by painful vesicular lesions that progress to ulcers in the genital region, caused by the herpes simplex virus (HSV). It is usually associated with localized tender lymphadenopathy.

Option D: Granuloma inguinale (also known as donovanosis) is caused by the bacterium Klebsiella granulomatis. It presents as a slowly progressive, painless, beefy-red ulcerative lesion in the genital region, often without lymphadenopathy.

Solution to Question 4:

The given image showing a black, heterogeneous, hairy palpable lesion in a child is suggestive of congenital melanocytic nevus. Malignant melanoma is more commonly associated with a congenital melanocytic nevus.

Congenital melanocytic nevi (CMN) are benign, pigmented, melanocytic nevi present at birth. They are most commonly brown, black, purplish, or red, usually a palpable lesion with heterogeneous color and texture. CMN often have thick hair at birth, which grows at a significantly greater rate than the surrounding scalp hair.

Congenital melanocytic nevi can be classified based on size: small (<1.5 cm), medium (1.5-19.9 cm), and large or giant (>20 cm). Larger CMN have a higher risk of developing malignant melanoma. Management of CMN depends on the size and location of the lesion. Smaller nevi may be observed, while larger nevi or those at a higher risk for complications may require surgical excision, particularly if there is a concern for malignant transformation. Complications of CMN include neurological abnormalities (intraparenchymal melanosis is the commonest finding), malignant melanoma, and other tumors like rhabdomyosarcoma.

The lifetime risk of malignant transformation in large or giant CMN ranges from 5% to 10%. Regular follow-up and monitoring for changes in size, color, or texture are essential for early detection of potential complications.

Note: More than 50% of patients with Sturge-Weber syndrome have glaucoma as an ocular complication. The image given below shows port-wine staining in a child with Sturge-Weber syndrome.


DAILYROUNDS

Dermatology INI-CET 2023

Question 1:

A patient on Anti PD1 immunotherapy(Nivolumab) comes with pruritus, eczema, and urticarial rashes. Histopathological examination reveals a subepidermal blister. The direct Immunofluorescence of the lesions showed Linear C3 and IgG on the basement membrane. What is the diagnosis?

- **a)** Pemphigus vulgaris
- **b)** Bullous pemphigoid
- **c)** Dermatitis herpetiformis
- **d)** Pemphigus foliaceous

Question 2:

A 24 year old young male presented with asymptomatic scaly lesions over back, symmetrical in distribution. Images of the lesions are given, Identify the disease and the pattern of lesions

- **a)** Collaratte scale-Christmas tree pattern-Pityriasis rosea
- **b)** Wickhams striae-Lichen planus
- **c)** Christmas tree pattern-Lichen planus
- **d)** Wornoff ring-Seborrhic dermatitis

Question 3:

Statement 1: Histopathological examination of a patient with Flaccid bulla and painful erosions showing suprabasal blister and inflammatory infiltrate in the blister cleft. Statement 2: The row of tombstones appearance is diagnostic of Pemphigus vulgaris.

a) Statement 1 is false and Statement 2 is true

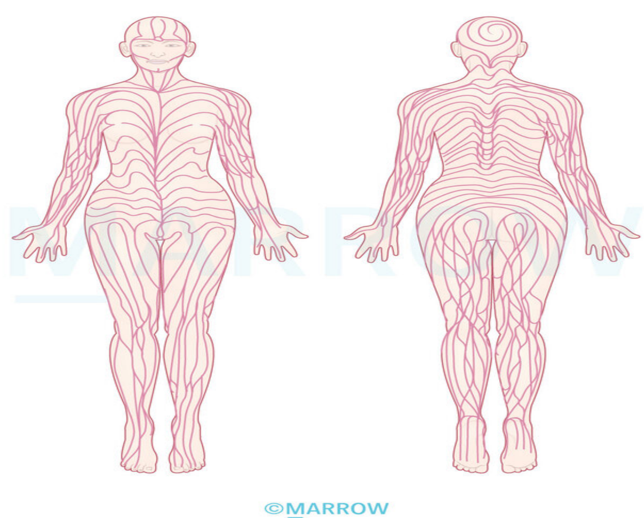
b) Statement 1 is true and Statement 2 is false

c) Both the statements are true but 2 is not the conclusion of 1

d) Both the statements are true but 2 is the conclusion of 1

Question 4:

Identify the following lines in dermatology



a) Hinderer lines

b) Marionettes line

c) Relaxed skin tension lines

d) Blaschko lines

Question 5:

Match the following wood lamp findings with the disease

a) 1-a,2-c,3-d,4-b

b) 1-e,2-c,3-d,4-b

c) 1-e,2-d,3-c,4-b

d) 1-a,2-d,3-c,4-b

Question 6:

Consider the causes of alopecia. Which among the following cause(s) non-scarring alopecia?

a) 3 and 4

b) 1, 2, and 3

c) 2, 3, and 4

d) Only 4

Question 7:

A farmer from Tamil Nadu presents with fever, body ache, and cough for 3 days. Lab studies showed thrombocytopenia. On examination, a black lesion is present, as in the picture below. IgM scrub typhus-positive. What drug will be used in the management?

a) Doxycyline

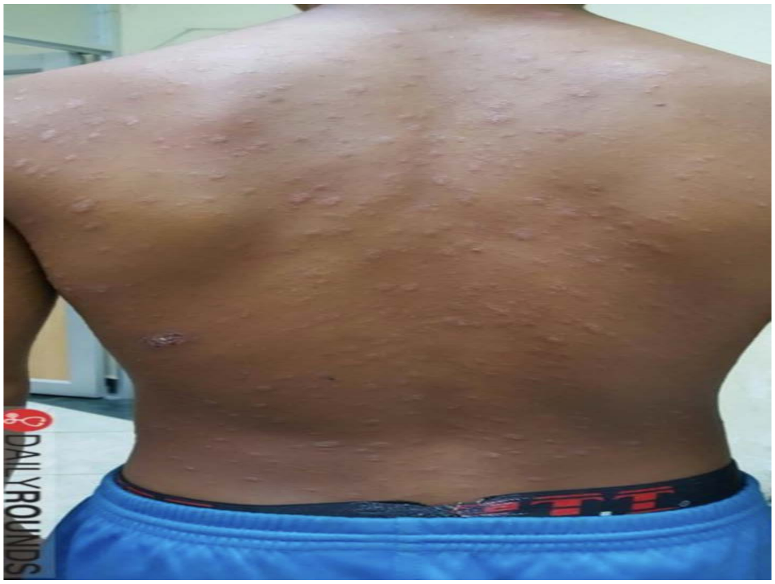
b) Ceftriaxone

c) Ceftriaxone+Doxycycline

d) Doxycycline + Azithromycin

Question 8:

A young male complains of an itchy rash over his back as seen below. What is the most likely sign and diagnosis?



a) Collarette scale-pityriasis rosea

b) Wickham's striae - lichen planus

c) Woronoff's sign - psoriasis

d) Christmas tree appearance - psoriasis

Question 9:

An overweight 12-year old girl has the following blackish discolouration on her neck, as shown below. What is this condition and its' association?

a) Acanthosis nigricans - Insulin resistance

b) Stria Albicans - Cushing syndrome

c) Xeroderma pigmentosum - Malignancy

d) Cafe au lait spots - McCune Albright syndrome

Question 10:

Which of the following are cause(s) of non-scarring alopecia?

a) 3 and 4

b) 1, 2 and 3

c) 2, 3 and 4

d) Only 4

Question 11:

A middle-aged Indian man came to your OPD with the lesion as shown in the image below. The most likely diagnosis is

a) Squamous cell carcinoma

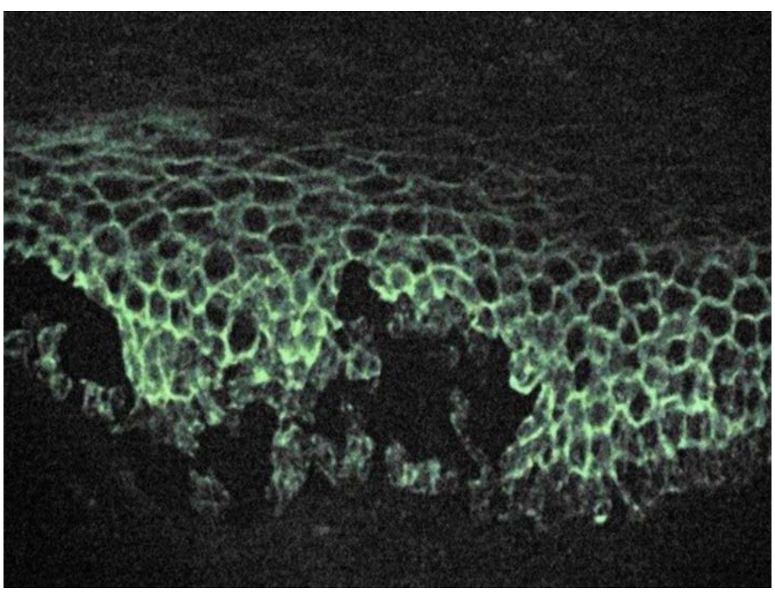
b) Basal cell carcinoma

c) Lupus vulgaris

d) Nevus

Question 12:

A patient with several skin lesions undergoes an investigation and an image of it is shown below. What is this test?



a) Indirect immunofluorescence

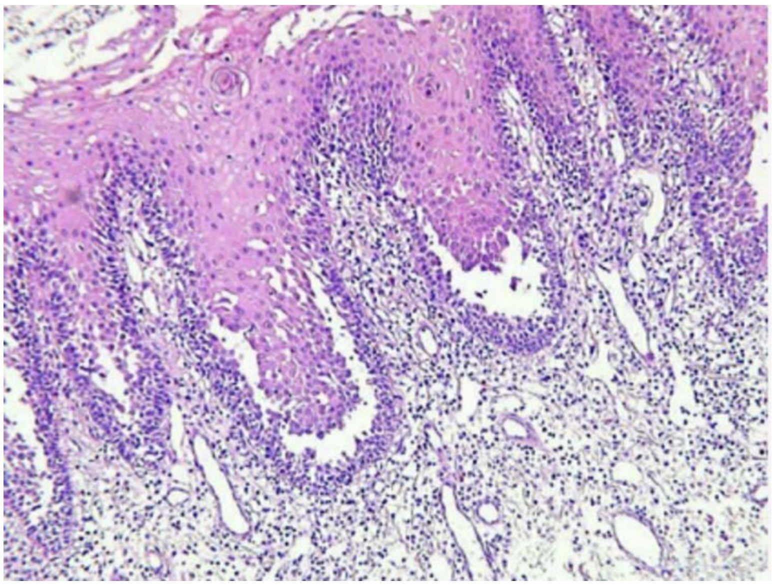
b) Direct immunofluorescence

c) Immunohistochemistry

d) Fluorescent in-situ hybridization

Question 13:

A patient presents with vesicles and oral lesions. Histopathological examination reveals suprabasal split. Which of the following is the most likely diagnosis?



a) Varicella

b) IgA disease

c) Pemphigus vulgaris

d) Pemphigus foliaceous

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	d
5	b

6	b
7	d
8	a
9	a
10	b
11	b
12	b
13	c

Detailed Explanations

Solution to Question 1:

In the given clinical scenario history of use of Anti-programmed cell death 1 (PD1) targeted immune checkpoint inhibitors with HPE showing subepidermal split and direct Immunofluorescence of the lesions showed Linear C3 and IgG on basement membrane is suggestive of bullous pemphigoid.

Anti-programmed cell death 1 (PD1) targeted immune checkpoint inhibitors such as nivolumab and pembrolizumab are associated with an increased risk of development of bullous pemphigoid. Histopathology will show the presence of subepidermal bullae with eosinophils. Immunofluorescent microscopy will show the presence of subepidermal split and deposition of IgG antibodies and C3 at the dermo-epidermal junction.

Bullous pemphigoid is one of the most common subepidermal immunobullous disorders. It is mostly a disease of the elderly. It is characterised by a subepidermal split, with tissue-bound and serum IgG antibodies against BP180, more than BP 230 antigens on hemidesmosomes, necessary to maintain the structure of.

Bullous pemphigoid is associated with psoriasis and treatments of psoriasis, like phototherapy and psoralen. There was an association seen with debilitating neurological conditions like stroke, Parkinsonism, cognitive impairment and multiple sclerosis, and haematological malignancies like lymphomas and myeloid leukaemia. It is also precipitated by drugs like diuretics, penicillamine and psycholeptics.

Clinical features include:

• Prodromal phase with excoriated papules, eczematous or urticarial skin lesions and haemorrhagic crusts may be seen.

• Bullous phase is characterised by:

• Intense pruritus is almost constantly present.

• Symmetrically distributed tense bullae and vesicles over the flexural aspects and abdomen of the body.

• Partly haemorrhagic crusts, erythematous plaques with an annular pattern are seen.

• Lesions heal without scarring, provided there are no superinfections.

• The bullae contain clear, sometimes haemorrhagic exudates.

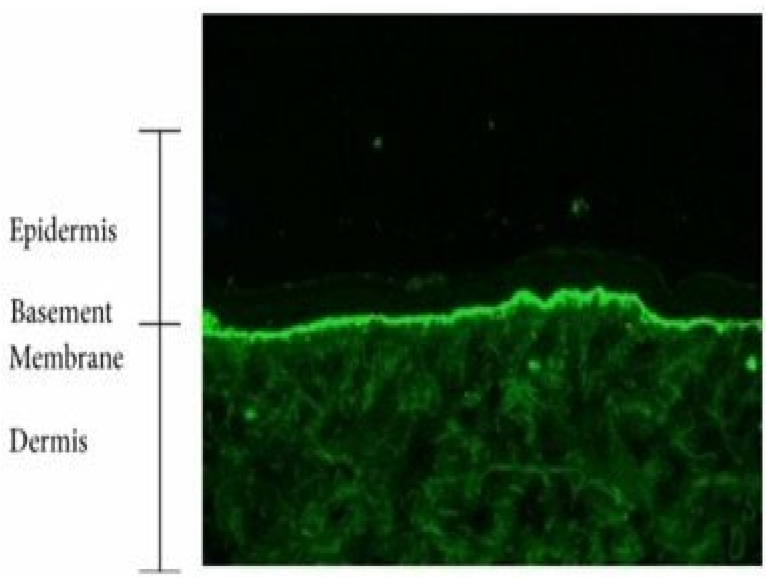
• Nikolsky's sign is negative.

• Oral or mucosal lesions and milia formation are rare.

The image given below shows the lesions seen in bullous pemphigoid.

Histopathology will show the presence of subepidermal bullae with eosinophils. Immunofluorescent microscopy will show the presence of subepidermal split and deposition of IgG antibodies and C3 at the dermo-epidermal junction.

The below image shows linear IgG deposits at dermo-epidermal junction seen in bullous pemphigoid.



Untreated bullous pemphigoid has a chronic, self-limiting course, with remissions and relapses. Complete remissions are possible with treatment.

Bullous pemphigoid is mainly treated with topical and systemic corticosteroids. Other drugs used like azathioprine, dapsone, doxycycline and immunomodulatory are also used.

Other options:

Option A-Pemphigus vulagaris is an intraepidermal immunobullous disease. It occurs commonly between the 4th to 6th decade. They present with flaccid blisters. Firm pressure over the lesion separates the epidermis from the dermis. Nikolsky's sign is positive. Mucosal involvement will be seen in the form of buccal and palatal erosions

Option C-Dermatitis herpetiformis typically appears in the 4th decade, and the patients present with itching and rash, most typically over the extensor surfaces of the elbows, knees, buttocks, and scalp. The majority of the patient have symptoms related to gluten-sensitive enteropathies, like bloating, diarrhea, and other gastrointestinal complaints. Immunofluorescence microscopy will show the deposition of IgA antibodies in the papillary dermis in a granular or fibrillar pattern. Complement 3 may also be found along with the IgA deposits.

Option D-Pemphigus foliaceous shows antidesmoglein-3 antibodies, anti-desmocollin antibodies and a predominance of anti-desmoglein-1 antibodies. Lack of mucosal involvement in pemphigus foliaceous, where anti-desmoglein-1 antibodies are prominent, is due to the low expression of desmoglein 1 in the mucosa. In pemphigus vulgaris, where anti-desmoglein-3 antibodies are prominent, mucosal involvement is extensive in early stages as desmoglein 3 is abundantly present in the mucosa.

Pemphigus Foliaceus

Solution to Question 2:

The above description with central detachment and peripheral attachment of scales with the images suggestive of collarette scales and lesions in christmas tree pattern is suggestive of pityriasis rosea.

Pityriasis rosea/pityriasis circinata/roseola annulata is an acute self-limiting papulosquamous disorder. It is commonly seen in children and young adults.

Clinical features include:

• Prodrome of a sore throat, gastrointestinal disturbance, fever or arthralgia.

• Herald patch - Large and conspicuous eruptions on the thigh, upper arm, trunk or neck covered by fine scales

The image given below shows a herald patch.

• After 5-15 days, it turns into discrete oval lesions, dull pink in color and covered by fine dry silvery-grey scales.

• These secondary lesions occur as macules and papules that are elliptical or ovular, symmetric, and commonly involve the thorax, back, and abdomen.

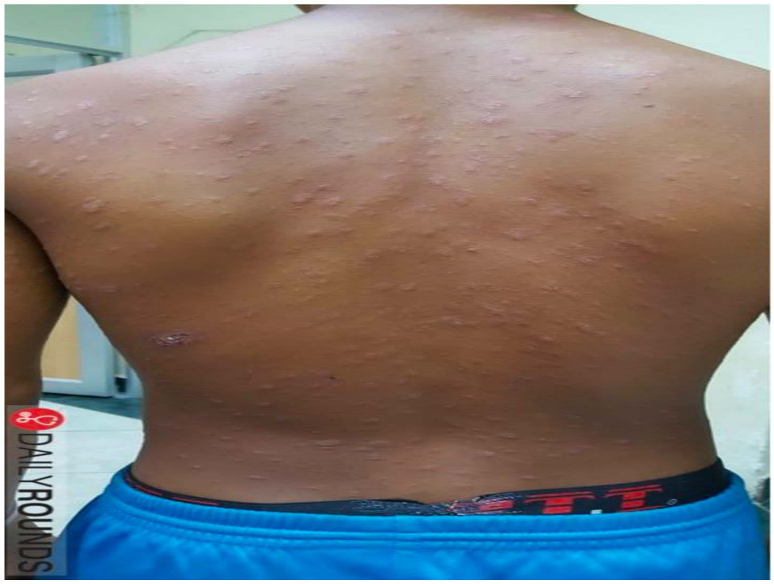
• It is followed by a marginal collarette of scale with a central wrinkling and atrophy. The collarettes of the scales are attached peripherally. It is also called a hanging curtain sign.

The image given below shows collarette scales.

Collarette scales in Pityriasis rosea

• The long axes of the lesions characteristically follow the lines of skin tension parallel to the ribs in a Christmas tree pattern on the upper chest and back.

The images given below show Christmas tree pattern along the lines of skin tension on the back and the chest.

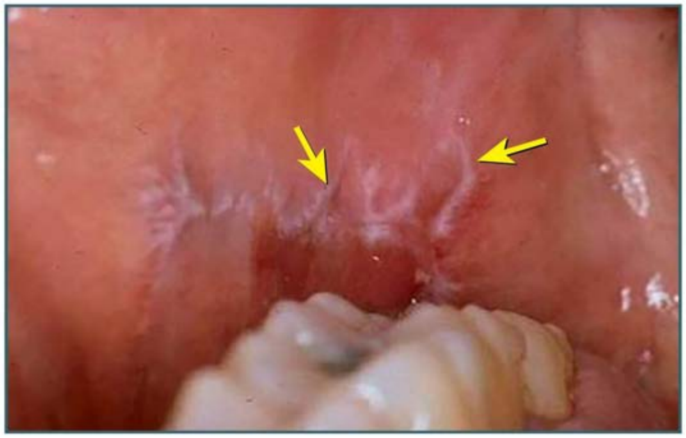


The rashes of pityriasis rosea usually resolve by 8 weeks.

Other options:

Lichen planus is an idiopathic inflammatory skin disease affecting the skin and mucous membranes, often with a chronic course with relapses and periods of remission.Patients present with pruritic, purple, polygonal, papules. The lesions have thin white lines on their surface known as Wickham's striae. They are commonly seen on the volar aspect of the wrists, the lumbar region, and around the ankles.The mucosal involvement is seen. The commonest mucosal sites are buccal mucosa and tongue, rarely it can also be found on the anus and genitalia.

Wickham's Striae on the buccal mucosa.



A hypopigmented ring surrounding individual psoriatic lesions is known as the Woronoff ring.It is usually associated with treatment, most commonly UV radiation or topical corticosteroids. The pathogenesis of the Woronoff ring is not well understood, but may result from the inhibition of prostaglandin synthesis.

Solution to Question 3:

Both statements 1 and 2 are correct but statement 2 is not the conclusion of statement 1. Statement 2 is not the conclusion of statement 1 because DIF is used for a definitive diagnosis.

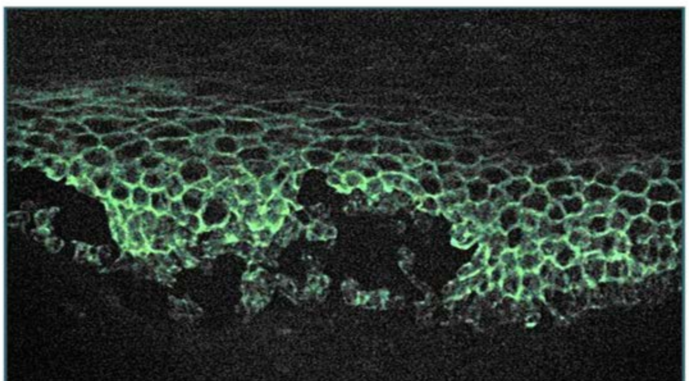
Pemphigus vulgaris is a chronic immunobullous disorder characterized by the presence of antibodies against desmosomal proteins expressed in the skin and mucosa. In the skin, desmoglein 1 is mostly seen in all layers, while desmoglein 3 is more in the lower layers. In the oral mucosa, desmoglein 3 is seen in all layers and desmoglein 1 is present in lower layers.

Clinically, the patients present with ill■defined, irregular, slow-to-heal oral lesions and erosions. Flaccid blisters filled with clear fluid may arise on normal skin or an erythematous base. The rupture of such blisters produces painful erosions and heals without scarring. Lesions may also involve the mucosa of the conjunctiva, nasopharynx, larynx, esophagus, urethra, vulva, or cervix.

Nikolsky's sign is positive which is elicited by applying firm tangential sliding pressure with a finger on lesional or perilesional skin. This leads to the separation of the normal■looking epidermis from the dermis, producing erosion.

Direct immunofluorescence is considered the most accurate confirmatory test of this condition. It shows intercellular IgG and C3 deposition on the surface of keratinocytes throughout the epidermis in normal, perilesional skin, in a characteristic fishnet/chickenwire appearance as seen below.

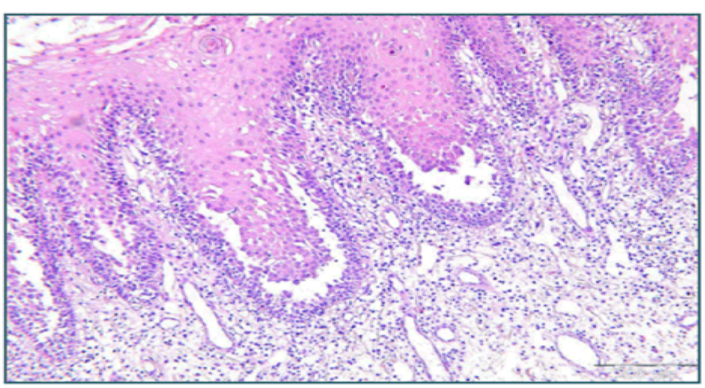
Direct immunofluorescence in Pemphigus Vulgaris



Intercellular deposition of immune complexes giving a "chicken wire/fish net" look

Histology shows a suprabasal split of the epidermis with a characteristic row of tombstone appearance as seen below.

Pemphigus Vulgaris -
Row of tombstones appearance

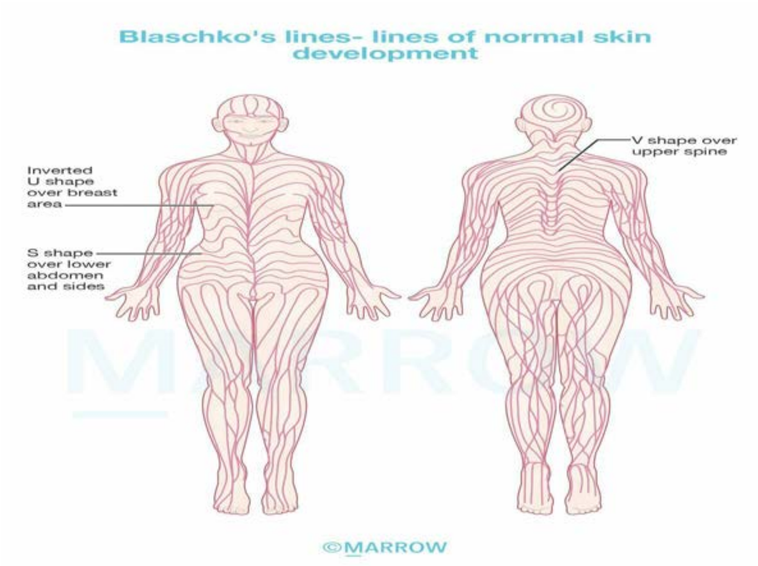


Other investigations include a Tzanck smear showing acantholytic cells. Anti-Dsg ELISA assays can be used to monitor the disease activity.

Systemic corticosteroids are the mainstay of treatment for controlling the disease, improving the quality of life and reducing disease activity.

Solution to Question 4:

The above image shows Blaschko lines as identified by the spiraling over the scalp. Blaschko's lines are lines of normal developmental growth patterns in the skin. They do not correspond to any known nervous, vascular, or lymphatic structures. Conditions that are distributed along these lines include epidermal nevi, and incontinentia pigmenti.



Other options:

Option A: Hinderer's lines are 2 intersecting lines—one drawn from the ala to the tragus and the other from the oral commissure to the lateral canthus.

Option B: Melomental folds, or marionette lines, are one of the consequences of facial aging. The curvilinear wrinkles form because of facial movements and the aging process extend downward from the oral commissures.

Option C: Relaxed skin tension lines run perpendicular to the direction of contraction of the underlying muscles and parallel to the dermal collagen bundles.To produce favourable aesthetic results, incisions should generally be designed to follow rhytids or the relaxed skin tension lines as the resulting scars will be stronger and less likely to stretch.

Solution to Question 5:

The correct match is

• Erythrasma e)Coral red fluorescence

• Pityriasis versicolor c)Yellow fluorescence

• Tinea Capitis d)Blue fluorescence

• Vitiligo b)Milky white florescence

Solution to Question 6:

Non-scarring alopecia is caused by androgenic alopecia, alopecia areata, and telogen effluvium (1, 2 and 3). Frontal fibrosing alopecia is a type of primary cicatricial (scarring) alopecia.

Alopecia areata is a chronic inflammatory disease that causes non-scarring hair loss. It is a T-cell-mediated inflammatory disease. It is associated with several autoimmune diseases, including thyroiditis, lupus erythematosus, vitiligo, and psoriasis. It mainly affects the scalp, but can also affect the beard, eyebrows, and eyelashes. In some cases, this progresses to a total loss of scalp hair (alopecia totalis) or a loss of all hair on the body (alopecia universalis).

Telogen effluvium is a type of nonscarring alopecia that is marked by diffuse shedding of normal hair. This occurs due to premature termination of the anagen phase of the hair cycle. It can be either acute or chronic.

• Acute telogen effluvium occurs 2–3 months after a stressful event such as high fever, surgery, starvation, and hemorrhage. Hormonal fluctuations during the postpartum period can trigger it (telogen gravidarum). Telogen effluvium can result in a diffuse reduction in hair density but never total baldness. It generally resolves spontaneously.

• Chronic telogen effluvium is used to describe the diffuse shedding of telogen hair for more than 6 months. Common causes include thyroid disorders, iron-deficiency anemia, and malnutrition.

Androgenetic alopecia is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. Dihydrotestosterone, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes frontal hairline recession and thinning of the vertex. Female-pattern hair loss involves the mid-frontal scalp diffusely. There is no or minimal involvement of the temporal region.

Solution to Question 7:

The above clinical scenario and the image show an eschar at the bite site along with positive IgM of scrub typhus caused by Orientia tsutsugamushi and transmitted by trombiculid mite. As per a study published in 2023, Doxycycline and Azithromycin combined therapy was found to be superior to Doxycycline monotherapy.

Scrub Typhus is also known as chigger-borne typhus, which is caused by Orientia tsutsugamushi. The vector is trombiculid mite. Humans are infected when they are bitten by the mite larvae known as chiggers. The incubation period is 1- 3 weeks. The patient typically develops a characteristic eschar at the site of the mite bite, with regional lymphadenopathy and maculopapular rash.

Diagnosis is by serologic assays.

Treatment is with a combination therapy of azithromycin and doxycycline.

Solution to Question 8:

Pityriasis rosea is the most likely diagnosis in this patient presenting with christmas tree lesions, and will likely have collarette scales.

The image given below shows collarette scales.

Pityriasis rosea/pityriasis circinata/roseola annulata is an acute self-limiting papulosquamous disorder. It is commonly seen in children and young adults.

Clinical features include:

• Prodrome of a sore throat, gastrointestinal disturbance, fever or arthralgia.

• Herald patch - Large and conspicuous eruptions on the thigh, upper arm, trunk or neck covered by fine scales

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The rashes of pityriasis rosea usually resolve by 8 weeks.

Other options:

Lichen planus is an idiopathic inflammatory skin disease affecting the skin and mucous membranes, often with a chronic course with relapses and periods of remission.Patients present with pruritic, purple, polygonal, papules. The lesions have thin white lines on their surface known as Wickham's striae. They are commonly seen on the volar aspect of the wrists, the lumbar region, and around the ankles.The mucosal involvement is seen. The commonest mucosal sites are buccal mucosa and tongue, rarely it can also be found on the anus and genitalia.

A hypopigmented ring surrounding individual psoriatic lesions is known as the Woronoff ring.It is usually associated with treatment, most commonly UV radiation or topical corticosteroids. The pathogenesis of the Woronoff ring is not well understood, but may result from the inhibition of prostaglandin synthesis.

Solution to Question 9:

The condition shown in the above image is likely acanthosis nigricans - that is strongly associated with insulin resistance/hyperinsulinemia

Acanthosis nigricans is a darkening of the skin of the intertriginous areas. The hyperpigmentation is generally velvety with poorly defined borders and thickening of the skin.

It can be grossly classified into benign and malignant.

Benign acanthosis nigricans is associated with obesity or insulin resistance. Insulin resistance is the most accepted hypothesis used to explain the pathophysiology of the condition. Elevated circulating fatty acids are considered a major factor in developing insulin resistance.

Malignant acanthosis nigricans has a rapid onset and is characterized by symmetrical, hyperpigmented, rugose, velvety plaques. It affects the axillae and flexures, nape of the neck, and mucosal surfaces. If it occurs de novo and is associated with weight loss, malignancy should be suspected. Generalized pruritus and skin changes of tripe palms (acanthosis palmaris) are also associated with malignancy. Associated malignancies are generally GI or lung carcinomas. In malignancy, tumor cells are believed to produce transforming growth factor-alpha or cytokines that activate insulin-like growth factors.

Acanthosis nigricans can only be treated by treating the underlying disease. In addition, treatment specific to acanthosis nigricans is done for aesthetic reasons. Medications like keratolytics, topical vitamin D, and melatonin can be used. Procedures like laser, dermabrasion, and chemical peels are also available.

Other options:

Option B: Stria Albicans is a stretch mark that is white or pearly in color. It is most common in pregnant women and people who have gained or lost weight quickly. Stria Albicans is not associated with Cushing syndrome, which is a hormonal disorder caused by excess cortisol production and produces a strechy, purplish rash most commonly over the abdomen/flanks.

Option C: Xeroderma pigmentosum - malignancy is a rare genetic disorder that makes the skin extremely sensitive to ultraviolet radiation from the sun. People with xeroderma pigmentosum have a greatly increased risk of developing skin cancer.

Option D: Cafe au lait spots - McCune Albright syndrome is a genetic disorder that causes multiple light brown, oval-shaped birthmarks on the skin. It is also associated with other symptoms, such as precocious puberty and bone abnormalities - polyostotic fibrous dysplasias.

Solution to Question 10:

Non-scarring alopecia is caused by androgenic alopecia, alopecia areata, and telogen effluvium.

Alopecia areata

It is a chronic inflammatory disease that causes non-scarring hair loss. It is a T-cell-mediated inflammatory disease. It is associated with several autoimmune diseases, including thyroiditis, lupus erythematosus, vitiligo, and psoriasis.

It mainly affects the scalp, but can also affect the beard, eyebrows, and eyelashes. In some cases, this progresses to a total loss of scalp hair (alopecia totalis) or a loss of all hair on the body (alopecia universalis).

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Androgenetic alopecia

It is a disorder characterized by a reduction of hair fiber production by follicles and miniaturization. Dihydrotestosterone, an androgenic metabolite, is implicated in the etiology. It is of two types: male-pattern and female-pattern hair loss. Male-pattern hair loss causes frontal hairline recession and thinning of the vertex. Female-pattern hair loss involves the mid-frontal scalp diffusely. There is no or minimal involvement of the temporal region.

CICATRICIAL/SCARRING ALOPECIA: Frontal fibrosing alopecia is a primary cicatrical condition that affects post-menopausal women in particular and frequently involves the eyebrows.

Solution to Question 11:

The image above shows a central ulcer with raised borders and rolled-out edges (rodent ulcer) suggesting a diagnosis of basal cell carcinoma.

Basal cell carcinoma is a slow-growing malignant tumor with locally invasive characteristics. It affects the pilosebaceous skin.

Ultraviolet radiation exposure is the strongest predisposing factor. Other risk factors include exposure to arsenic, coal tar, aromatic hydrocarbons, and ionizing radiation.

Basal cell carcinoma mainly affects elderly or middle-aged men with excessive skin exposure, commonly in the age group of 40–80 y and in white-skinned people.

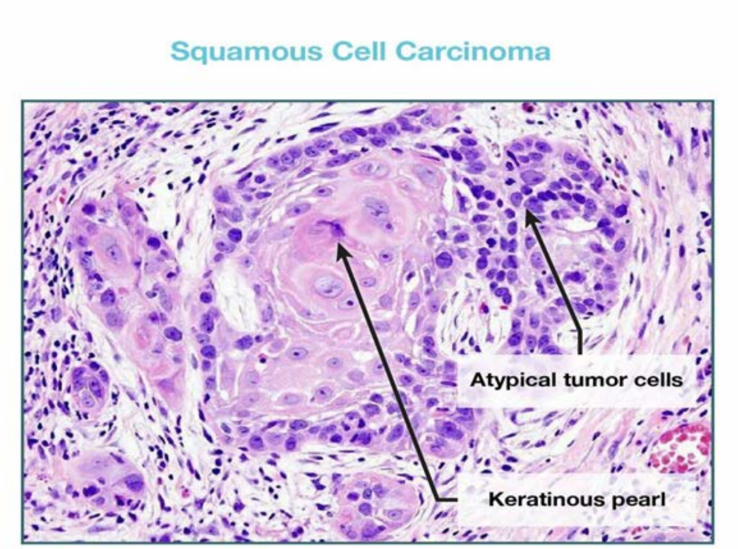
Macroscopically, it is categorized as localized and generalized. Nodular, nodulocystic (nodular and nodulocystic together account for 90% of basal cell carcinoma), pigmented, and nevoid are the localized cancers, while superficial multifocal and superficial spreading, infiltrative, and cicatrizing are the generalized types. The nodular type is described as pearly waxy nodules at the margins. A central depression (umbilication) or ulceration and rolled-out edges are characteristic of basal cell carcinoma.

Sites near the eye, nose, and ear and of size >2 cm are considered to be high risk. Direct invasion at these sites can reach the cranium.

Management is by Mohs microscopic surgery. Non-surgical measures, such as the administration of topical 5-fluorouracil and imiquimod, are useful. Metastasis is extremely rare.

Other Options:

Option A: Squamous cell carcinoma is the second most common malignant tumor, seen more commonly in men. Microscopically, the proliferation of irregular masses of squamous epithelium is seen. The image is shown below.



Option C- Lupus vulgaris originates from an underlying focus of tuberculosis, typically in a bone, joint, or lymph node, or maybe from the hematogenous and lymphatic spread. It presents as a slowly enlarging plaque with an elevated border and central atrophy. The edges of the lesion gradually extend in some areas and heal with scarring in others. In India, the buttocks, and trunk are more frequently involved. Apple jelly nodules are demonstrated on diascopy.

Option D: Nevus are aggregates of melanocytes at the dermis or dermo-epidermal junction. It can get transformed into malignant melanoma and the features suggestive of malignant transformation are a change in size, shape, color, thickness, satellite lesions, and tingling/ itching/serosanguinous discharge.

Solution to Question 12:

Direct immunofluorescence test, that is shown above, is the most accurate test to diagnose Pemphigus Vulgaris.

Direct Immunofluorescence uses anti-IgG antibodies which detects the anti-desmoglein IgG antibodies deposited on the Desmosomes. These desmosomes are present at cell junctions; this leads to the fishnet pattern.

Direct immunofluorescence (DIF) is a type of immunofluorescence that is used to diagnose autoimmune blistering diseases, such as pemphigus. In DIF, antibodies are applied directly to a skin biopsy specimen. If the patient has pemphigus, the antibodies will bind to the desmosomes, which are the proteins that hold skin cells together. This binding will produce a fish net pattern of fluorescence under a microscope.

Other options:

Indirect immunofluorescence (IIF) is a type of immunofluorescence that is used to diagnose autoimmune diseases. In IIF, antibodies are first applied to a patient's serum, and then the serum is applied to a skin biopsy specimen. The presence of antibodies in the patient's serum will cause the skin biopsy specimen to fluoresce under a microscope. Seen below is a centromeric pattern on IF in Sjogren's disease -1 and Diffuse/homogenous staining in SLE - 2.

Immunohistochemistry is a type of microscopy technique used to visualize the presence of specific proteins/receptors in cells or tissues. It uses antibodies that are labeled with enzymes or fluorescent dyes to bind to specific antigens. An example is receptor identification in breast cancer specimens.

Fluorescent In Situ Hybridisation (FISH) is a simple technique for the detection of specific DNA sequences in cells/tissues. It uses fluorescent labeled probes to bind to complementary DNA FISH can be performed on prenatal samples, peripheral blood cells, touch preparations from cancer biopsies, and even fixed archival tissue sections. It is used to detect

- numeric abnormalities of chromosomes (aneuploidy)
- subtle microdeletions and rearrangements.
- complex translocations that are not demonstrable by routine karyotyping
- gene amplification (e.g., HER2 in breast cancer or NMYC amplification in neuroblastomas)

Solution to Question 13:

The image given shows the row of tombstone appearance seen in Pemphigus vulgaris.

Pemphigus vulgaris is an autoimmune blistering disease caused by the deposition of IgG antibodies against desmoglein 3, a protein that holds skin cells together. It is characterized by widespread flaccid bullae, which are large, fluid-filled blisters that rupture easily. The lesions typically start on the trunk and spread to the extremities. Oral lesions are common in pemphigus vulgaris. Histologically, pemphigus vulgaris is characterized by a suprabasal split with acantholysis (loss of cohesion between skin cells).

Acantholysis and cell apoptosis are the key pathological processes in pemphigus. Histological changes include:

- Intercellular oedema with loss of intercellular attachments in the basal layer is the earliest change
- Suprabasal epidermal cells separate from the basal cells to form clefts and blisters.
- Basal cells remain attached to the basement membrane, in a ‘row of tombstones’ on the floor of the blister.
- Blister cavities contain rounded■up acantholytic cells, also seen in Tzank smears.

IgA disease (Linear IgA bullous dermatosis): Autoimmune blistering disease caused by the deposition of IgA antibodies at the dermal-epidermal junction. It is characterized by vesicles and bullae, which are large, fluid-filled blisters. The lesions typically start on the trunk and spread to the extremities. Oral lesions are not common in IgA disease. Histologically, IgA disease is characterized by a suprabasal split with linear IgA deposition at the dermal-epidermal junction. Anti-transglutaminase antibodies are positive in dermatitis herpetiformis. Patients present with erythematous papules and vesicles on the extensor aspect of elbows, knees, buttocks, and scalp. Histopathology demonstrates subepidermal blister formation with neutrophils located at the tips of the dermal papillae. Direct immunofluorescence will reveal the deposition of IgA in the papillary dermis in a granular or fibrillar pattern.

Pemphigus foliaceus

Pemphigus foliaceus is an autoimmune blistering disease caused by the deposition of IgG antibodies against desmoglein 1, a protein that holds skin cells together. It is characterized by superficial, flaccid bullae that are prone to rupture and erosion. The lesions typically start on the face and scalp and then spread to the trunk and extremities. Oral lesions are not common in pemphigus foliaceus. Histologically, pemphigus foliaceus is characterized by a subcorneal split with acantholysis and minimal inflammation.

The image given below shows the lesions seen in pemphigus foliaceous.

Type of pemphigus	Cutaneous distribution	Mucosal involvement	Lesions
Pemphigusvulgaris	- Scalp, face, chest, upper trunk- May be generalised	- Oro■pharynx, conjunctiva, genital	- Flaccid blisters- Erosions - Flexural vegetations
Pemphigusvegetans	- Flexural aspects	- Oro■pharynx, conjunctiva, genital	- Vegetating plaques- Vesicles- Pustules- Erosions
Pemphigusfoliaceous	- Scalp, face, chest, upper back, seborrheic areas- May be generalised	- None	- Scaly papules- Crusted erosions- Erythroderma
Paraneoplasticpemphigus	- Upper body- Palmoplantar	- Severe mucositis	- Polymorphous bullae- Erosions- Target lesions

Dermatology INI-CET 2024

Question 1:

Which of the following is not associated with annular lesions?

a) Tinea corporis
b) Pityriasis rosea
c) Urticaria
d) Porokeratosis

Question 2:

Match the following

a) a-3, b-4, c-2, d-1
b) a-1, b-4, c-2, d-3
c) a-3, b-2, c-4, d-1
d) a-2, b-4, c-3, d-1

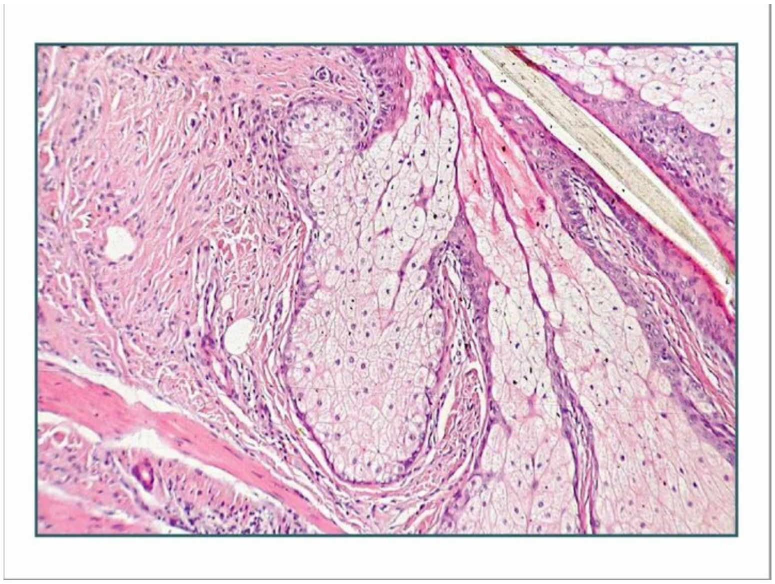
Question 3:

A hypopigmented patch which grows initially and stops progressing after sometime is seen in which type of vitiligo?

a) Focal
b) Segmental
c) Mixed
d) Acrofacial

Question 4:

What is the mechanism responsible for the release of secretions from the gland seen in the image below?



a) Apocrine

b) Merocrine

c) Eccrine

d) Holocrine

Question 5:

All are a part of NAME syndrome, a subset of Carney's complex, except?

a) Atrial myxoma

b) Nevi

c) Ebstein anamoly

d) Myxoid neurofibroma

Question 6:

A 60-year-old woman presented with a great toe infection. She had significant discoloration of the toenail with a subungual hyperkeratosis that caused color alteration and a rough surface area. She was suspected to have onychomycosis. Which topical drug is given for toenail onychomycosis?

a) Caspofungin

b) Itraconazole

c) Griseofulvin

d) Tavaborole

Question 7:

A male patient presents with urethritis. It can be caused by all except?

a) Neisseria Gonorrhea

b) Gardenella vaginalis

c) Trichomonas vaginalis

d) Mycoplasma

Answer Key

Question No.	Correct Option
1	c
2	a
3	b
4	d
5	c
6	d
7	b

Detailed Explanations

Solution to Question 1:

Urticaria is not associated with annular lesions. It is characterized by wheals.

Urticaria is a heterogeneous group of disorders characterized by itchy wheals, which develop due to evanescent edema of dermis.

A weal is a descriptive term for transient, well■demarcated, superficial pink or pale swellings of the dermis due to reversible exudation of plasma in the skin that fade, usually within hours, without leaving a mark . Weals are usually very itchy and associated with a surrounding red flare when they arise. Lesions begin as erythematous macules, which rapidly evolve into pale pink edematous wheals with a surrounding flare.

Option A: Tinea corporis/ ringworm infection is caused by the dermatophyte Trichophyton rubrum. It results from invasion and proliferation of the causal fungi in the stratum corneum with lesions on the trunk and limbs. Characteristic lesions are circular, usually sharply marginated with a raised edge with central resolution.

Option B: Pityriasis Rosea (PR) is an unknown etiology, self-limiting condition that typically lasts between 2 to 10 weeks. It is characterized by a herald patch, followed by a secondary eruption with typical lesions that are oval, annular, erythematous, and plaque-like with a collarette of scales. These lesions commonly appear on the trunk in a "fir tree" pattern along lines of cleavage. Differential diagnoses include guttate psoriasis, secondary syphilis, and drug rash. There is no specific treatment for PR; symptomatic treatment is usually sufficient.

Option D: Porokeratosis is a clonal expansion of keratinocytes which differentiate abnormally. But they are not hyperproliferative. It may present as single or multiple lesions and may be localized or disseminated. Porokeratoses present with single or multiple papules or plaques which develop into annular lesions with a thin raised border.

Solution to Question 2:

The correctly matched option is a-3, b-4, c-2, d-1.

Lupus vulgaris is one of the prevalent forms of cutaneous tuberculosis that occurs in previously sensitized individuals with high immunity. It is characterized by reddish-brown papules and slow-growing plaques with elevated borders and central atrophy. Diagnosis is made by biopsy. Caseating granulomas are pathognomonic of tuberculosis.

Tinea corporis/ ringworm infection is caused by the dermatophyte Trichophyton rubrum. It results from invasion and proliferation of the causal fungi in the stratum corneum with lesions on the trunk and limbs. Characteristic lesions are circular, usually sharply marginated with a raised edge with central resolution.

KOH mount is used for the diagnosis. Branching hyphae are seen in Tinea corporis.

Leprosy is a chronic granulomatous condition caused by Mycobacterium leprae. It affects the skin, cutaneous and peripheral nerves, and mucous membranes.

Cardinal signs for the diagnosis of leprosy:

• Characteristic skin lesions- hypopigmented or erythematous lesions with definite impairment of sensation.

• Peripheral nerve involvement- thickening of the nerve with sensory impairment. Nerve tenderness can also be seen.

• Detection of acid-fast bacilli in slit-skin smear, skin biopsy, or positive biopsy PCR.

Allergic contact dermatitis is an eczematous reaction that occurs as an immunological response that occurs due to exposure to a substance towards which the immune system has previously been sensitized.

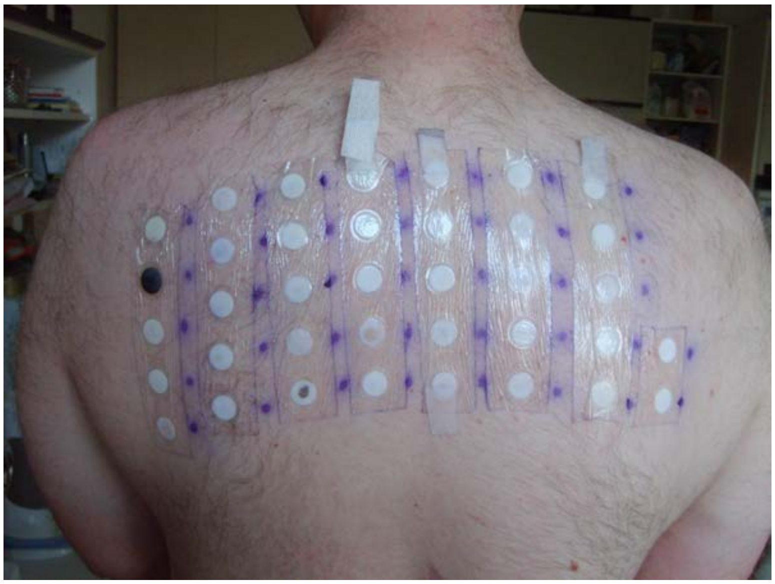
The patch test is used to detect delayed (type IV) hypersensitivity to antigens that come in contact with the skin, by applying the suspected antigen under occlusive patches to the skin. Patches are removed after 48 hours (or sooner if there is itching and burning).

• 48 hours: Patch removal for initial reading to detect delayed hypersensitivity reaction.

• 72-96 hours: Final reading done for confirmation.

An image showing the patch test:

a. Patch test	3. Contact dermatitis
b. KOH mount	4. Tinea corporis
c. Biopsy	2. Lupus vulgaris
d. Slit skin smear	1. Leprosy



Solution to Question 3:

In segmental vitiligo, lesions increase initially (6–12 months) and then remain static.

Vitiligo is a localized depigmentation that occurs due to a progressive loss in the melanocytes in the basal layer of skin. It is classified as

• Segmental vitiligo- which appears as unilateral maculae in a segmented or band-shaped pattern

• Non-segmental vitiligo- which appears as bilateral maculae distributed in an acrofacial pattern or scattered symmetrically over the entire body. Non-segmental can either be focal vitiligo or undetermined/unclassified vitiligo.

Acrofacial vitiligo is a type of generalized vitiligo, predominantly seen periorificially (around eyes and on lips on the face) and acral parts (periungual area, palms, and soles). This type of vitiligo is more resistant to therapy due to absence of hair in the affected parts.

Mixed vitiligo: the coexistence of non■segmental and segmental vitiligo in one patient is called mixed vitiligo

According to a recent Vitiligo Global Issue Consensus Conference, the term ‘vitiligo’ can be used as an umbrella term for all non■segmental forms of vitiligo (including several variants: acrofacial, mucosal, generalized, universal, mixed, and rare variants of vitiligo). The classification is shown below.

Clinical features:

The lesions become apparent between 20-30 years of age.

Hypopigmented macules in vitiligo are commonly found in the hips, dorsum of the hands/fingers, feet, elbows, knees, and ankle areas. These are the areas of repeated friction. Koebner phenomenon is seen, which is the development of lesions at the site of trauma on the uninvolved skin.

Vitiligo universalis, which is complete vitiligo, or trichrome vitiligo where there are areas of depigmentation, hyperpigmentation, as well as areas of normal skin may be seen rarely.

Several autoimmune conditions can be associated with vitiligo. They are:

• Thyroid disease (hyperthyroidism and hypothyroidism)

• Pernicious anemia

• Addison disease

• Diabetes

• Hypoparathyroidism

Examination:

Under the Wood's lamp examination the depigmented areas emit a bright blue-white fluorescence and appear sharply demarcated.

Treatment:

Treatment involves cosmetic camouflage and the use of sunscreen. The topical application of corticosteroids is the first line of treatment. Topical calcineurin inhibitors like pimecrolimus, tacrolimus have also shown to be successful. Psoralen photochemotherapy, khellin, selective ultraviolet B therapy are used as second-line treatment. In patients who do not respond to this treatment, surgical grafting techniques can be used, where the melanocytes from a normal pigmented area are transferred to a non pigmented area. In patients with extensive vitiligo where only a few pigmented areas are left, depigmentation can be done.

Vitiligo (Non segmental)	AcrofacialMucosal (>1 one mucosal site)GeneralizedUniversalMixed (associated with SV)Rare variants
Segmental vitiligo	Uni-, bi-, or pleuri-segmental
Undetermined/unclassified vitiligo	FocalMucosal (one site in isolation)

Solution to Question 4:

The gland depicted in the image is a sebaceous gland, which releases its secretions through the holocrine mechanism.

Identification features of sebaceous gland:

• Lobular structure with clusters of cells.

• Foamy cytoplasm due to lipid content.

• Holocrine secretion with disintegrated cells.

• Found near hair follicles in the dermis.

Sebaceous gland cells synthesize and accumulate lipid-rich sebum within their cytoplasm.

Holocrine secretions are produced in the cytoplasm of the cell and released by the rupture of the plasma membrane, which destroys the cell and results in the secretion of the product into the lumen.

Examples of holocrine glands are

• Sebaceous glands of the skin and

• Meibomian glands of the eyelid

Other options:

Option A: Apocrine: involves partial loss of the apical portion of the cell (e.g., mammary glands).

Option B: Merocrine: involves secretion via exocytosis (e.g. sweat glands).

Option C: Eccrine glands are a type of sweat gland responsible for thermoregulation through the secretion of sweat. They use the merocrine mechanism for secretion.

Solution to Question 5:

Ebstein anomaly is not a part of NAME syndrome.

Carney Complex is a rare, autosomal dominant genetic disorder characterized by multiple neoplasms and lentigines. It includes features like cardiac myxomas, endocrine tumors, spotty skin pigmentation, and other benign or malignant tumors, caused by mutations in the PRKAR1A gene. It includes:

• Myxomas - cardiac, skin, and/or breast

• Lentigines and/or pigmented nevi

• Endocrine overactivity - primary nodular adrenal cortical disease, testicular tumors, and/or pituitary adenomas with gigantism or acromegaly

NAME syndrome and LAMB syndrome are synonymous with the Carney complex:

• LAMB - Lentigines, Atrial Myxoma, and Blue nevi.

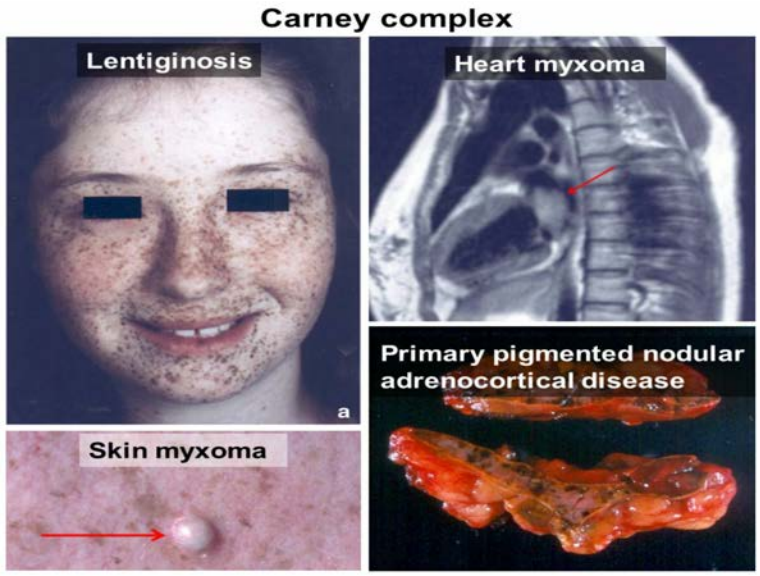
• NAME syndrome is a subset of Carney's complex, which is a rare autosomal dominant disorder characterized by multiple neoplasms and pigmentation abnormalities. The acronym NAME stands for:

• Nevi

• Atrial myxomas

• Myxoid neurofibromas

• Ephelides (freckle-like pigmented lesions)



Solution to Question 6:

Tavaborole is the topical drug given in toenail onychomycosis.

Tavaborole is a topical antifungal. It is an oxaborole antifungal that inhibits fungal protein synthesis by targeting leucyl-tRNA synthetase.

Onychomycosis is defined as the infection of the nail plate by fungus.

Most common cause is Trichophyton rubrum, but many fungi may be causative.

The clinical scenario of Tinea unguium presents with yellowish discolouration, which spreads proximally as a streak in the nail. Later, subungual hyperkeratosis becomes prominent and spreads until the entire nail is affected. Gradually, the entire nail becomes brittle and separated from its bed.

Diagnosis is made by KOH mount examination of dystrophic subungual debris.

Management of onychomycosis involves Oral terbinafine as the treatment of choice for Tinea unguium or onychomycosis. It acts as a non-competitive inhibitor of squalene epoxidase, which is essential in the synthesis of fungal ergosterol. Oral terbinafine at 250 mg/day is given for 6 weeks for fingernail infection and 3 months (12 weeks) for toenail infection. An oral Griseofulvin dose of 1 g/day for 4–8 months (longer for toenails) is used as a second-line treatment.

Other Options:

Option A: Caspofungin: Caspofungin is an echinocandin antifungal that works by inhibiting fungal cell wall synthesis. It is used in intravenous form for systemic fungal infections like candidiasis and aspergillosis.

Option B: Itraconazole: Itraconazole is an oral antifungal from the triazole class, used for onychomycosis due to its penetration into keratinised tissues.

Option C: Griseofulvin: Griseofulvin is an oral antifungal used for dermatophyte infections, including onychomycosis.

Solution to Question 7:

Gardenella vaginalis does not cause urethritis in male patients.

Urethritis is classified into two types:

• Gonococcal urethritis - caused by Neisseria gonorrhoeae(Option A)

• Non-gonococcal urethritis - caused by:

• Chlamydia trachomatis (most common)

• Ureaplasma urealyticum

• Trichomonas vaginalis (rare) (Option C)

- Herpes simplex virus (rare)
- Mycoplasma genitalium (Option D)

The most likely causative organism for acute urethritis in males is Neisseria gonorrhoeae.

Dermatology NEET 2024

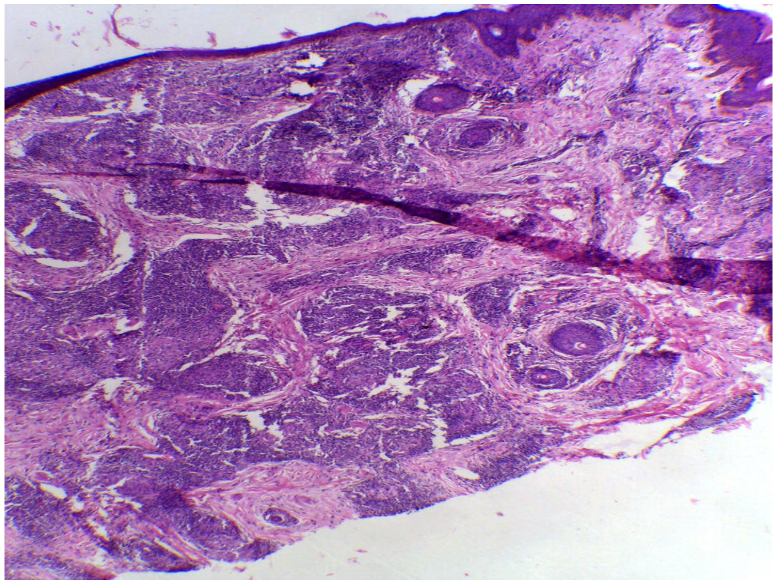
Question 1:

A child presents with orangish hue skin with rough and scaly skin as shown in the image. On histopathological examination, orthokeratosis and parakeratosis are seen. What is the most likely diagnosis?

- **a)** Follicular psoriasis
- **b)** Lichen planus
- **c)** Pityriasis rubra pilaris.
- **d)** Keratosis follicularis

Question 2:

A patient presented with a slight loss of sensation along the ulnar nerve and a solitary nodule in the forearm. Intradermal antigen was injected. The histopathology image is shown below. Which of the following is true regarding this condition?



a) Lepromatous leprosy with dermal test positive

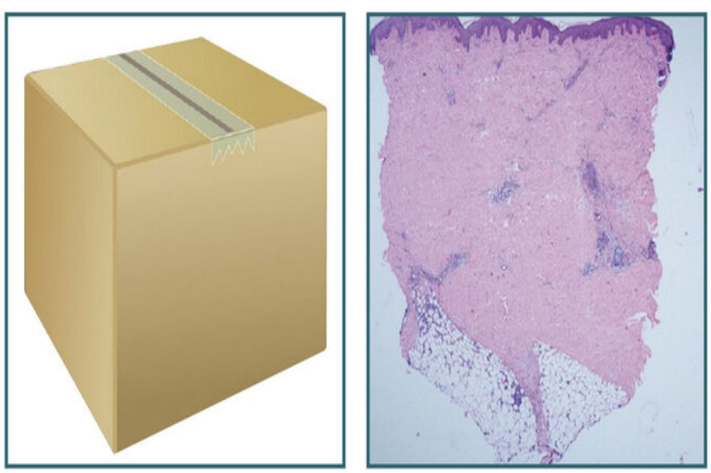
b) Tuberculoid leprosy with dermal test positive

c) Lepromatous leprosy with dermal test negative

d) Erythema nodosum leprosum with negative skin test

Question 3:

Which of the following is associated with the" box sign"?



a) Lichen planus

b) Lichen nitidus

c) Morphea

d) Amyloidosis

Question 4:

A 32-year-old man presents with painful vesicular lesions over the genital area. He has a history of multiple sexual partners and inconsistent condom use. What is the most likely diagnosis?

a) Chancroid

b) Genital Herpes

c) Primary syphillis

d) Granuloma inguinale

Question 5:

An adult male presented with asymptomatic macules over the chest, moist grey, perianal lesions, and findings on the scalp as shown. What is the most likely causative organism of this condition?

a) Malassezia

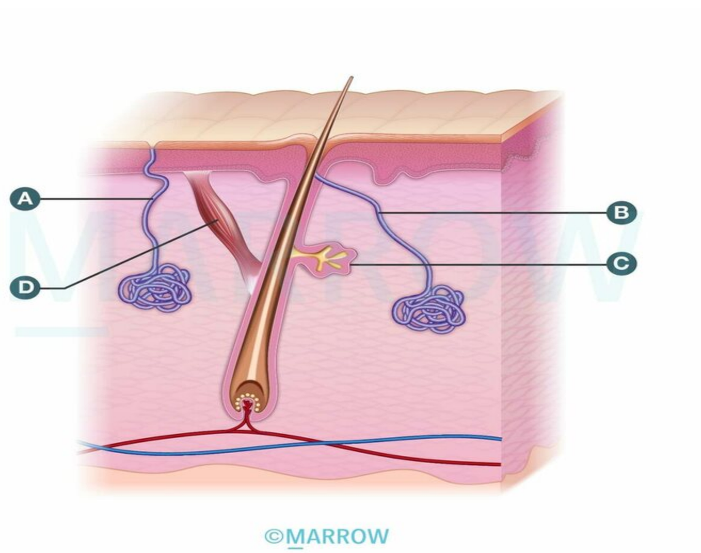
b) Trichophyton

c) Treponema pallidum

d) Microsporum

Question 6:

Match the structures shown in the image:



a) A- Apocrine sweat gland; B- Eccrine sweat gland; C- Sebaceous gland; D- Erector pili muscle

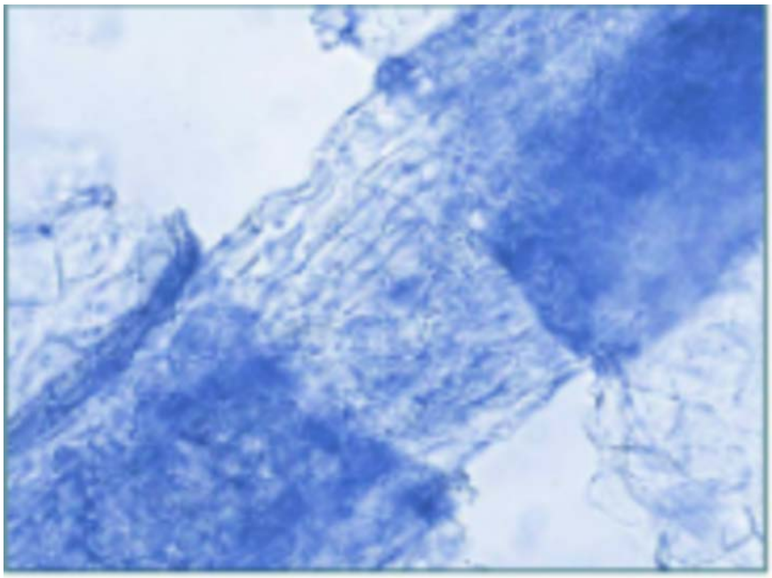
b) A- Eccrine sweat gland; B- Apocrine sweat gland; C- Sebaceous gland; D- Erector pili muscle

c) A- Sebaceous gland; B- Apocrine sweat gland; C- Eccrine sweat gland; D- Erector pili muscle

d) A- Eccrine sweat gland; B- Sebaceous gland; C- Apocrine sweat gland; D- Erector pili muscle

Question 7:

Which of the following causes the condition shown in the image below?



a) Candida albicans

b) Trichophyton species

c) Epidermophyton floccosum

d) Microsporum species

Question 8:

A 65-year-old treated leprosy patient came with complaints of a well-demarcated, non-tender ulcer on his foot with hyperkeratotic borders as given below for the last 3 months. What should be the appropriate next step?

a) Restart MDT

b) Rest the limb, non weight bearing splints and foot care

c) Admit the patient and aggressive debridement and antibiotics

d) Amputation

Question 9:

A man presents with multiple lesions on his face and upper back, diagnosed as acne vulgaris. Which of the following is the most effective treatment for his symptoms?

a) Systemic Minocycline

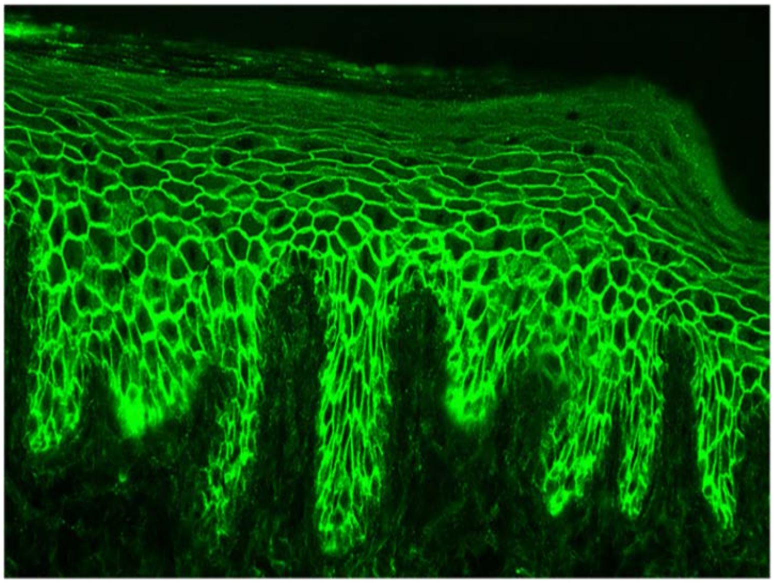
b) Systemic Isoretinoin

c) Systemic Doxycycline

d) Topical Dapsone

Question 10:

A 42-year-old man comes with vesicular eruptions in the buccal mucosa. Direct immunofluorescence is done and the image is shown below. What is the most likely diagnosis?



a) Lupus Vulgaris

b) Pemphigus vulgaris

c) Bullous Pemphigoid

d) Borderline Leprosy

Question 11:

A 3-year-old boy had patches of partial hair loss with mild itching and scales. The picture of his scalp is shown below. His sister also has a similar condition. Which of the following is the most likely diagnosis?

a) Alopecia areata

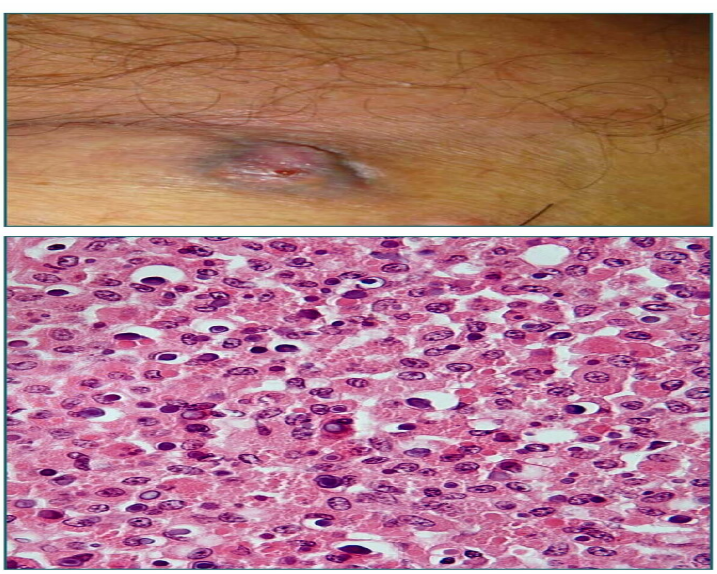
b) Tinea capitis

c) Pediculis Capitis

d) Pyoderma Scalp

Question 12:

A middle aged man comes with complaints of painless progressively enlarging lesions on his scalp and perianal region for the last 3 months. He has had liver and kidney transplant 2 years back and is on Mycophenolate Mofetil, Tacrolimus and Prednisolone for his immunosupression. Which of the following is the most likely diagnosis?



a) Drug induced eruption

b) Malakoplakia

c) Cutaneous TB

d) Pyoderma Gangrenosum

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	b
5	c
6	b
7	b
8	b
9	b
10	b
11	b
12	b

Detailed Explanations

Solution to Question 1:

The image shows prominent erythema and scale on the palms of the hands and wrists with marked orange■yellow palmoplantar keratoderma which is a feature of pityriasis rubra pilaris.

Features of pityriasis rubra pilaris:

• Salmon red or orange red-coloured dry scaly plaques are seen characteristically, which coalesce together, leaving islands of normal skin, typically known as 'islands of sparing' in between the lesions

• Most common site - trunk

• On the elbow and wrist, nutmeg grater papules are seen due to follicular hyperkeratosis

• Palmoplantar keratoderma - keratoderma on the sole is very thick that it appears as a sandal. (keratotic sandal)

• Nails - thickened, discoloured distally, showing splinter haemorrhages

Histologically, pityriasis rubra pilaris is characterized by hyperkeratosis with follicular plugging, alternating ortho- and parakeratosis, dilated but non-tortuous dermal capillaries, sparse lymphocytic infiltrate, and possible acantholysis in the adnexal epithelium.

The image below shows islands of sparing in pityriasis rubra pilaris.

The image below shows keratotic sandal.

Other options:

Option A: Follicular psoriasis affects hair follicles on the trunk and limbs and can occur alone or alongside plaque psoriasis. The lesions are smaller than those seen in guttate psoriasis and may appear clustered or spread out. In children, it can be mistaken for pityriasis rubra pilaris.

Option B: Lichen planus is a papulosquamous lesion presenting as pruritic, purple, polygonal, planar, papules and plaques. These papules are often highlighted by white dots or lines called

Wickham striae, which are created by areas of hypergranulosis.

Option D: Keratosis follicularis or Darier's disease or dyskeratosis follicularis, is an autosomal dominant condition arising from mutations in the ATP2A2 gene which affects the SERCA2 calcium pump. It presents with greasy hyperkeratotic papules and plaques in seborrheic regions. Neuropsychiatric features such as depression, psychosis and mental retardation may also occur.

Solution to Question 2:

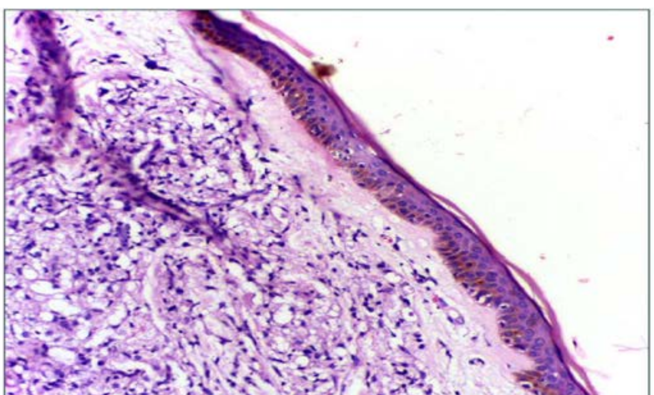
The patient’s slight loss of sensation along the ulnar nerve, solitary forearm nodule, and histopathology showing well-defined granulomas indicate tuberculoid leprosy with a dermal test positive.

In tuberculoid leprosy, dermal granulomas, consisting of groups of epithelioid cells with giant cells, are seen. The granulomas are elongated and generally run parallel to the surface, following neurovascular bundles. The granulomas extend up to the epidermis, thereby obscuring the Grenz zone. It also shows high positivity of the lepromin skin (dermal) test.

Leprosy is a chronic granulomatous condition caused by Mycobacterium leprae. It affects the skin, cutaneous and peripheral nerves, and mucous membranes. The most commonly affected peripheral nerves are the posterior tibial nerve, followed by the ulnar, median, and lateral popliteal nerves. The most frequently involved cranial nerve is the facial nerve. It has a long incubation period, usually 2 to 12 years. It commonly presents as an area of numbness on the skin or a visible skin lesion, which could be a hypopigmented or erythematous macule.

The Grenz zone is most prominent in lepromatous leprosy, as there are scarce granulomas to fill up the space. It is also seen in the borderline spectrum.

Grenz Zone in Lepromatous leprosy



Narrow area of uninvolved papillary dermis seen

Cardinal signs for the diagnosis of leprosy:

• Characteristic skin lesions- hypopigmented or erythematous lesions with definite impairment of sensation.

• Peripheral nerve involvement- thickening of the nerve with sensory impairment. Nerve tenderness can also be seen.

• Detection of acid-fast bacilli in slit-skin smear, skin biopsy, or positive biopsy PCR.

The presence of two of the three signs can establish the diagnosis of leprosy. Other tests done for diagnosis of leprosy include nerve biopsy in pure neural leprosy and PGL-1 antibody in cases with low bacterial load.

Solution to Question 3:

The "box sign" is associated with morphea, also known as localized scleroderma.

Morphea is a condition characterized by localized skin sclerosis and fibrosis due to increased collagen deposition. This can lead to well-defined, box-like patches of thickened, hardened skin. The "box sign" refers to the sharp, boxy edges of these lesions, which can vary in depth and extent, sometimes involving deeper tissues such as muscle and bone.

Biopsies taken during the indurative stage often display squared-off edges, resulting in a "box-shaped" or "boxed dermis" appearance under the microscope, which can vary in depth and extent, sometimes involving deeper tissues such as muscle and bone. This characteristic sign occurs because of extensive collagen sclerosis throughout the reticular dermis and extending into the subcutaneous fat septa. The pronounced sclerosis of the collagen fibers causes this distinct boxy appearance. Similar squared-off edges may also be observed in morphea-lichen sclerosus overlap and scleredema, due to comparable changes in the dermal and subcutaneous tissue structure.

The epidermis in morphea can exhibit varying changes, from normal to flattened with loss of rete ridges, or slightly acanthotic. In the inflammatory phase, the reticular dermis shows edema and a dense perivascular infiltrate, which gradually gives way to thickened, horizontally oriented collagen bundles. The loss of adnexal structures, such as hair follicles and eccrine glands, is common, and the subcutaneous fat may be replaced by collagen, contributing to the appearance of a "box sign."

Management of morphea focuses on controlling inflammation and preventing disease progression. Early intervention is crucial and may include topical treatments like corticosteroids or tacrolimus for localized lesions. For more extensive or resistant cases, phototherapy and systemic treatments, such as methotrexate or oral corticosteroids, may be considered. The choice of therapy often depends on the extent and activity of the disease, with the goal of minimizing skin damage and improving functional outcomes.

Other options:

Option A: Lichen planus is a papulosquamous lesion presenting as pruritic, purple, polygonal, planar, papules and plaques. These papules are often highlighted by white dots or lines called Wickham striae, which are created by areas of hypergranulosis.

The characteristic lesions in lichen planus is shown below:

Option B: Lichen nitidus is a rare variant of lichen planus, where both are histologically similar in early lesions. It is mostly seen in children and young adults. Lesions of lichen nitidus are typically described as a minute pinpoint to pinhead-size, flat or dome-shaped papules with a shiny surface. They are most commonly found over the forearm, dorsum of the penis, abdomen, chest, and buttocks.

Lichen nitidus

Pinhead-sized flesh-colored papules seen.

Option D: Amyloidosis refers to extracellular deposits of fibrillar proteins. The kidney is the most common organ involved in amyloidosis. Abnormal fibrils are produced by the aggregation of misfolded proteins. The fibrillar deposits bind a wide variety of proteoglycans and glycosaminoglycans, including heparan sulfate and dermatan sulfate, and plasma proteins, notably serum amyloid P component (SAP). Amyloid appears as an amorphous, eosinophilic, hyaline, extracellular substance.

Solution to Question 4:

The most likely diagnosis for this patient with painful vesicular lesions over the genital area is genital herpes.

The diagnosis of genital herpes is suggested by the presence of painful, multiple vesicular ulcers clustered on the genitalia, along with inguinal lymphadenopathy. These symptoms are indicative of an infection caused by the Herpes Simplex Virus-2 (HSV-2), which is the primary cause of genital herpes.

Genital herpes is characterized by vesiculoulcerative lesions, which appear as grouped vesicles filled with clear fluid on the penis in males or on the cervix, vulva, vagina, and perineum in females. The painful nature of these lesions, combined with the history of multiple sexual partners and inconsistent condom use, supports the diagnosis. Recurrences of genital herpes are common and generally tend to be milder compared to the initial outbreak.

Other options:

Option A: Chancroid caused by Haemophilus ducreyi may also present with soft wide indurated painful genital ulcers and painful genital lymphadenopathy.

Option C: In primary syphilis, the typical primary chancre usually begins as a single painless papule that rapidly becomes eroded and usually becomes indurated, with a characteristic cartilaginous consistency on palpation of the edge and base of the ulcer.

Option D: Granuloma inguinale (also known as donovanosis) is caused by Klebsiella granulomatis. It presents with a painless papule or nodule that erodes into a beefy red granulomatous ulcer with rolled edges and often no lymphadenopathy.

Solution to Question 5:

The most likely causative organism for this condition is Treponema pallidum, which caused the asymptomatic macules on the chest, moist grey, perianal lesions, and moth-eaten pattern alopecia seen in secondary syphilis.

The causative organism for this condition is Treponema pallidum, a spirochete bacterium. In secondary syphilis, the macular or roseolar rash is often the earliest sign, which may fade, leaving hypopigmented areas around the neck, known as leucoderma syphiliticum or "necklace of Venus." This stage also features a papular rash that can appear on the palms and soles.

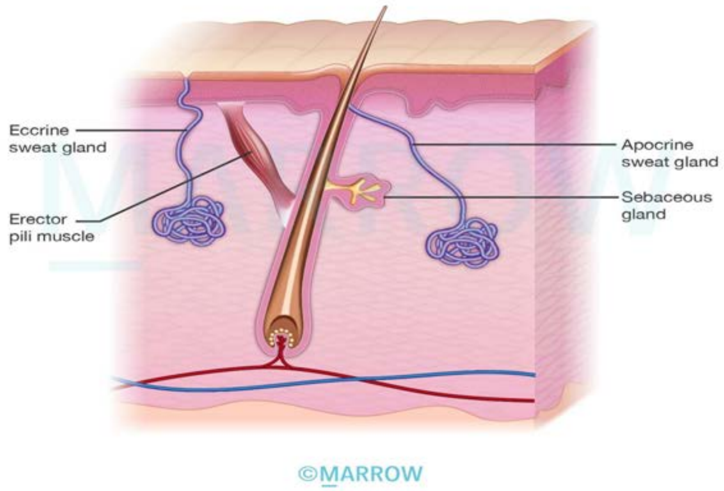
Additionally, secondary syphilis can present with nummular lesions that resemble psoriasis, often with easily removable scales due to serous discharge. On the oral mucosa, papules may form grey oval patches, leading to "snail-track" ulcers (as shown below).

Condyloma lata, which are moist, well-demarcated papules and plaques, are commonly seen in the genital area and are highly infectious. Syphilitic alopecia, characterized by non-scarring patchy hair loss, and generalized lymphadenopathy are also notable features.

The systemic manifestations of secondary syphilis include panuveitis, periostitis, joint effusions, and involvement of various organs such as the kidneys, liver, and heart. The lesions in secondary syphilis typically resolve spontaneously, transitioning into the latent stage of the disease.

Solution to Question 6:

The correctly matched option (B)- A- Eccrine sweat gland; B- Apocrine sweat gland; C- Sebaceous gland; D- Erector pili muscle.



Human skin contains various structures, including sweat and sebaceous glands. Eccrine sweat glands, which are crucial for temperature regulation, are distributed across most of the body and are identified early in fetal development. These glands have a coiled secretory portion in the lower dermis and a duct that opens directly onto the skin surface. They are most concentrated on the palms, soles, and forehead.

Apocrine sweat glands, found in areas like the axillae, genital, and mammary regions, differ from eccrine glands in that they release secretion via decapitation, where parts of the cell pinch off. They are connected to hair follicles and have a larger lumen compared to eccrine glands. Their primary function remains unclear, and they have a lower secretory output.

Sebaceous glands secrete sebum, providing skin lubrication and waterproofing. These glands are associated with hair follicles and are absent in non-hairy skin.

The erector pili muscle, a smooth muscle bundle, extends from the dermis to the follicle wall and is involved in hair erection.

Solution to Question 7:

The above image shows numerous fungal spores within the hair shaft suggestive of endothrix type of tinea capitis, which is caused by Trichophyton tonsurans.

Tinea capitis is of four types:

Other options:

Option A: Candida is the most common cause of opportunistic mycoses in the world. Immunocompromised individuals are at an increased risk of systemic candidiasis. Clinically, infection with candida can present as cutaneous & mucosal candidiasis, systemic candidiasis, or rarely, as chronic mucocutaneous candidiasis.

Morphologically, they are yeast-like fungi. It appears as budding gram-positive yeast cells with pseudohyphae

Option C: Epidermophyton floccosum: Distinguished by absent microconidia and club-shaped macroconidia, which are smooth, thick-walled, and occur in clusters. It is a dermatophyte responsible for superficial fungal infections of the skin, particularly tinea pedis (athlete’s foot), tinea cruris (jock itch), and tinea unguium (nail infection).

Option D: Tinea cruris, a superficial fungal skin infection that affects the groin area, is primarily caused by Epidermophyton floccusum. Typical symptoms include redness and inflammation in the groin region, accompanied by sensations of itching, burning, or stinging. The characteristic lesions exhibit raised borders surrounding a central clearing, thus imparting a ring-like appearance.

Type of Infection	Causative fungus
Endothrix (infection inside shaft)	Trichophyton tonsuransT.violaceumT. schoenleiniiT.yaoundeiT. soudanense
Ectothrix(infection outside shaft)	MicrosporumaudouiniiM.canisM.equinumM.ferrugineum

Type of Infection	Causative fungus
Favus	T. schoenleinii
Kerion	T. verrucosumT. mentagroph ytes

Solution to Question 8:

The given scenario and the image indicate a neuropathic ulcer. This can be managed by limb rest, utilising non-weight-bearing splints, and administering foot care.

Leprosy, a condition affecting the skin and peripheral nerves, results in peripheral neuropathy and subsequent loss of sensation. This can lead to the development of chronic trophic or neuropathic ulcers characterised by punched-out appearance, regular margins, and calluses due to repeated trauma in the presence of anaesthesia of feet.

Management primarily focuses on relieving pressure through the application of non-weight bearing splints for even pressure distribution across the entire foot, limb elevation to mitigate swelling, and regular foot inspection to promptly identify trauma, in addition to the use of microcellular rubber footwear.

Other options:

Option A: Trophic ulcers are not an indication for restarting MDT in an already treated leprosy patient.

Option C: Due to the absence of debris or signs of infection in the ulcer, aggressive debridement and antibiotics are not indicated.

Option D: As there are no signs of vascular compromise or gangrene, amputation of the limb is not required.

Solution to Question 9:

The presence of multiple nodular and cystic lesions over the face and upper back implies that this is a case of acne vulgaris. Systemic Isotretinoin is the most effective treatment for acne.

Systemic retinoids are a very efficient treatment for acne vulgaris. Their mechanism of action si by inhibiting the formation of keratin, a protein that can clog pores and cause acne. Oral retinoids are usually reserved for severe cases of acne that do not respond to topical creams and gels.

Acne is a chronic inflammatory disease of the pilosebaceous unit. Comedonal lesions are the earliest lesions of acne. They represent the keratin plugs of the pilosebaceous duct.

When the keratin plug is inside the follicle, it is called a closed comedone (whitehead). When the keratin plug is exposed to the environment, it gets oxidized by the air and turns black which is known as open comedone (blackhead).

Solution to Question 10:

The given image shows a characteristic fish-net appearance on direct immunofluorescence, which is diagnostic of pemphigus vulgaris.

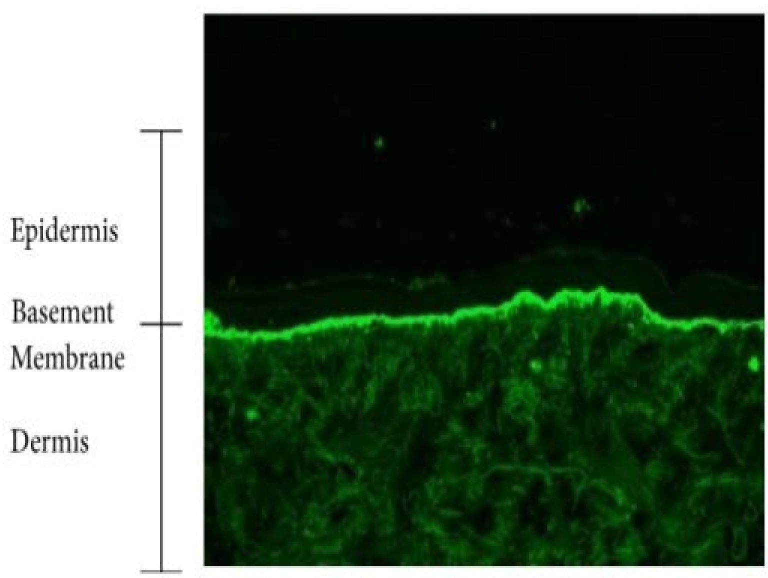
This pattern is due to intraepidermal intercellular deposition of IgG and C3.

Direct or indirect immunofluorescence cannot be used to differentiate between pemphigus vulgaris and pemphigus foliaceus, as they present similarly.

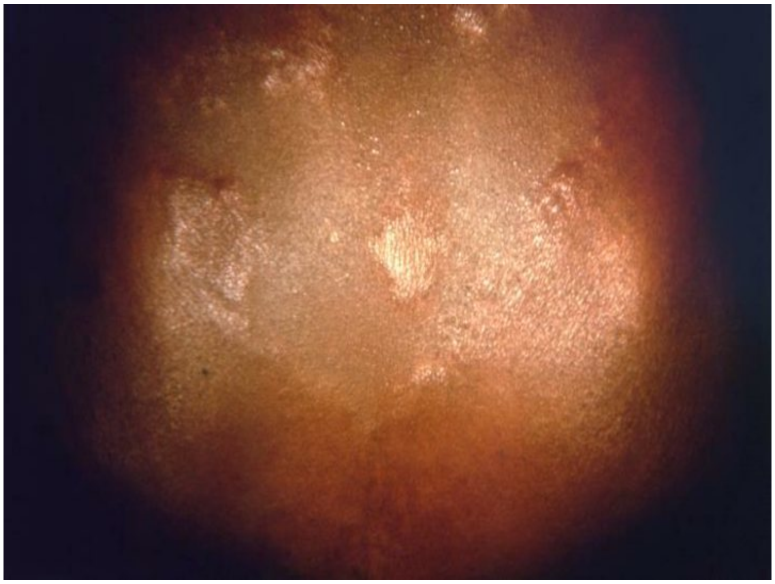
Other options:

Option A: Lupus vulgaris(Cutaneous TB) is the most common form of cutaneous tuberculosis in adults. The characteristic lesion is a plaque, composed of soft, reddish brown papules, the appearance on the diascopy being said to resemble apple jelly. The image below shows the characteristic lesion:

Option C: Bullous pemphigoid present linear or shoreline pattern is seen due to IgG and C3 deposits along the dermo-epidermal junctions, as seen in the following image.



Option D: In borderline leprosy, inverted-saucer-shaped lesions are seen and are characterized by a dome-shaped appearance with central infiltration and a gradually sloping peripheral edge.



Solution to Question 11:

Patchy hair loss and multiple black dots within the areas of alopecia along with fine scaling and itching of the scalp point towards the diagnosis of tinea capitis.

Tinea capitis is a dermatophyte infection primarily occurring in children. Clinical features of tinea capitis are patchy hair loss, scaling, tender lymphadenopathy, and in severe cases, there might be pustules and crusting of the affected area along with sinus formation.

The clinical variants of tinea capitis are:

Endothrix - a non-inflammatory type in which multiple black dots are present within the areas of alopecia. It is most commonly caused by T. tonsurans and T. violaceum.

Ectothrix - a mild inflammatory type in which single or multiple scaly patches with hair loss are seen. Microsporum audouinii, M.canis, M.equinnum, and M.ferrugineum cause ectothrix infection.

Kerion - a severe inflammatory type that presents with an inflammatory mass with thick crusting and matting of hair with pus and sinus formation. It is commonly caused by T. verrucosum and T. mentagrophytes.

Favus - is characterized by the presence of yellowish, cup■shaped crusts known as scutula that develop around hair follicles. It is caused by T. schoenleinii.

Other options:

Option A: Alopecia areata is a chronic inflammatory disease causing non-scarring hair loss. It is characterized by a circumscribed, smooth patch of hair loss. Exclamation mark hairs are often seen at the margins of the bald patches during active phases of the disease.

The image below shows alopecia areata.

Option C : Pediculis Capitis (head louse infestation) is very common in children. The majority of the transmission is by head-to-head contact, which is frequently observed with overcrowding (as in schools), poverty, and poor hygiene.

Symptoms include severe itching of the scalp. On examinations, nits are seen, which are firmly attached to hair shafts on which they can be freely glided.

Option D : Pyoderma of the scalp is an idiopathic cutaneous disorder. It is characterized by progressive painful ulceration which can cause destruction of underlying structures. It is a benign lesion, which typically develops after truama to that area.

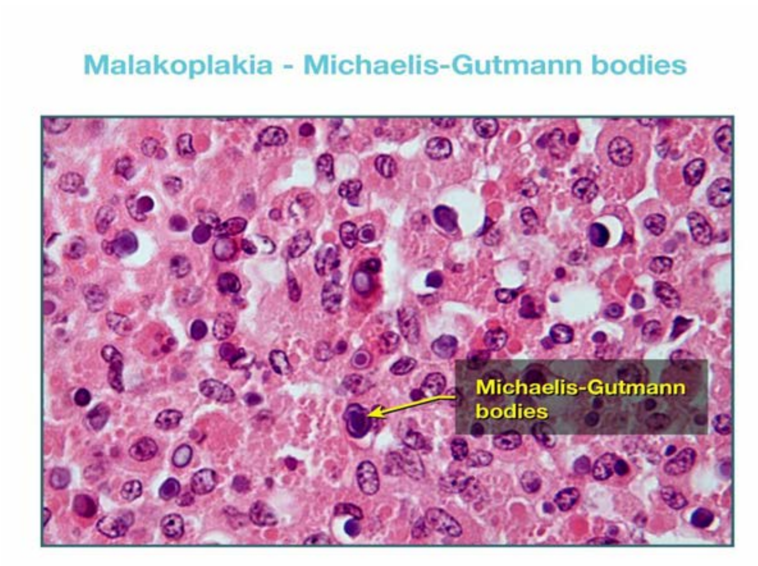
Solution to Question 12:

Clinical scenario of painless progressive nodular, ulcerated lesions in an immuno-compromised patient with the clinical image showing yellow-erythematous-purple plaque and histopathology image showing Michaelis–Gutmann bodies are consistent with the diagnosis of cutaneous malakoplakia.

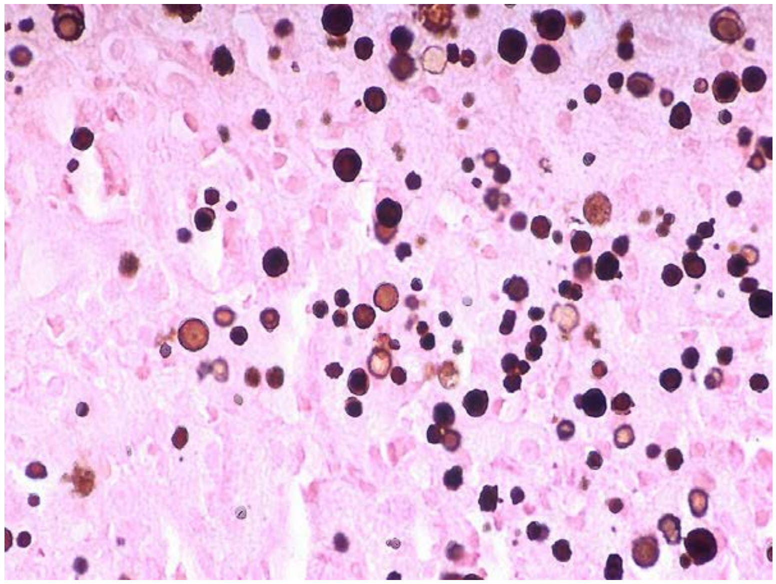
Malakoplakia is a rare inflammatory condition that typically occurs in the urinary tract. The cutaneous form of malakoplakia is less prevalent and it most commonly occurs in the perianal or genital regions. . It is caused due to acquired defects in phagocyte function, chronic bacterial infection by Escherichia coli or Proteus species, and immunosuppression (renal transplant patients).

The term malakoplakia means soft plaques. Grossly, malakoplakia appears as soft, yellow, slightly raised mucosal plaques.

Microscopically, they are composed of large, foamy macrophages mixed with occasional multinucleate giant cells and lymphocytes. The macrophages in malakoplakia have abundant granular cytoplasm due to phagosomes stuffed with undigested bacterial products and intracellular deposition of calcium and iron within enlarged lysosomes form laminated basophilic concretions. These are known as Michaelis–Gutmann bodies, pathognomonic of malakoplakia (shown below).



The image given below shows the Von Kossa stain demonstrating calcium and iron-containing Michaelis-Gutman bodies (MG bodies) in malakoplakia.



Other options :

Option A : Drug induced eruption can mimic a wide range of dermatoses. They can develop due to traditional pharmaceutical agents, biologic agents and targeted immunotherapeutic agents, food additives, and dyes. The morphologies of dryg induced eruptions can be morbilliform, urticarial, papulosquamous, pustular, and bullous.

Option C : Cutaneous TB (Lupus vulgaris) is the most common form of cutaneous tuberculosis in adults. The characteristic lesion is a plaque, composed of soft, reddish brown papules, the appearance on diascopy being said to resemble apple jelly.

Option D : Pyoderma gangrenosum is a non■infectious neutrophilic dermatosis commonly associated with inflammatory bowel disease. Central necrosis and ulceration of the epidermis and dermis surrounded by a strong inflammatory cell infiltrate are typical findings.